Krill oil supplementation with and without resistance training lowers the burden of the metabolic syndrome but does not affect glucose tolerance in sedentary adults: a randomized controlled trial

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1. Introduction

This document includes supplementary data and computational details related to the paper "Krill oil supplementation with and without resistance training lowers the burden of the metabolic syndrome but does not affect glucose tolerance in sedentary adults: a randomized controlled trial".

1.1 Project organisation

This notebook is built from source code available at the github repository trainome/alpha-omega-trial. The data used in analyses is organized as an R data package available in a separate repository trainome/AOData.

1.1.1 R scripts

- `R/fig-opt-fun.R` - Helper functions and settings for data visualisations.
- `R/helpers.R` - Helper functions for modelling.
- `R/packages.R` - Loading required packages.
- `R/data-prep.R` - Prepares data for the Metabolic syndrome analysis.
- `R/sim-fun.R` - Functions related to simulation studies.

1.1.2 Stan models

All Stan model scripts are stored in `/models/`. If model code is executed locally, model executable are also stored here. Model fits are stored in `fits/` but not pushed to github due to their large size.

1.2 Packages and computational environment

Stan (CmdStan) version 2.39.0 was used in compiling this notebook. R packages are listed below together with additional session info.

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26300)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=Norwegian Bokmål_Norway.utf8
[2] LC_CTYPE=Norwegian Bokmål_Norway.utf8
[3] LC_MONETARY=Norwegian Bokmål_Norway.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=Norwegian Bokmål_Norway.utf8

time zone: Europe/Oslo
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
```

```
 [1] Gmisc_3.3.0       htmlTable_2.5.0   corrr_0.4.5       loo_2.9.0
 [5] posterior_1.7.0   cmdstanr_0.9.0    ggh4x_0.3.1       gt_1.3.0
 [9] patchwork_1.3.2   ggrepel_0.9.8     ggtext_0.1.2      lubridate_1.9.5
[13] forcats_1.0.1     stringr_1.6.0     dplyr_1.2.1       purrr_1.2.2
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[21] tidyverse_2.0.0   AOData_0.0.1.9000

loaded via a namespace (and not attached):
 [1] gtable_0.3.6         tensorA_0.36.2.1     xfun_0.57
 [4] htmlwidgets_1.6.4    processx_3.9.0       lattice_0.22-9
 [7] tzdb_0.5.0           vctrs_0.7.3          tools_4.6.0
[10] generics_0.1.4       parallel_4.6.0       cluster_2.1.8.2
[13] pkgconfig_2.0.3      data.table_1.18.4    checkmate_2.3.4
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[43] scales_1.4.0         backports_1.5.1      timechange_0.4.0
[46] rmarkdown_2.31       matrixStats_1.5.0    nnet_7.3-20
[49] otel_0.2.0           gridExtra_2.3        ragg_1.5.2
[52] hms_1.1.4            evaluate_1.0.5       knitr_1.51
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[58] glue_1.8.1           xml2_1.5.2           rstudioapi_0.18.0
[61] jsonlite_2.0.0       R6_2.6.1             systemfonts_1.3.2
[64] fs_2.1.0
```

2. Analytic approach and data description

2.1 Data analysis

For inferential statistics, Bayesian models were constructed using the Stan probabilistic programming language (1) and fitted using the `cmdstanr` interface (2) in R (3). The choice of a Bayesian modeling framework allow us to naturally handle missing data in a wide class of models. This was important as it enabled us to model data from the project as *intention-to-treat* with the uncertainty associated with missing values incorporated into the posterior estimates.

A general workflow was adopted where models were incrementally built and tested for validity through simulations (4). First, implications of prior distributions were evaluated using prior-predictive simulations. Priors were selected to constrain model predictions and parameter values to *a priori* plausible ranges. This approach encourages the use of weakly informative priors on model parameters, which regularize models and shrink estimates and predictions without distorting the influence of the data on inference (5–7). In relevant cases, synthetic data were used to confirm that model inferences were not biased by model misspecification. Samples from the joint posterior distribution were used to create posterior predictive checks, in which model implications are simulated and compared with characteristics of the observed data (7). Selected models were also assessed for inferential capabilities using synthetic data simulations where parameter values are known. The sensitivity of model specifications to inference was assessed by comparing predictions across different models.

All sampling from prior and posterior distributions were done using the Hamiltonian Monte Carlo (HMC) no-U-turn sampler (Stan Reference Manual: MCMC sampling). Performance of the HMC algorithm was assessed through chain convergence statistics (R-hat) and visually inspecting converging and stationary sampling across multiple HMC chains.

Descriptive data are presented as mean and standard deviation (SD). Inferential statistics are presented as full posterior densities, and percentile intervals (PI), or probability statements derived from full posterior distributions.

2.2 Description of the data set and participants

The overall objective of the present analysis is to determine the effect of krill oil supplementation in combination with resistance training on glucose metabolism and indicators of metabolic health. Outcome variables in the analysis are:

- Whole blood fatty acid composition, including omega-3 and omega-6 fatty acids and the omega-3 index
- Blood glucose in response to a oral glucose tolerance-test
- Metabolic syndrome (derived from sub-scores)
  - Triglycerides
  - Waist circumference
  - Blood pressure
  - HDL Cholesterol
  - Fasting glucose
- Body mass index
- Relative fat mass
- HOMA2-IR
- Adiponectin
- LDL Cholesterol
- HbA1c

At enrollment, participants were randomized into two groups, an active treatment-group (krill-oil fatty acids, 1 mg day^{-1}) and a placebo control group (Placebo; Figure 2.1, Table 2.1).

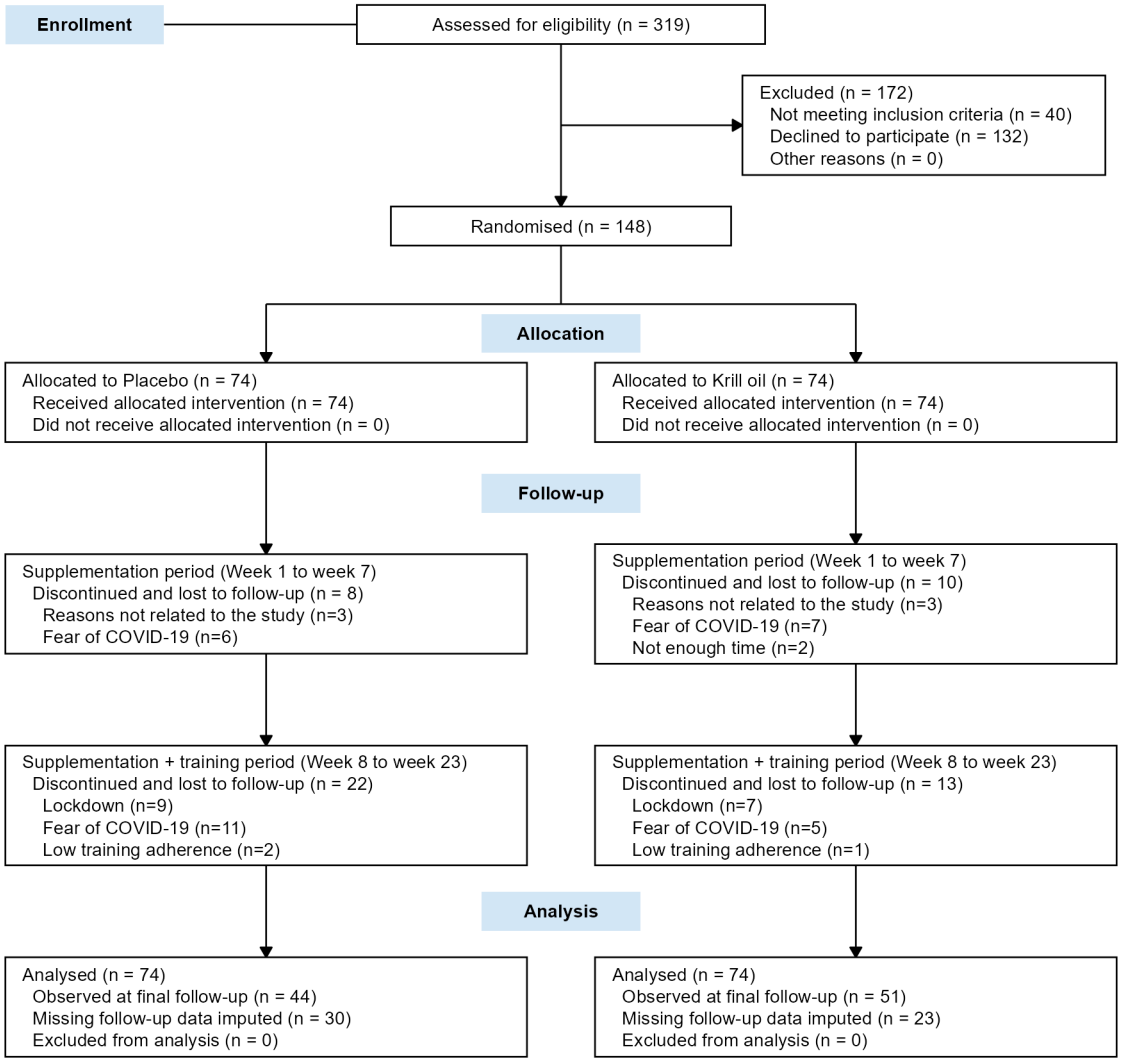

Figure 2.1: Flow of participants through the trial, presented according to the CONSORT 2025 statement. All randomized participants were included in the analysis of outcomes, with missing follow-up data imputed.

Table 2.1: Participant characteristics.

	Krill oil(♂ n = 32; ♀ n = 42)	Placebo(♂ n = 36; ♀ n = 38)
	Male	
Age (years)	47.3 (7.7) [29.0 - 60.0]	48.5 (5.8) [36.0 - 59.0]
Stature (cm)	180.5 (8.1) [153.3 - 193.2]	180.9 (6.3) [165.2 - 191.7]
Body mass (kg)	95.5 (17.1) [68.6 - 151.9]	97.8 (17.4) [71.2 - 142.2]
Body mass index (kg×m^{-2})	29.4 (5.1) [20.5 - 45.1]	30.0 (6.0) [23.1 - 52.1]
	Female	
Age (years)	44.5 (8.1) [30.0 - 59.0]	46.6 (9.4) [23.0 - 60.0]
Stature (cm)	168.9 (6.0) [158.0 - 181.9]	167.6 (6.4) [152.1 - 179.9]
Body mass (kg)	83.6 (16.5) [57.7 - 135.7]	79.3 (14.2) [52.9 - 113.8]
Body mass index (kg×m^{-2})	29.4 (5.8) [18.4 - 44.9]	28.2 (4.8) [20.4 - 39.9]

Data are presnted as Mean (SD) [Min-Max]

Participants were given krill oil or placebo in a supplementation-only period followed by a lead-in and combined period (supplementation and resistance training; Figure 2.2)

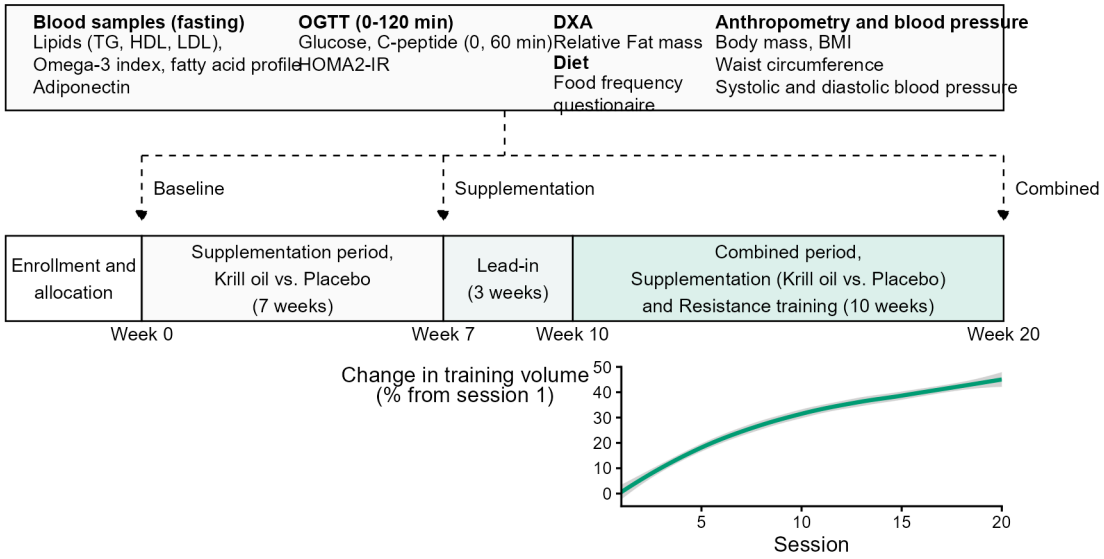

Figure 2.2: Study design. After baseline testing (week 0), participants received krill oil or placebo for 7 weeks, followed by 13 weeks of supervised resistance training, comprising a 3-week familiarization period (weeks 8–10) and 10 weeks of progressive resistance training (weeks 11–20), with supplementation continued throughout. The measurements listed were taken at each of the three time points: baseline (week 0), after supplementation alone (end of week 7) and after the combined period (end of week 20). The lower panel shows the change in training volume across the 20 training sessions, relative to session 1 and summed across all upper- and lower-body exercises. Line and shaded area show the smoothed trend with 95% confidence interval. BMI, body mass index; DXA, dual-energy X-ray absorptiometry; HDL, high-density lipoprotein cholesterol; HOMA2-IR, updated homeostasis model assessment of insulin resistance; LDL, low-density lipoprotein cholesterol; OGTT, oral glucose tolerance test; TG, triglycerides.

3. Effects of krill oil supplementation of serum fatty acids

Fatty acids, including including EPA (C205n3) and DHA (226n3) where measured in red blood cells. We will assess the effect of krill oil supplementation on group level fatty acid profiles. As preliminary models indicated heteroscedasticity of residuals, log-transformed and standardized data was used in modelling. This allows for the same weakly regularizing priors for all fatty acids (FA). In addition to each fatty acid, the Omega 3 index (8) is included in these models as the EPA and DHA content relative to the total fatty acid content of red blood cells. The omega 3 index is included in models from model `m3` (see below).

The models were designed to estimate differences between treatments (TX) at post-treatment time-points (T2 and T4) using pre-treatment ($\text{Pre}_{i[\text{FA}]}$) values as a covariate. The intercept in the model ($\alpha_{[\text{FA, TI, TX}]}$) gives us mean estimates for fatty-acid FA, time-point TI, and treatment TX.

$$
\begin{aligned}
y_{i[\text{FA}]} &\sim \text{Normal}\ (\mu_{i[\text{FA}]}, \sigma_{\text{FA}}) \\
\mu_{i[\text{FA}]} &= \alpha_{[\text{FA, TI, TX}]} + \beta_{\text{FA}} \times \text{Pre}_{i[\text{FA}]} \\
\alpha_{[\text{FA, TX}]} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{FA}} &\sim \text{Normal}\ (1, 1) \\
\sigma_{\text{FA}} &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{3.1}
$$

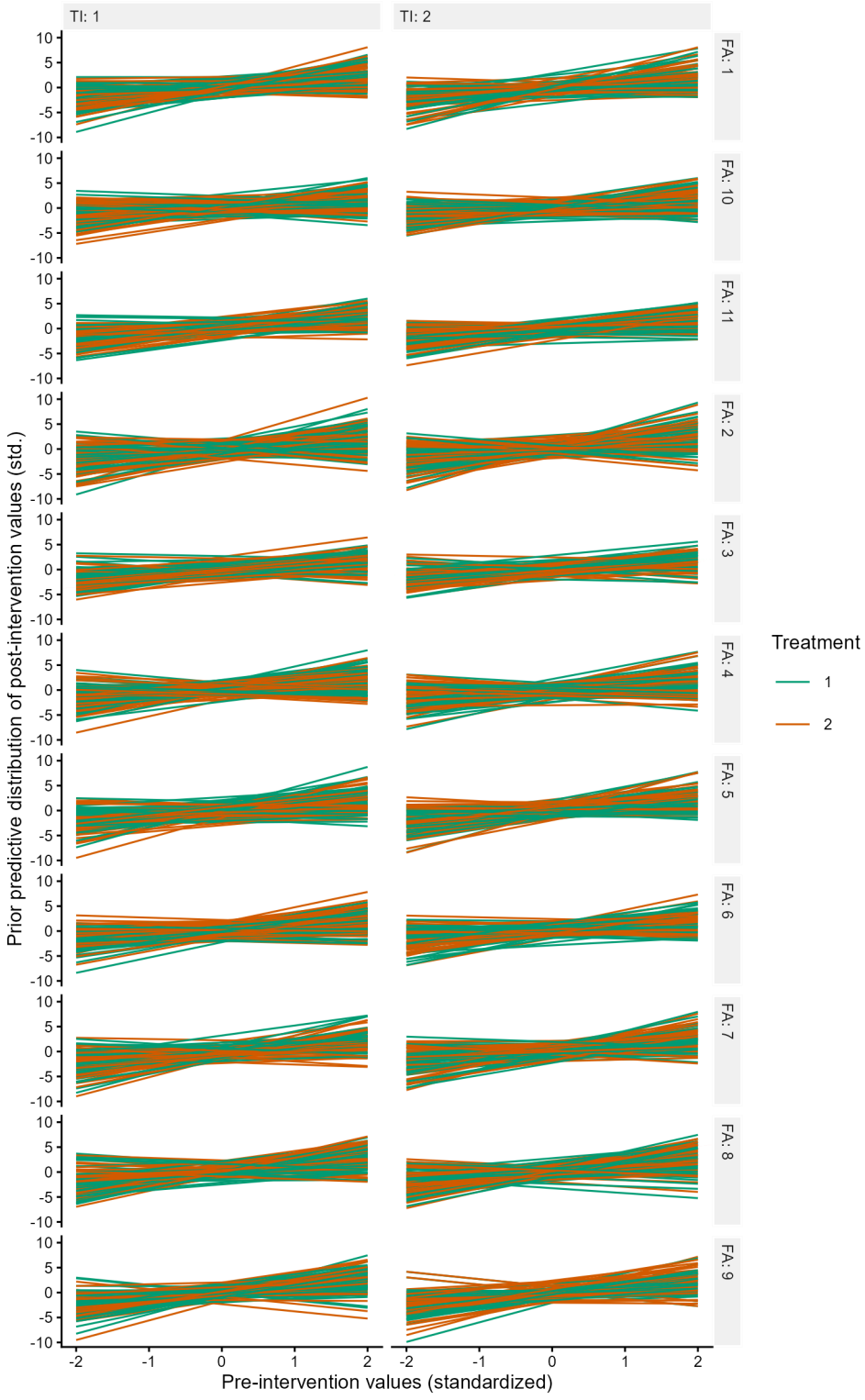

Figure 3.1: Prior predictive check for target-wise models of post-treatment fatty acid concentrations conditional on pre-treatment values. The model expects log-transformed standardized values and the priors suggest no difference between treatment groups and a proportional relationship between th pre-intervention values and post-intervention measurements.

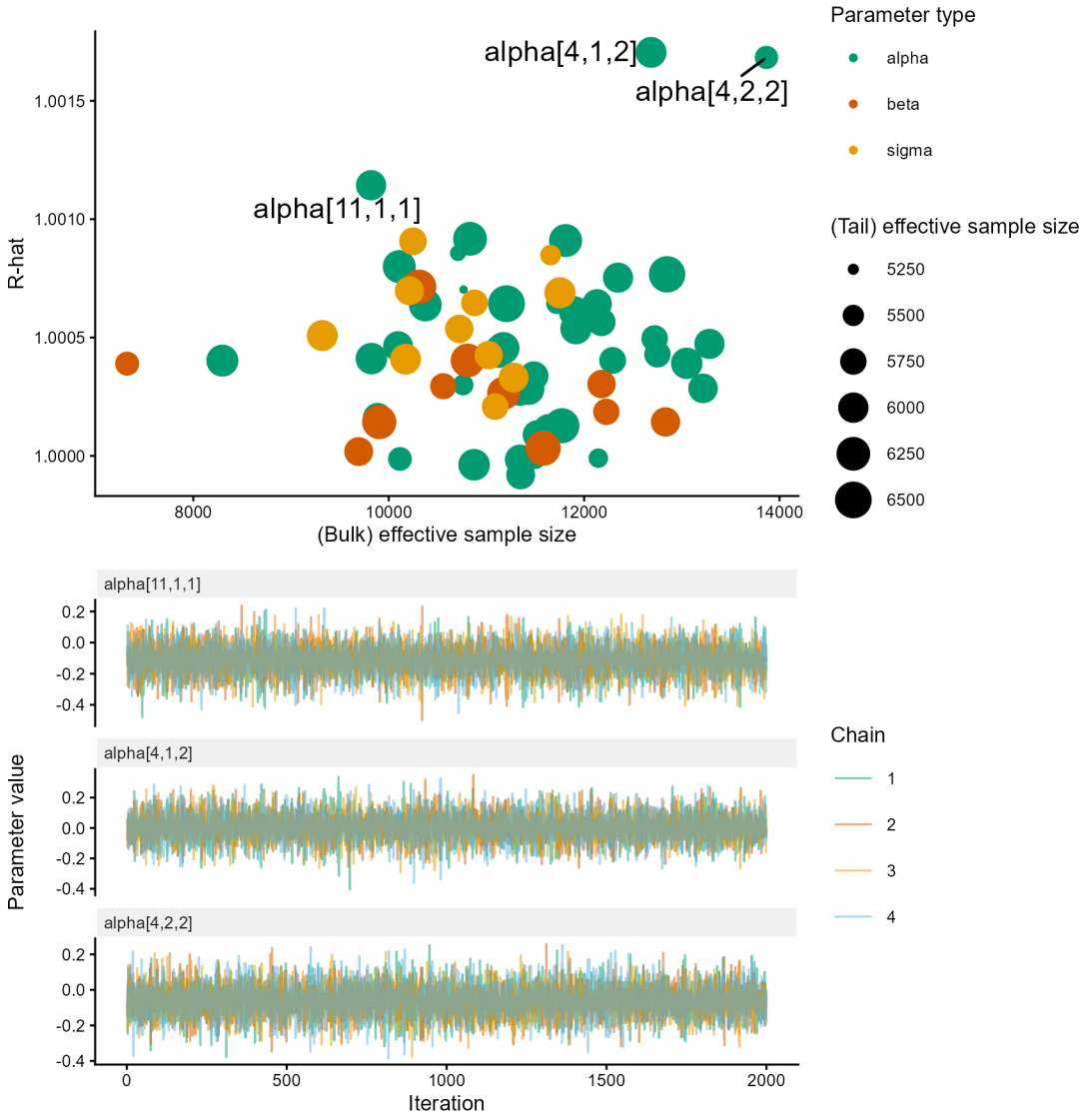

Figure 3.2: Model performance metrics from Model 2.

The first model uses only complete-cases. The prior predictive check confirms that the model expects eleven fatty acids, a positive relationship between pre-intervention values and post-treatment values, and no differences in treatments (Figure 3.1). Sampling from the posterior was non-problematic as R-hat values approached 1 and chains mixed and converged well (Figure 3.2).

Posterior predictive checks showed satisfactory patterns between replicated data (yrep; thin lines, one per draw in Figure 3.3) and observed data (thick lines in Figure 3.3).

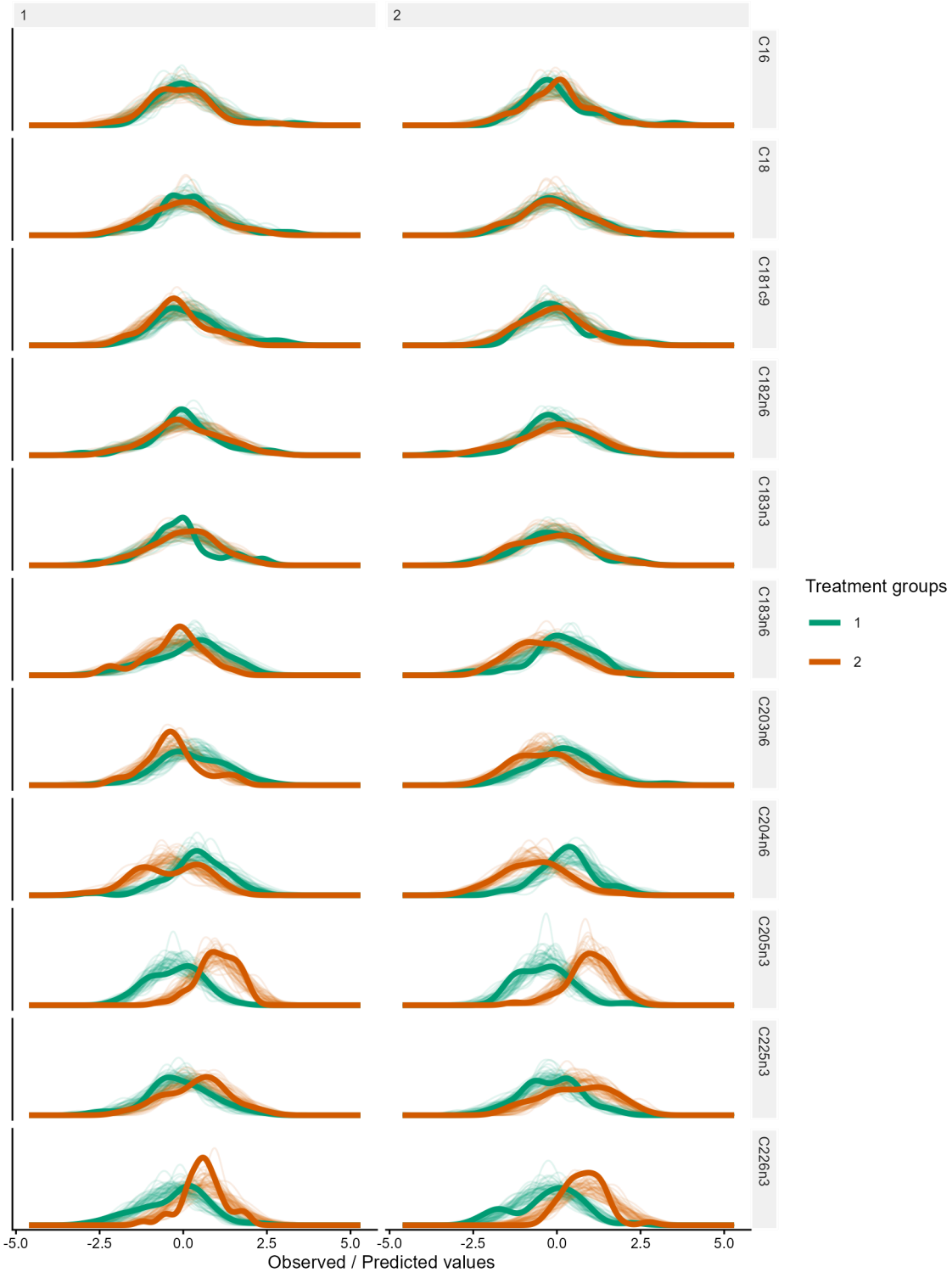

Figure 3.3: Posterior predictive check of Model 2.

Krill oil supplementation robustly increased whole blood fatty acid composition levels of C205n3, C225n3 and C226n3 fatty acids compared to placebo (Figure 3.4 and Figure 3.5). Post-treatment values were lower in the krill oil group for C181c9, C183n6, C203n6 and C204n6 with high probability at both T2 and T4 (Figure 3.5).

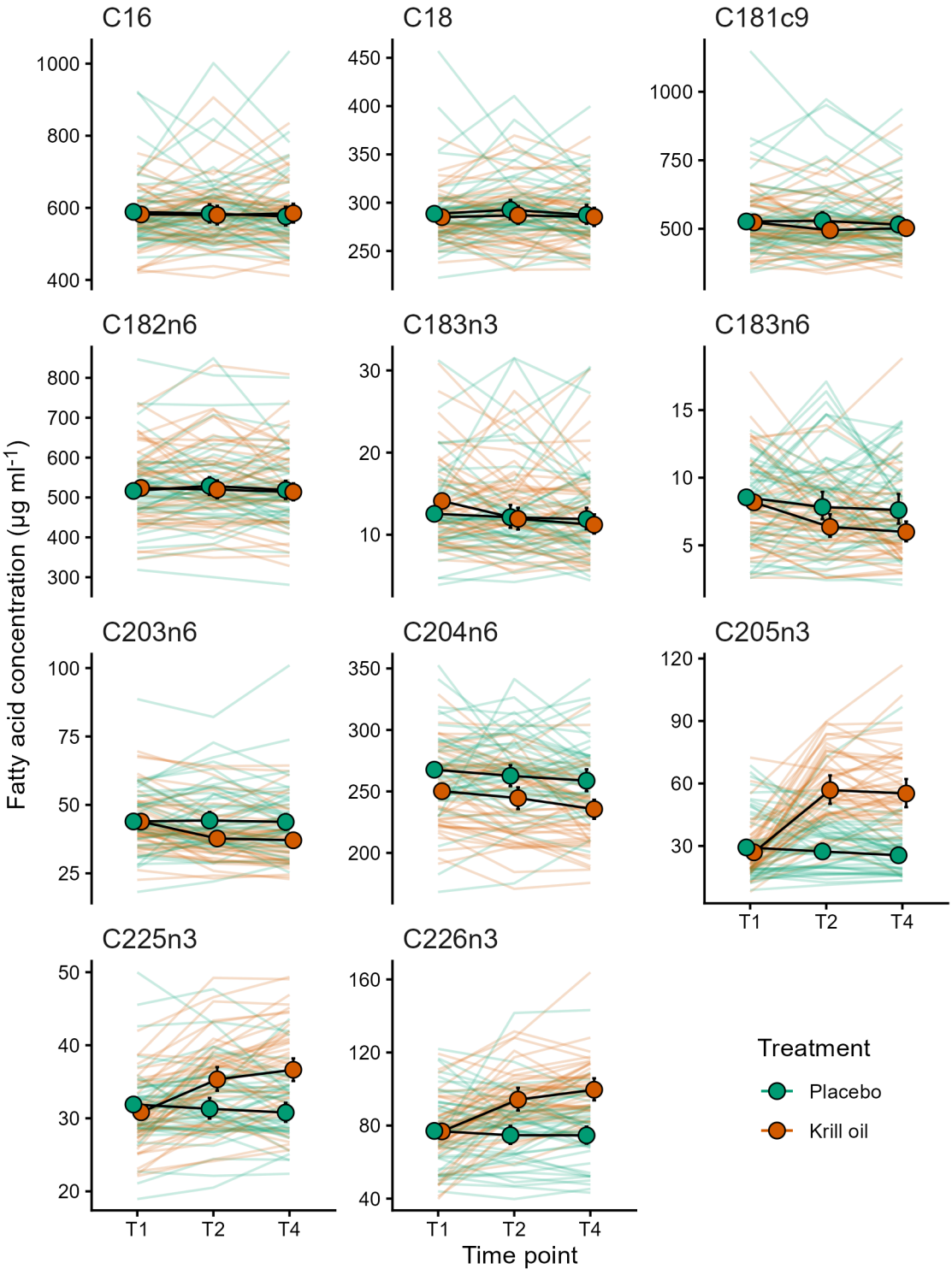

Figure 3.4: Predictions of fatty acid concentration in response to treatment. Pre-intervention values are means per treatment group, post-intervention means are predicted from the model with 99% PI.

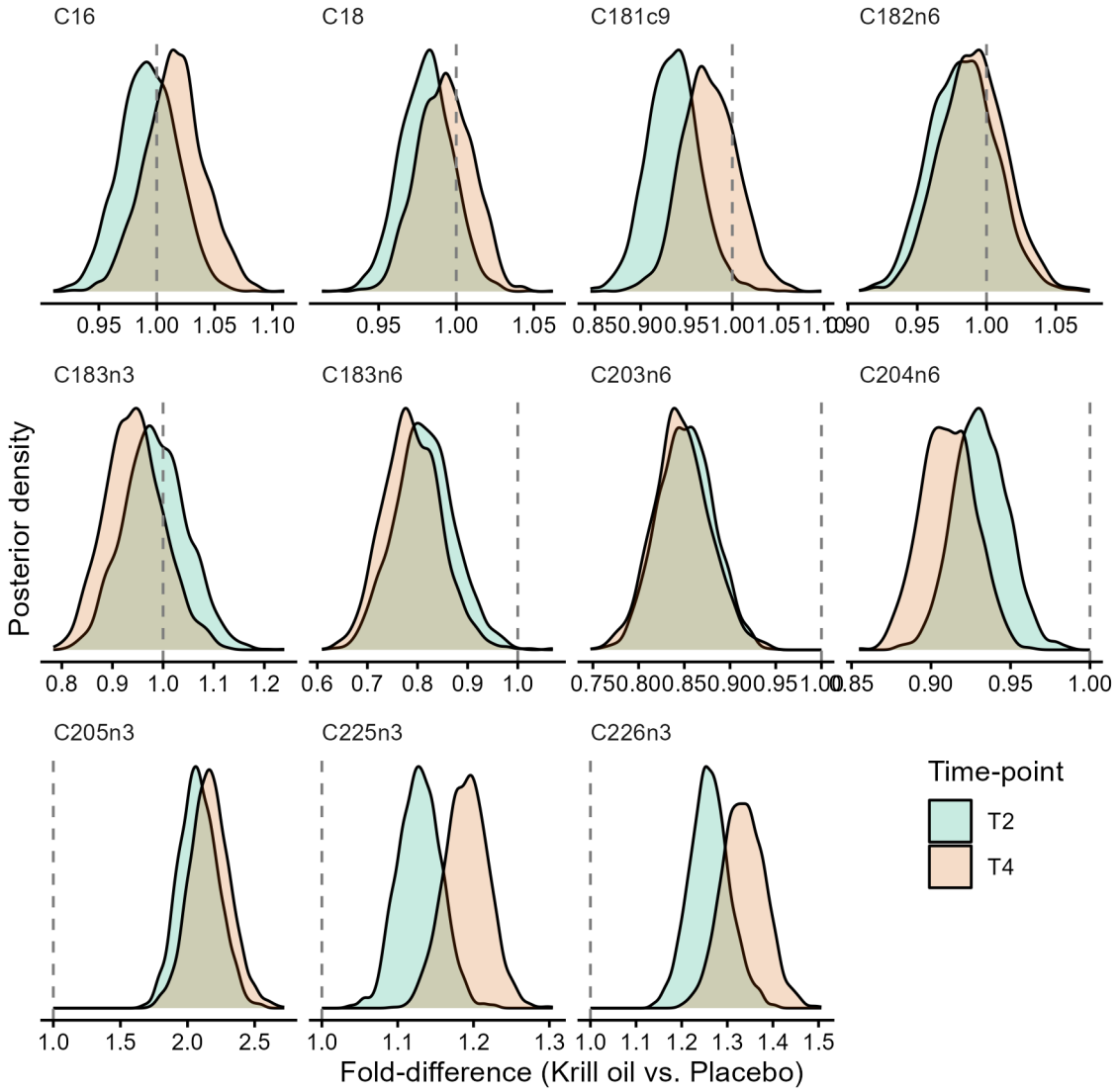

Figure 3.5: Posterior densities for differences between treatment groups in fatty acid concentrations.

Extending the first model with missing data imputations allows us to estimate the effect of treatment on fatty acid profiles with intention-to-treat (ITT) by averaging over the uncertainty associated with the missing values. Missing data imputation is done on post-treatment scores only with the total sample size being n = 144 with 14 and 36% of participants missing at time points T2 and T4, respectively.

The imputation mechanism replaces missing values with parameters estimated from observed data (pre-treatment values and treatment allocation) through the likelihood.

The imputation model is defined as:

$$
\begin{aligned}
y^{\text{all}}_{i[\text{FA}]} &\sim \text{Normal}\ (\mu_{i[\text{FA}]}, \sigma_{\text{FA}}) \\
\mu_{i[\text{FA}]} &= \alpha_{[\text{FA, TI, TX}]} + \beta_{\text{FA}} \times \text{Pre}_{i[\text{FA}]} \\
y^{\text{all}}_{i[\text{FA}]} &= \begin{cases} y^{\text{obs}}_{i[\text{FA}]} \text{ if observed} \\ y^{\text{mis}}_{i[\text{FA}]} \text{ if missing} \end{cases} \\
y^{\text{mis}}_{i[\text{FA}]} &\sim \text{Normal}\ (0, 2) \\
\alpha_{[\text{FA, TX}]} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{FA}} &\sim \text{Normal}\ (1, 1) \\
\sigma_{\text{FA}} &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{3.2}
$$

Prior predictive simulations confirm the general structure of the model (Figure 3.6). Model performance metrics from posterior sampling also indicates a functioning model (Figure 3.7).

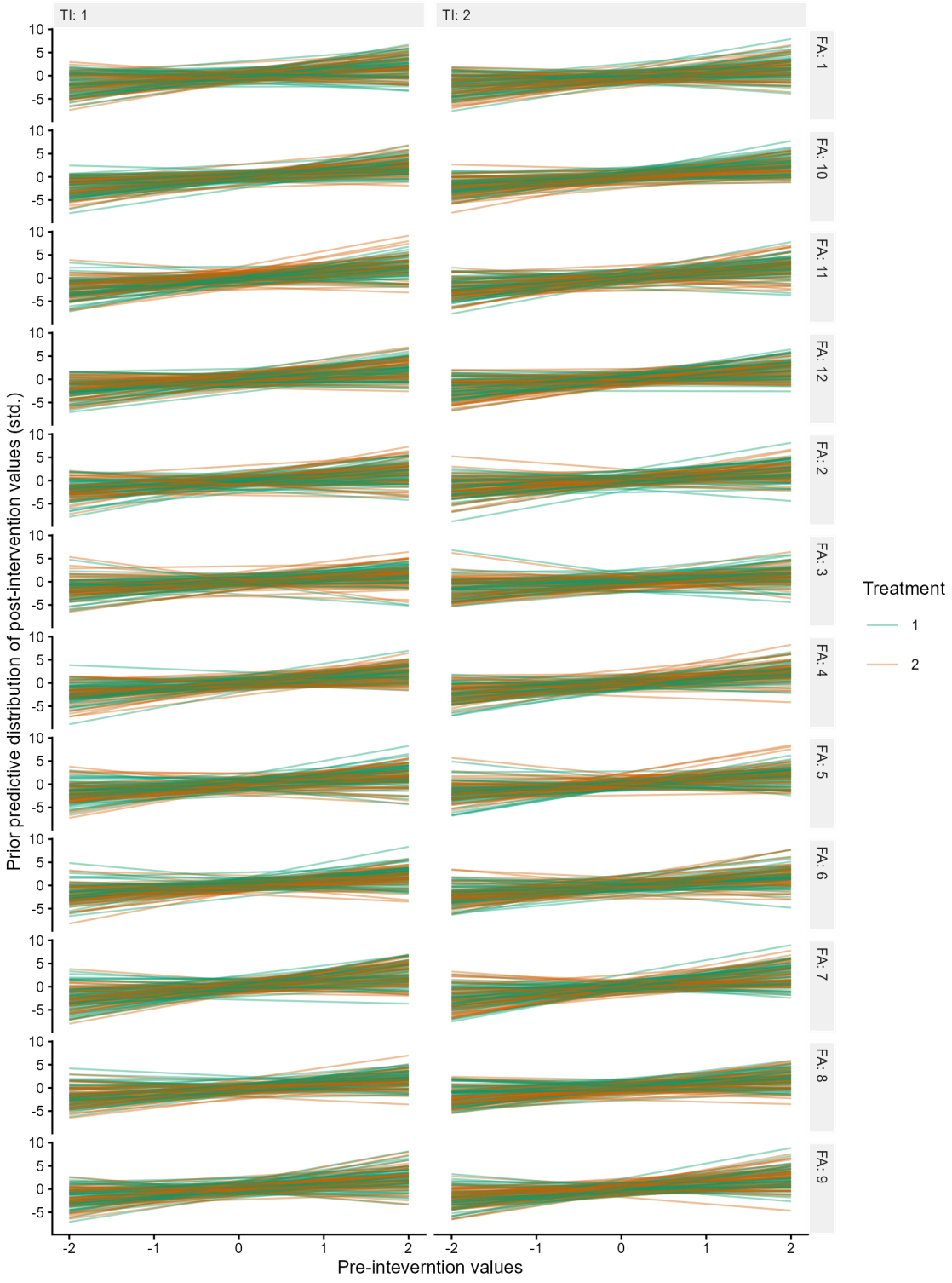

Figure 3.6: Prior predictive check model 3 (data imputation).

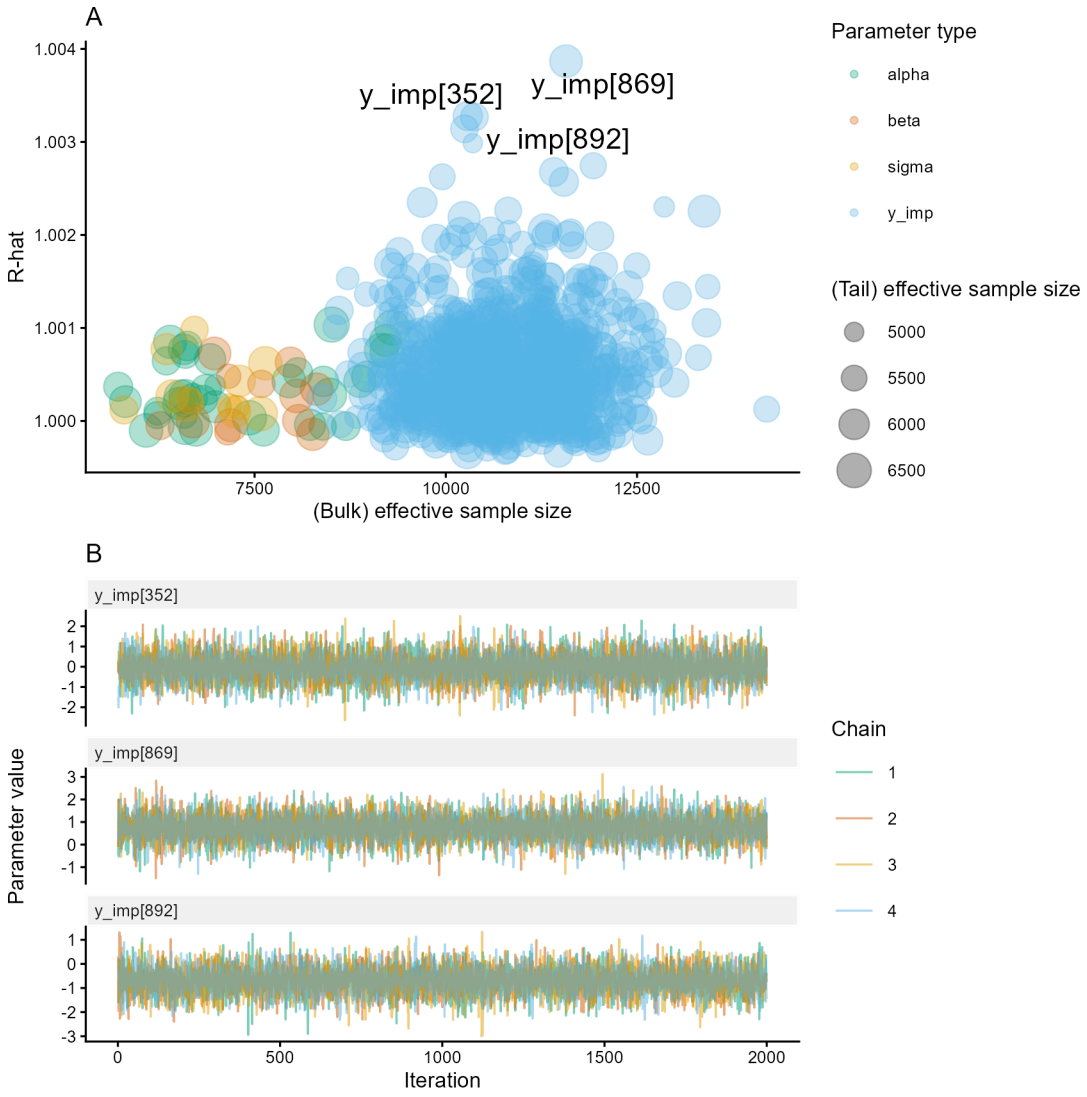

Figure 3.7: Model performance metrics model 3.

Posterior predictive checks indicated that the model capture general patterns in the data (Figure 3.8).

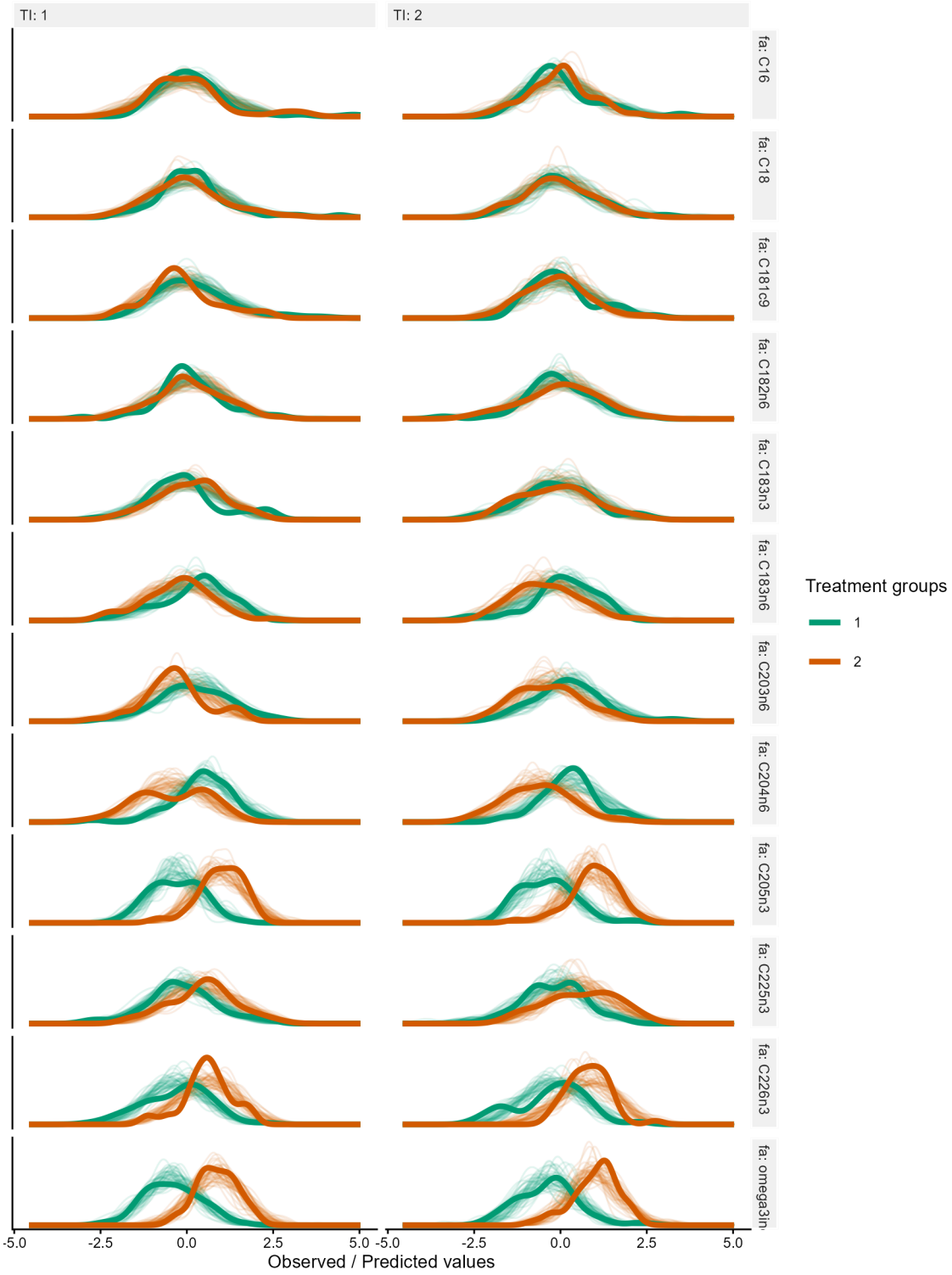

Figure 3.8: Posterior predictive check of Model 3. Thin lines represent model simulations (n = 50), thick lines represent observed data.

The imputation model generally agreed with the complete-case analysis on the effect of treatment of serum fatty acid profiles (Figure 3.9, Figure 3.10, Figure 3.11, and Figure 3.12).

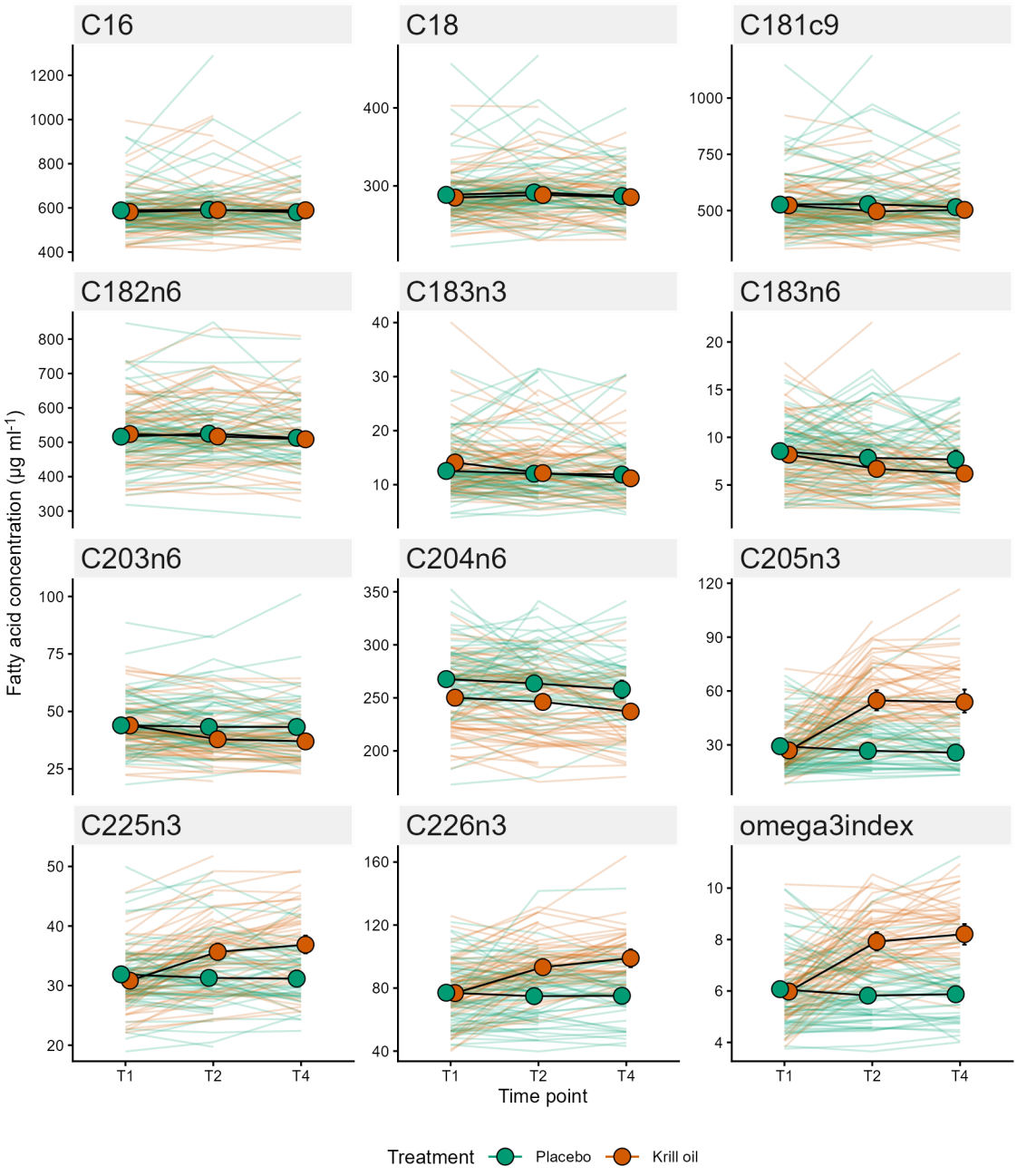

Figure 3.9: Model predictions. Pre-intervention values are observed means, post-treatment (T2, T4) values are model predictions conditional on pre-intervention value with 99% percentile intervals. The omega-3 index is EPA and DHA content of erythrocytes expressed as a percent of total identified fatty acids.

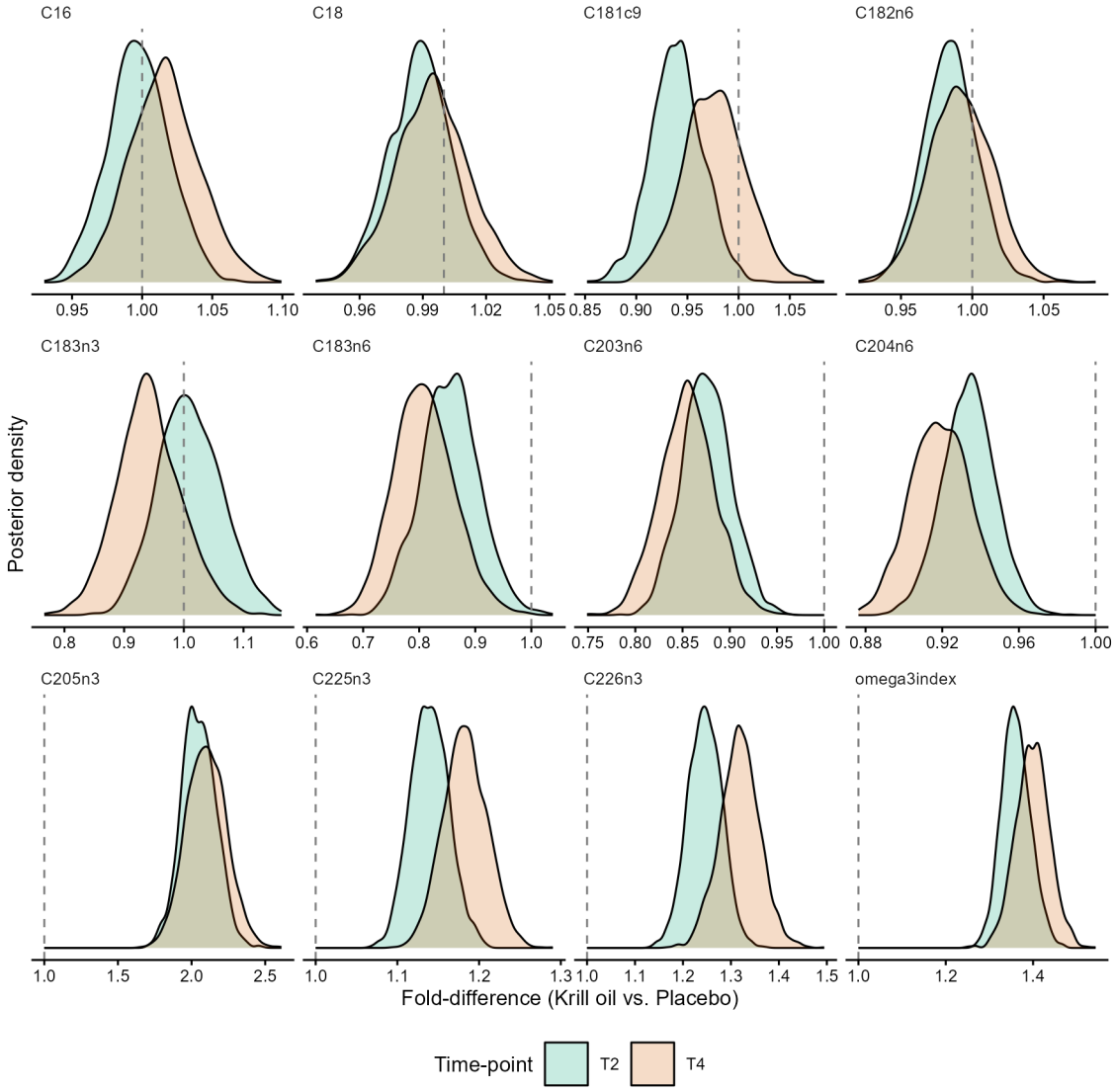

Figure 3.10: Posterior densities for differences between treatment groups in fatty acid concentrations (data imputation model).

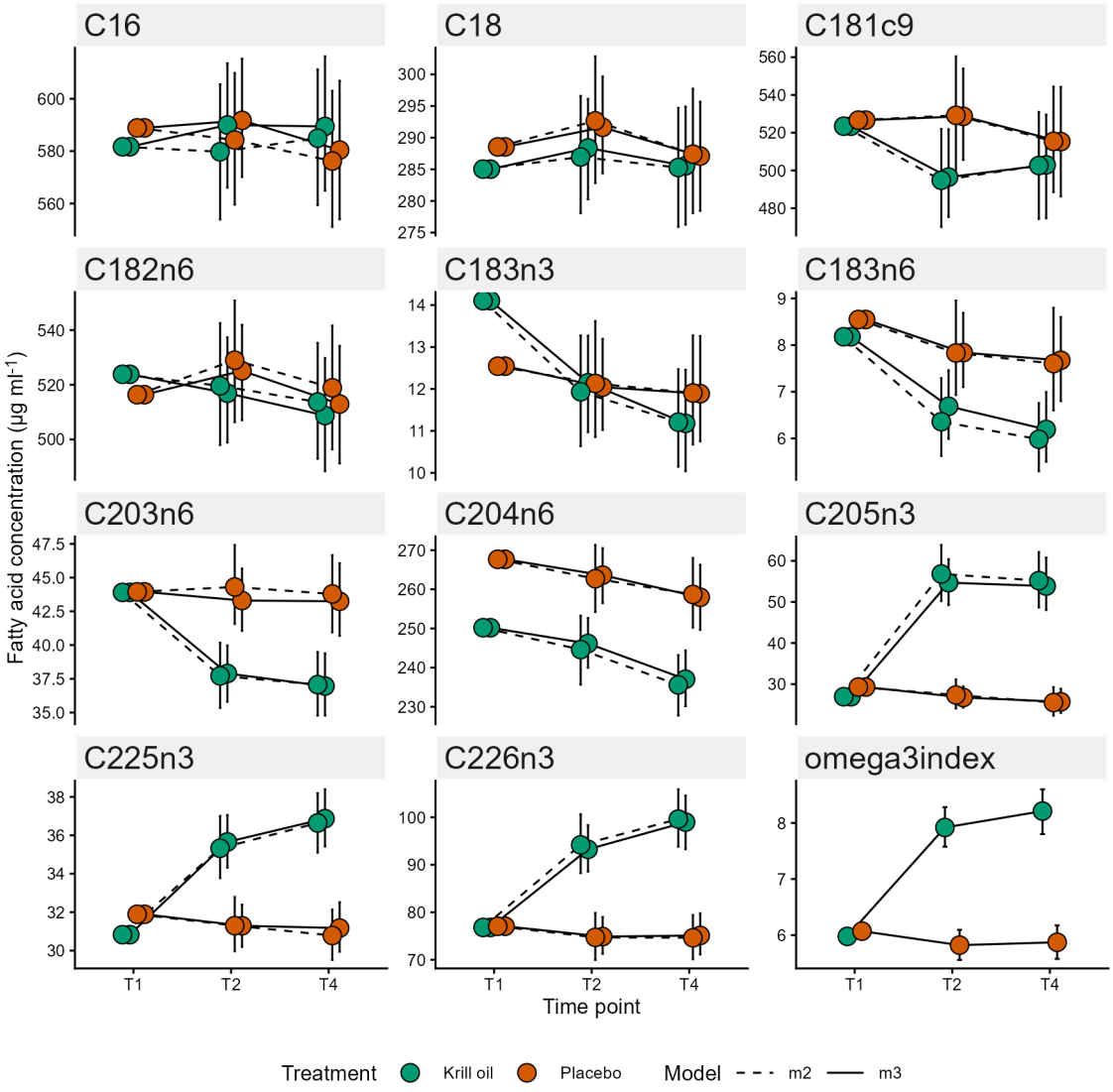

Figure 3.11: Posterior predictions for fatty acid concentrations in complete-case analysis (model m2) and data imputation model (model m3).

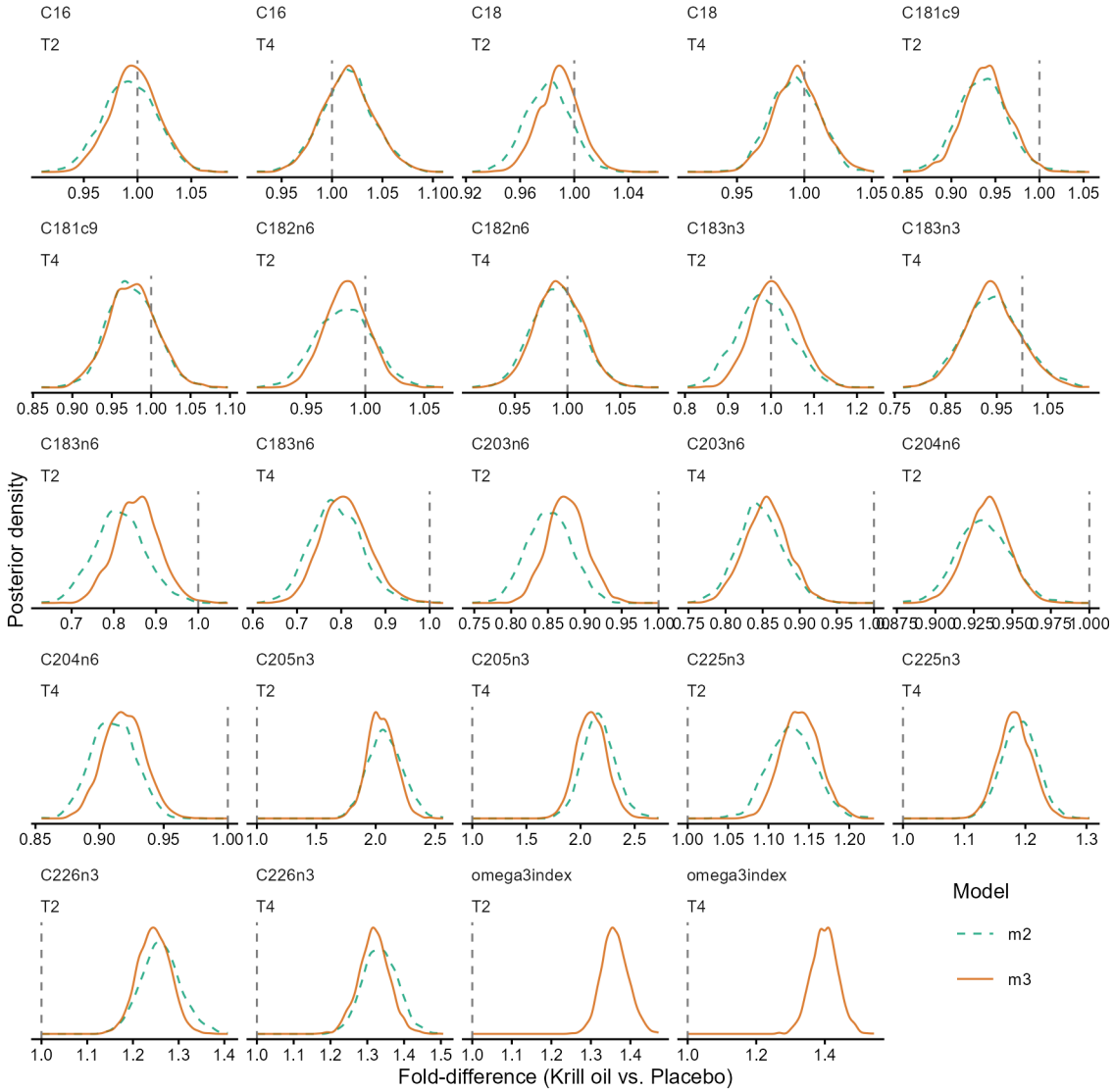

Figure 3.12: Comparing inference from complete-case and imputation models.

3.0.1 Model expansion: Heterogeneous variance

Model `m3` does not enable treatment- and time-wise heterogeneous variation. In model `m4` we therefore add $\sigma_{[\text{FA, TX}]} \sim \text{Exponential}\ (1)$, in model `m5` we add $\sigma_{[\text{FA, TI}]} \sim \text{Exponential}\ (1)$, and model `m6` adds $\sigma_{[\text{FA, TI, TX}]} \sim \text{Exponential}\ (1)$. Model performance metrics confirmed that all three extended models provided reliable chain convergence characteristics (Figure 3.13).

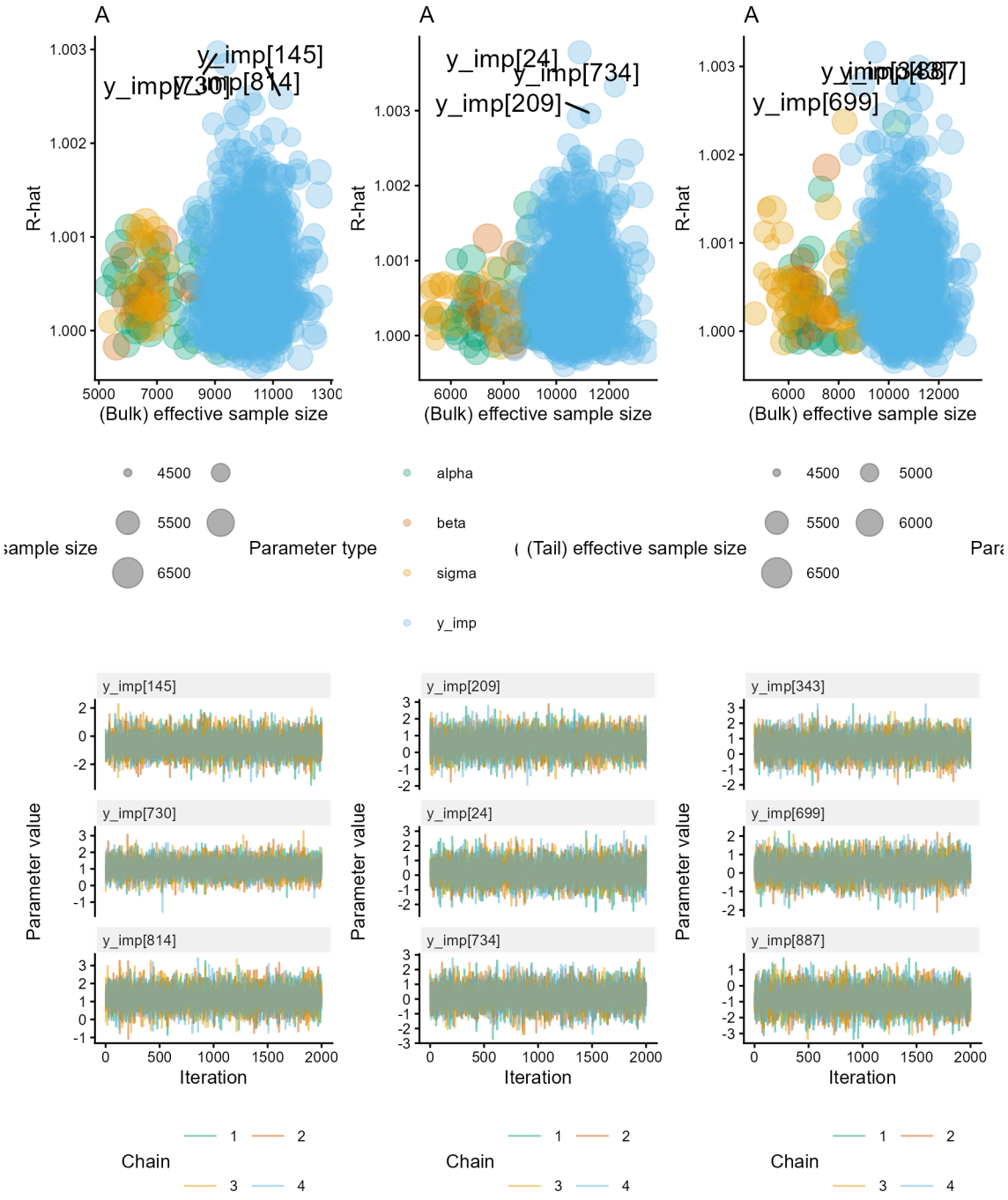

Figure 3.13: Model performance metrics models 4, 5, and 6.

Comparing the models in posterior predictive simulations of condition-specific (time, treatment and fatty-acid) SD across models indicated that no model displayed extreme deviations in replicated values compared to observed variation in the data (Table 3.1). This comparison agreed with the out of sample predictive performance where models `m3` to `m5` where comparable. Model `m6`, the most heavily parameterized model showed four unreliable Pareto k values but was also slightly worse in in out of sample predictions (Table 3.2). Together this suggest that model `m4` is the preferred model with model `m6` being the worst model.

Table 3.1: Posterior predictive p-values comparing models m3 to m6.

Model	Proportion extreme p(< 0.05 | > 0.95)	Median p	Min p	Max p
m3	0.083	0.375	0.024	0.997
m4	0.021	0.363	0.078	0.954
m5	0.062	0.358	0.017	0.999
m6	0.021	0.429	0.051	0.963

Table 3.2: Expected pointwise predictive density across models.

Model	ΔELPD	SE(ΔELPD)	ELPD LOO
m4	0	0	−2,589.086
m3	−0.235	6.591	−2,589.321
m5	−3.156	9.228	−2,592.242
m6	−17.098	6.38	−2,606.184

3.1 Inference

We draw inference from model `m4` regarding the effect of treatment on whole blood fatty acid profiles.

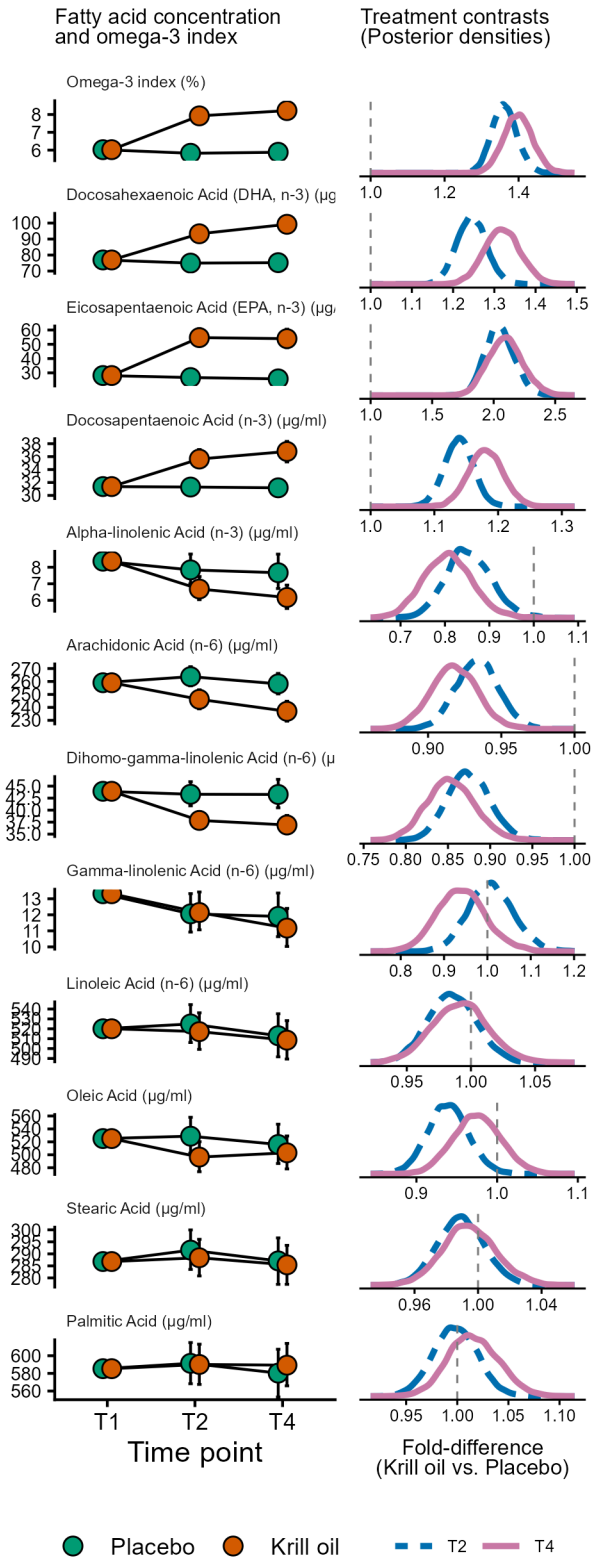

Figure 3.14: Adjusted group means and treatment contrasts of fatty acids concentrations and the omega-3 index in whole blood. Error bars in treatment-group means represents 95% CrI, posterior contrasts are displayed as the full posterior distribution ratio between krill oil and placebo estimates at time-point T2 and T4 (1 indicating equivalent concentrations).

Krill oil supplementation led to robust increases in Docosahexaenoic (DHA; time-point T2 krill oil vs. placebo, 24.5% [18.7, 30.8]; T4, 31.6% [24.8, 40.2]), Eicosapentaenoic (EPA; T2, 103.0% [85.8, 125.3]; T4, 109.4% [87.4, 134.8]), and Docosapentaenoic acid (T2, 14.1% [10.4, 18.2]; T4, 18.3% [13.8, 22.2]; Figure 3.14, Table 3.3). At the same time, omega-6 fatty acids were reduced in the krill oil group compared to placebo at both time-points (Alpha-linolenic acid T2, −15.9% [−24.6, −6.9]; Alpha-linolenic acid T4, −19.4% [−30.2, −9.9]; Arachidonic acid T2, −6.5% [−9.0, −3.7]; Arachidonic acid T4, −8.8% [−10.8, −5.7]; Dihomo-gamma-linolenic acid T2, −12.7% [−17.3, −7.9]; Dihomo-gamma-linolenic acid T4, −15.6% [−20.4, −9.2]; Figure 3.14, Table 3.3). While gamma-linolenic, linoleic, steraic and palmitic acids remained stable unchanged in krill oil compared to placebo (Figure 3.14, Table 3.3), Oleic acid was reduced in krill oil compared to placebo at T2 (-6.1% [−11.8, −1.4]) but not at T4 (-2.8% [−7.3, 4.2]).

Table 3.3: Descriptive data, adjusted mean estimates, prediction intervals and treatment contrasts for fatty acids measured in whole blood before the intervention (time-point T1), after supplementation-only (T2), and after supplementation and resistance training (T4). All values are reported on the measurement scale (fatty acids, µg ml-1; omega-3 index, %).

	Time-point	Observed $\bar{x}$ (SD)		Estimate μ [95%]		Contrast [95%]
		Placebo	n-3	Placebo	n-3	n-3 vs. placebo
Omega-3 index (%)	T1	6.1 (1.5)	6.0 (1.2)			
	T2	5.9 (1.2)	8.0 (1.1)	5.8 [5.6, 6.0]	7.9 [7.7, 8.2]	2.2 [1.8, 2.6]
	T4	6.0 (1.3)	8.3 (1.3)	5.9 [5.6, 6.1]	8.2 [7.9, 8.5]	2.4 [2.0, 2.8]
Docosahexaenoic Acid (DHA, n-3)	T1	77 (20)	77 (19)			
	T2	77 (19)	95 (17)	75 [72, 78]	93 [90, 96]	19.1 [14.3, 23.8]
	T4	78 (20)	100 (16)	75 [71, 79]	99 [95, 103]	25.0 [19.1, 30.5]
Eicosapentaenoic Acid (EPA, n-3)	T1	30 (15)	27 (13)			
	T2	29 (12)	58 (18)	27 [25, 29]	55 [50, 59]	31.1 [25.7, 36.8]
	T4	29 (15)	58 (20)	26 [23, 28]	54 [49, 59]	31.3 [25.2, 37.7]
Docosapentaenoic Acid (n-3)	T1	32 (5)	31 (5)			
	T2	32 (6)	36 (6)	31 [30, 32]	36 [35, 37]	4.5 [3.1, 5.8]
	T4	32 (5)	37 (6)	31 [30, 32]	37 [36, 38]	5.8 [4.2, 7.4]
Alpha-linolenic Acid (n-3)	T1	8.5 (3.1)	8.0 (3.6)			
	T2	8.8 (3.6)	7.1 (3.3)	7.8 [7.2, 8.6]	6.7 [6.2, 7.2]	−1.1 [−2.0, −0.3]
	T4	8.2 (3.1)	6.6 (3.1)	7.7 [6.9, 8.5]	6.2 [5.7, 6.7]	−1.4 [−2.4, −0.6]
Arachidonic Acid (n-6)	T1	268 (33)	250 (34)			
	T2	272 (30)	242 (34)	264 [258, 269]	246 [241, 252]	−17.3 [−25.0, −9.6]
	T4	264 (29)	234 (31)	258 [252, 264]	237 [231, 243]	−21.1 [−29.4, −12.7]
Dihomo-gamma-linolenic Acid (n-6)	T1	44 (12)	43 (11)			
	T2	45 (13)	39 (10)	43 [41, 45]	38 [37, 39]	−5.4 [−7.6, −3.1]
	T4	45 (13)	38 (10)	43 [41, 46]	37 [35, 38]	−6.3 [−8.9, −3.6]
Gamma-linolenic Acid (n-6)	T1	13 (5)	14 (6)			
	T2	13 (7)	14 (5)	12 [11, 13]	12 [11, 13]	0.1 [−1.1, 1.3]
	T4	13 (6)	13 (6)	12 [11, 13]	11 [10, 12]	−0.7 [−2.1, 0.6]
Linoleic Acid (n-6)	T1	522 (94)	531 (91)			
	T2	534 (96)	534 (98)	525 [511, 539]	517 [503, 531]	−7.9 [−27.5, 12.0]
	T4	521 (94)	529 (103)	513 [497, 530]	509 [494, 524]	−4.4 [−27.4, 17.8]
Oleic Acid	T1	534 (138)	521 (116)			
	T2	551 (151)	507 (121)	529 [509, 550]	496 [479, 514]	−32.3 [−59.6, −5.2]
	T4	530 (121)	510 (109)	516 [493, 540]	503 [484, 523]	−12.8 [−43.0, 16.7]
Stearic Acid	T1	289 (36)	285 (29)			
	T2	295 (38)	289 (35)	292 [285, 298]	288 [283, 294]	−3.2 [−11.7, 5.4]
	T4	290 (33)	286 (31)	287 [280, 294]	285 [279, 292]	−1.7 [−11.2, 8.1]
Palmitic Acid	T1	591 (96)	577 (105)			
	T2	606 (125)	595 (124)	591 [574, 609]	590 [572, 607]	−1.3 [−26.7, 23.6]
	T4	592 (109)	584 (87)	580 [560, 601]	589 [571, 609]	8.7 [−18.8, 37.1]

4. Effects of krill oil supplementation on indications of metabolic syndrome - Introduction

Here we aim to model the components of the metabolic syndrome definition (9), along with related outcomes (Table 4.2). The goal is to build a model that imputes missing data to enable an intention-to-treat (ITT) analysis of all randomized participants. Imputation will also allow for assessing the probability of meeting metabolic syndrome definition thresholds in each group while accounting for uncertainty associated with missing observations. The strategy will be to chose a model that will allow for sharing of information between variables to enable more efficient data imputation. We use the harmonised metabolic syndrome definition, where any three of the five criteria (waist circumference, triglycerides, HDL-cholesterol, blood pressure and fasting glucose) beyond their threshold values are sufficient for a diagnosis, and central obesity is not required (10).

Figure 4.1 shows distributions of the variables selected for joint modeling, including variables used to define metabolic syndrome (waist circumference, raised triglycerisdes, reduced HDL-cholesterol, raised blood pressure, raised fasting glucose).

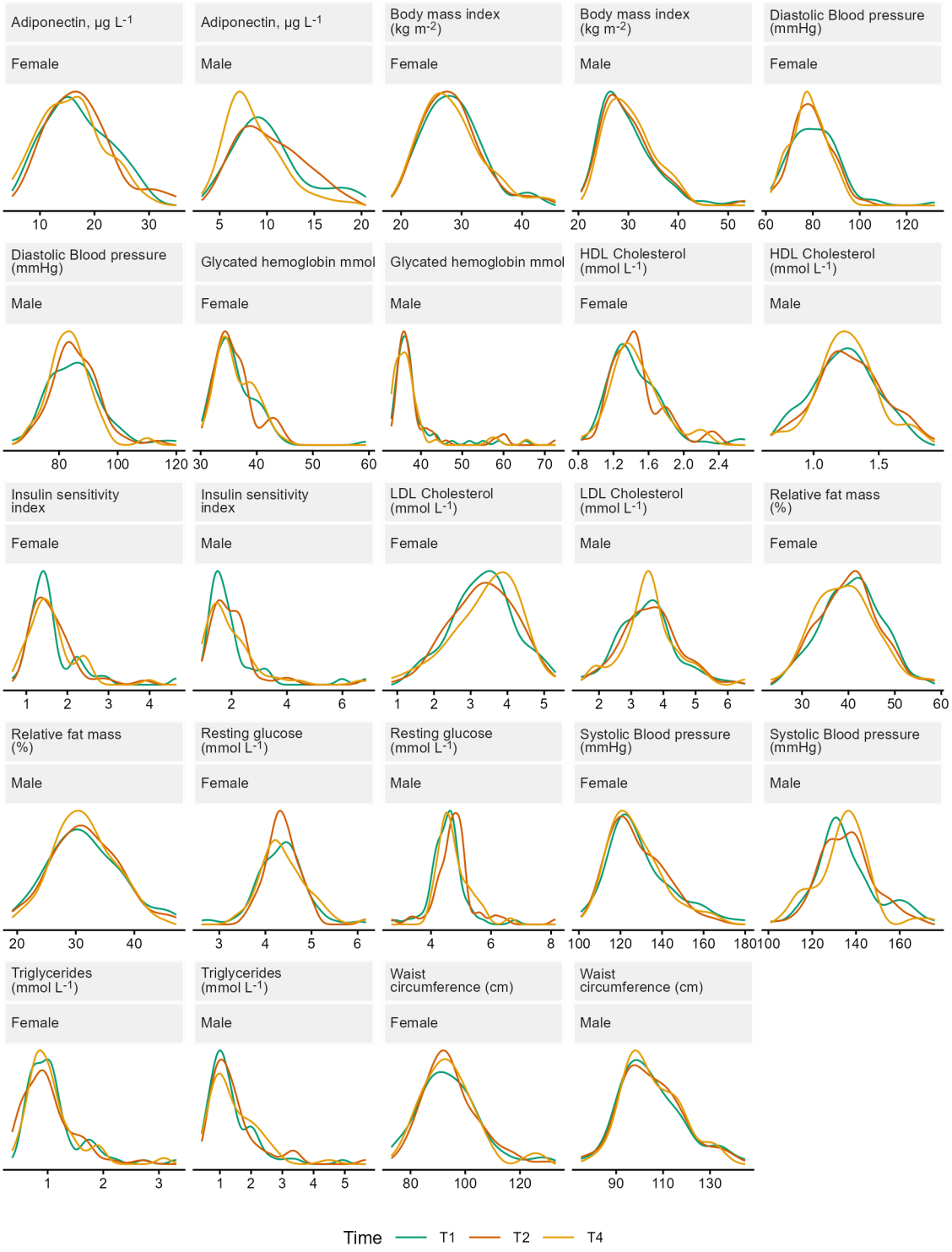

Figure 4.1: Observed data distributions of main outcomes.

Descriptive analysis of complete cases indicates variation in the proportion of participants meeting the criteria thresholds across treatment groups and post-intervention time points. Table 4.1 and Figure 4.2 show the proportions of participants who have three or more diagnostic criteria for metabolic syndrome among complete cases. We will expand this analyses later when incorporating the diagnosis in modelling of subscores in a joint model.

Table 4.1: Participants in the complete-case analysis of metabolic syndrome indications.

	Time		
	T1	T2	T4
	Male		
Placebo	21 (19%)	21 (29%)	21 (38%)
Krill oil	19 (32%)	19 (16%)	19 (32%)
	Female		
Placebo	16 (12%)	16 (12%)	16 (19%)
Krill oil	21 (19%)	21 (10%)	21 (14%)
	Total		
Placebo	37 (16%)	37 (22%)	37 (30%)
Krill oil	40 (25%)	40 (12%)	40 (22%)

Data are presented as n (Metabolic syndrome proportion).Time T1, Baseline;T2 Supplementation; T4, Combined.

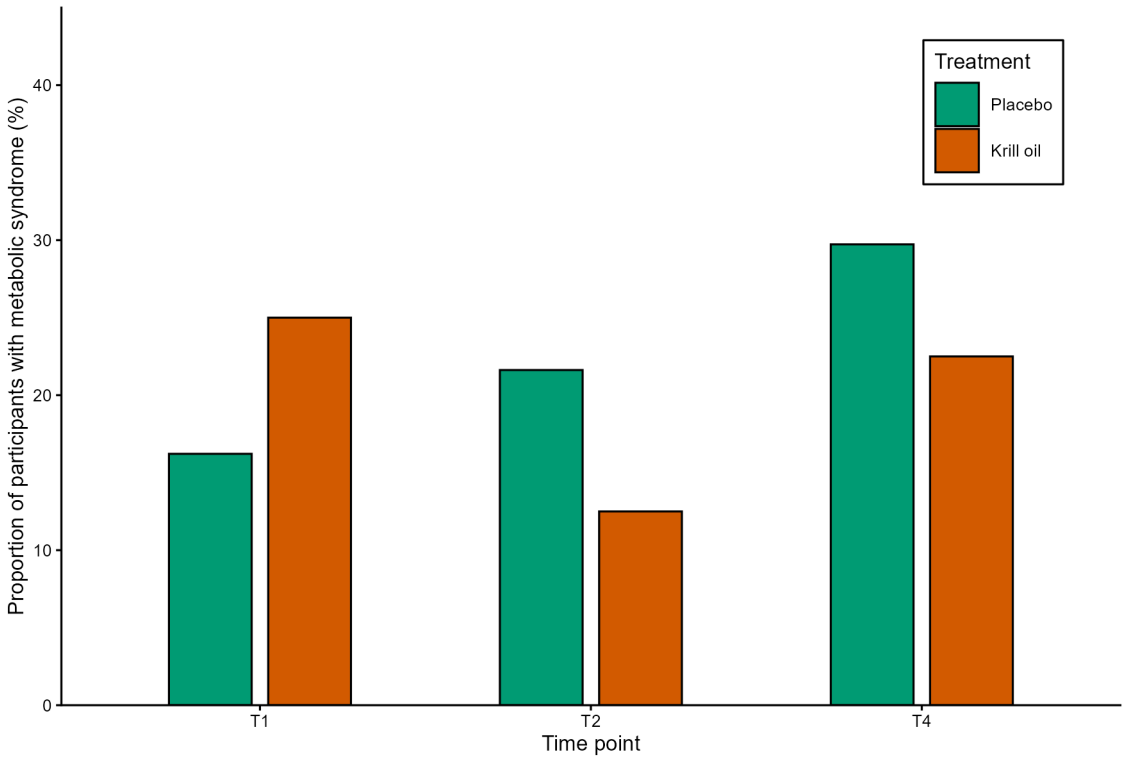

Figure 4.2: Proportion of participants with metabolic syndrome (at least three criteria thresholds).

Table 4.2: Components of the metabolic syndrome diagnostic criteria and related measurements used in modelling.

Subscore	Threshold
Waist circumference	> 102 cm for men, > 88 cm for women
HDL Cholesterol	< 40 mg/dL for men, < 50 mg/dL for women
Fasting glucose	100 mg/dL or higher (or taking medication)
Triglycerides	150 mg/dL or higher (or taking medication)
Blood pressure	130/85 mm Hg or higher (or taking medication)
Relative fat mass	NA
Body mass index	NA
Insulin resistance index	NA
Adiponectin	NA
LDL Cholesterol	NA
HbA1c	NA

Metabolic syndrome is defined as at least three of the five criteria with thresholds (harmonised definition)

For the joint model we also include measures related to metabolic health. Among these variables, HOMA2-IR is a derived variable based on C-peptide and fasting glucose calculated using the HOMA2 calculator. HOMA2-IR was calculated when C-peptide and glucose was available and any missing data was primarily seen in the C-peptide variable. Based on the strong correlation between C-peptide and HOMA2-IR vs. the more moderate correlation between glucose and HOMA2-IR (**?@fig-cpeptidehoma**) we chose to use the derived variable HOMA2-IR and drop C-peptide from the joint modelling to avoid duplicationg information in the same model.

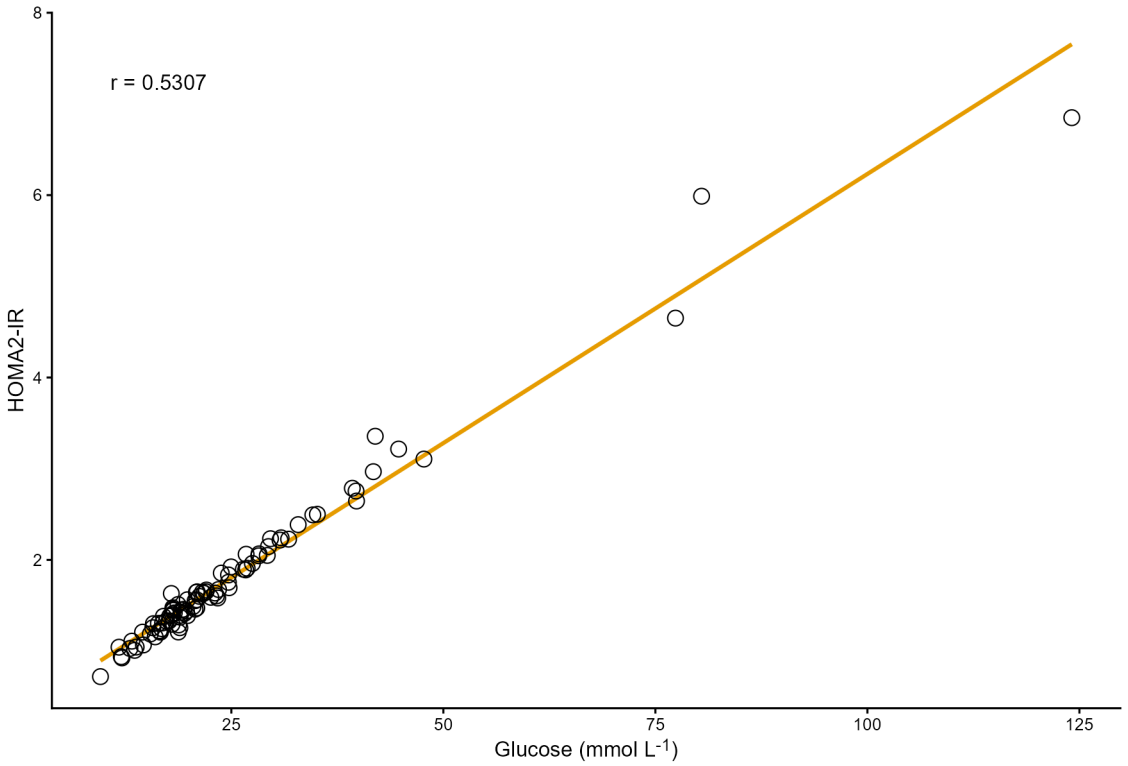

Figure 4.3: Associations between C-peptide or glucose and HOMA2-IR in samples collected at baseline. These associations indicate that HOMA2-IR more strongly captures variation in C-peptide.

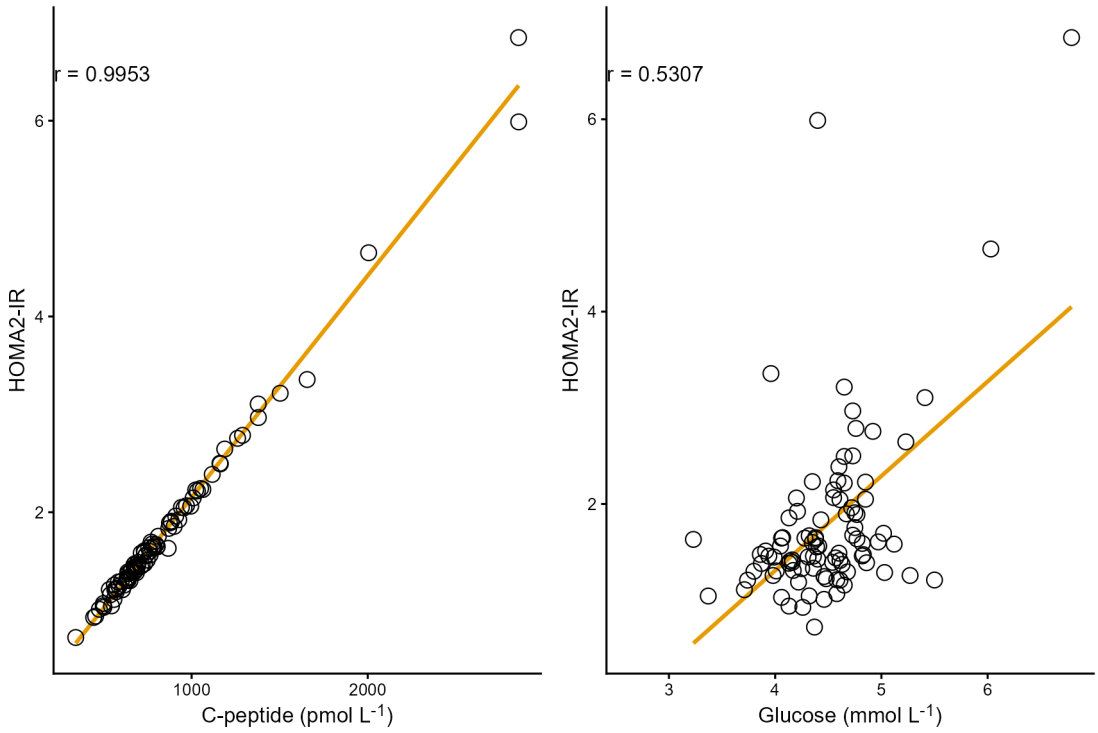

Figure 4.4: Associations between C-peptide or glucose and HOMA2-IR in samples collected at baseline. These associations indicate that HOMA2-IR more strongly captures variation in C-peptide.

Variables in the observed data are correlated within time points within variables and between variables across time points. To describe approximate variation and co-variation patterns, "naive" linear models were fitted with sex as a single predictor at time-point T1, sex, baseline values, and treatment as predictors for values at time-points T2 and T4. Residuals were extracted from the models, and estimates were plotted in Figure 4.5.

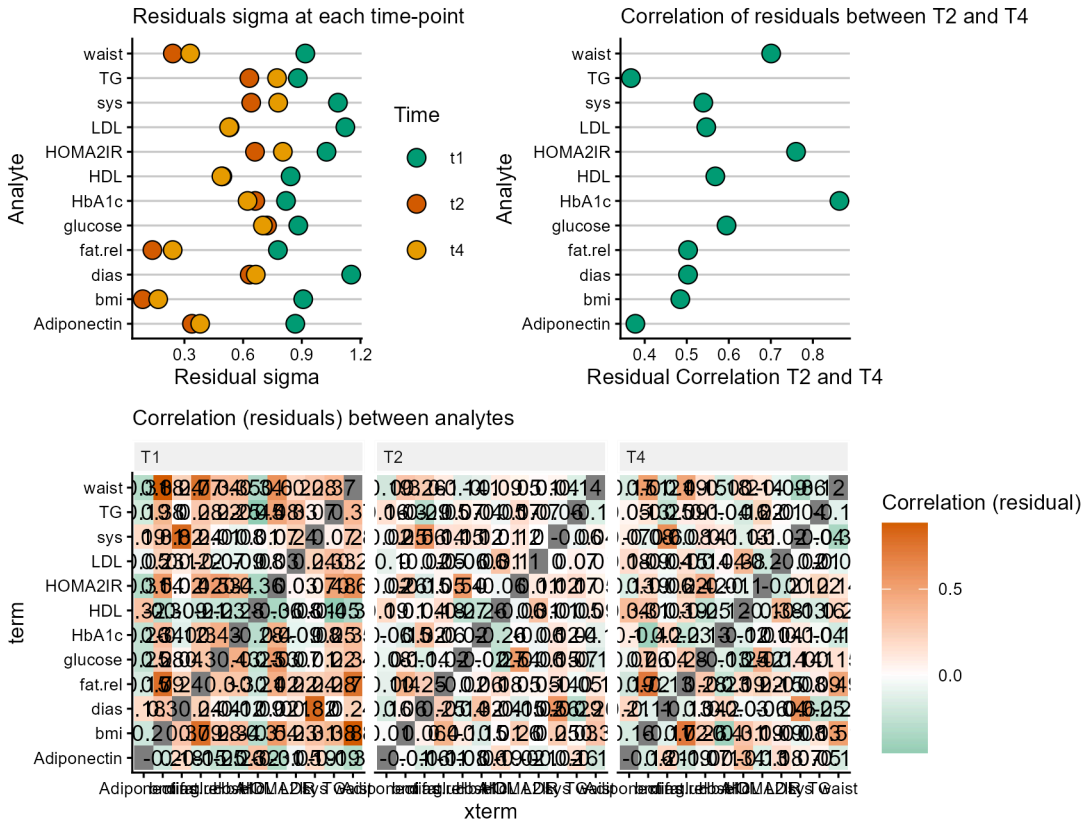

Figure 4.5: Residual variation and correlation pattern within and between variables in the observed data.

Correlation patterns among variables will be used to inform simulations (see Section 4.0.1). Correlations among variables also suggest that uncertainty associated with missing data can be reduced by exploiting the covariance among variables. The missing data patterns in the data set indicate that some missing data occurs in a single variable at a single time point whilst some participants have missing data in all variables at specific time points (Figure 4.6).

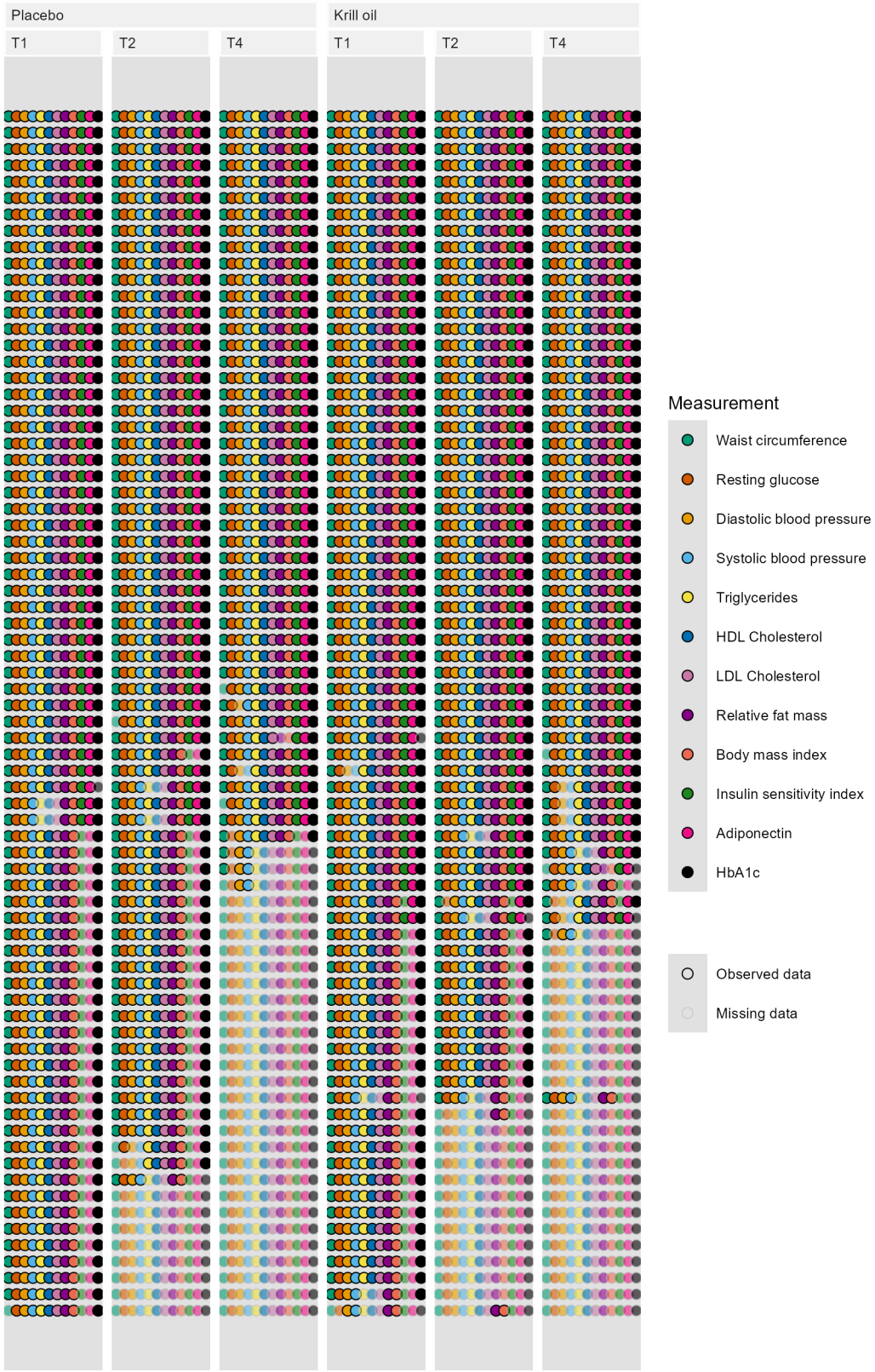

Figure 4.6: Observed and missing data across treatment groups and time-points. Rows are participants, columns are data points.

To assess if the outcome variables predict drop-out we fit a logistic regression model with drop-out (D_i) at time Time T2 or T4, coded as 0/1 as the outcome variable. The association between the standardized measurement type value ($X_{i[A]}$) and drop-out was determined while controlling for sex (S) within each treatment group (T).

$$
\begin{aligned}
D_{i[\text{Time}]} &\sim \text{Bernoulli } (p_i) \\
\text{Logit } (p_i) &= \alpha_{[A]} + \beta_{X[A]} X_{i[A]} + \beta_{S[A]} S_i + \beta_{T[A]} T_i + \beta_{TX[A]} T_i X_i \\
\alpha_{[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{X[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{S[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{T[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{TX[A]} &\sim \text{Normal } (0, 0.5)
\end{aligned}
\tag{4.1}
$$

In addition to measurements included in the metabolic syndrome diagnostic criteria (Fasting glucose, waist circumference, HDL, triglycerides, blood pressure), we also included relative fat mass, body mass index, age and insulin resistance index (HOMA2IR), LDL, and adiponectin. Prior predictive checks (Figure 4.7) confirmed that the model expects eleven measurement types in two treatment groups and that the combined priors weakly regularize away from all/non drop-out.

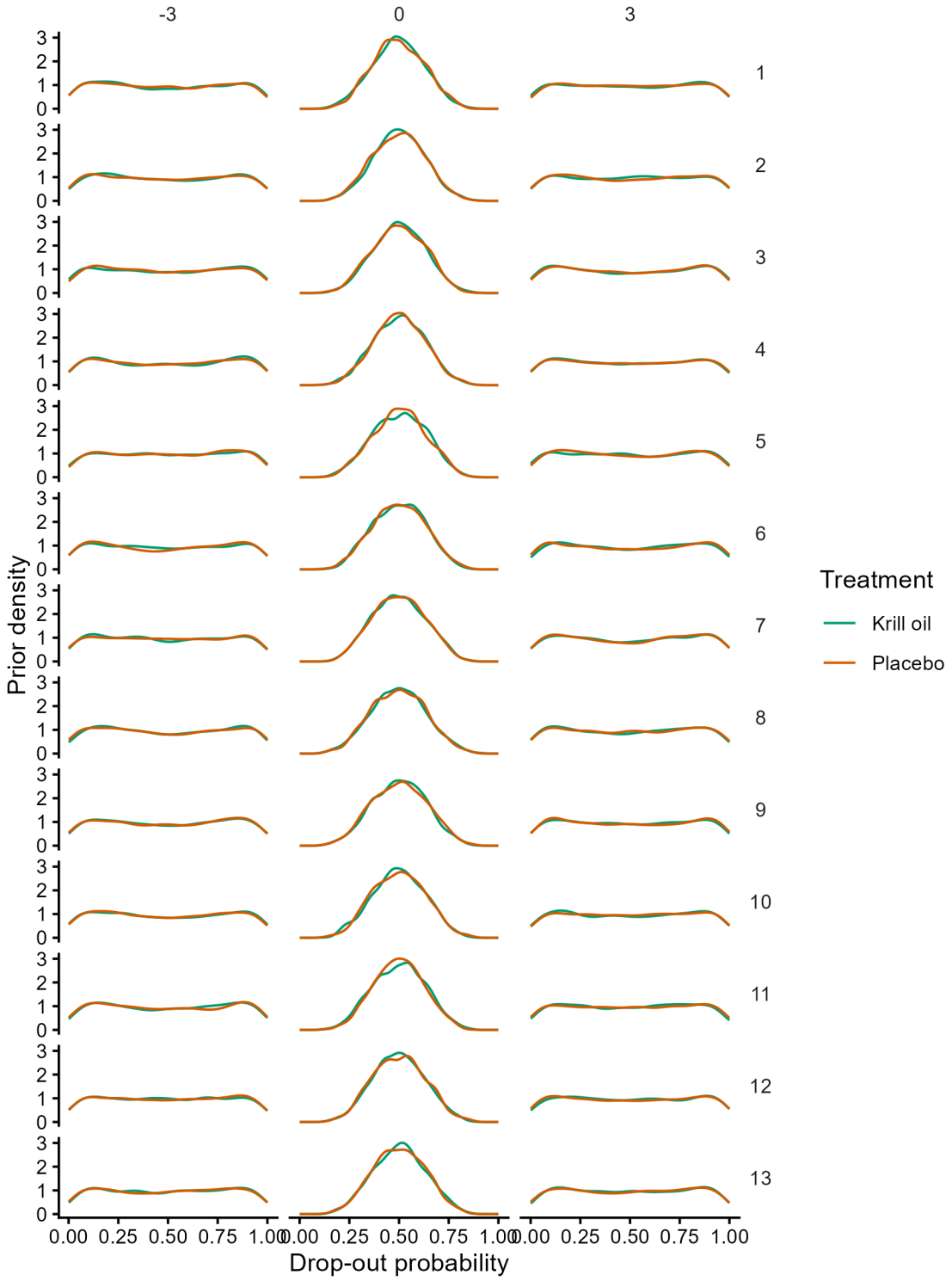

Figure 4.7: Prior predictive simulation for model 1. Columns are values of the predictor (SD-units) rows represents each predictor.

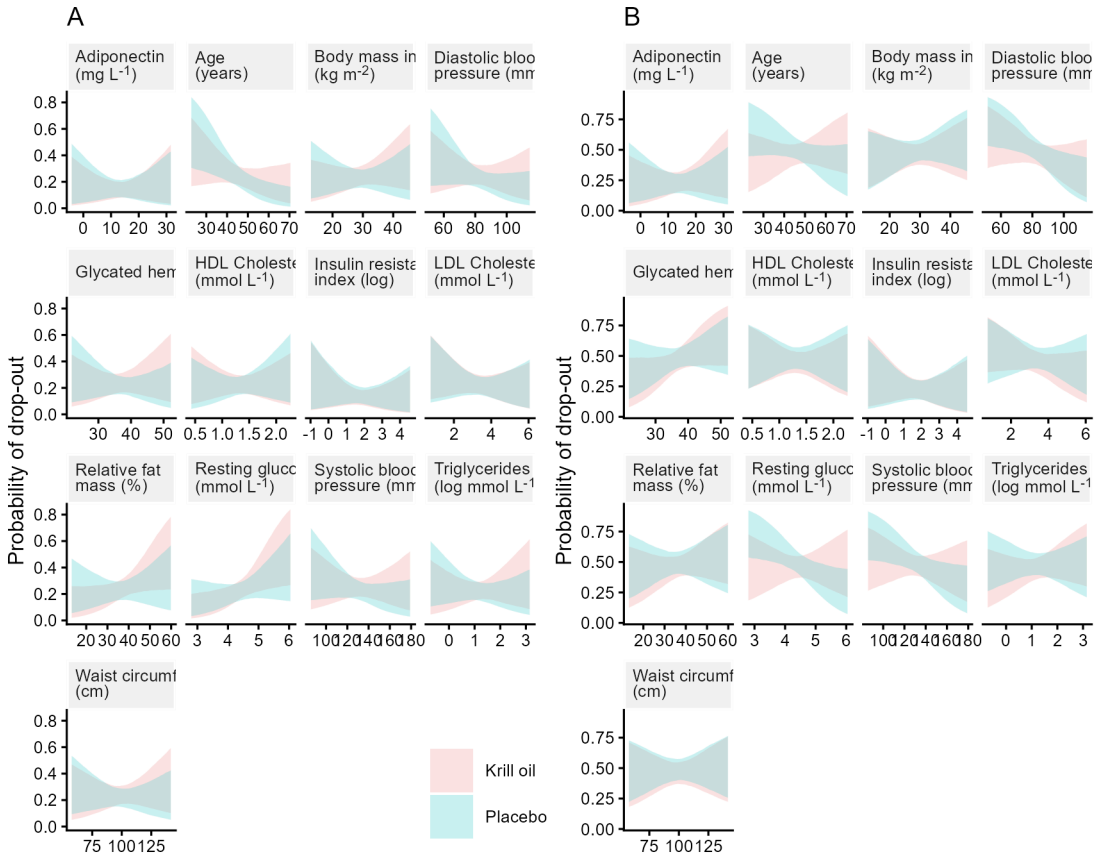

Figure 4.8: Probability of drop-out across values of baseline co-variates on per treatment group to time-point 2 (A) and 4 (B). Estimates are adjusted for sex, intervals are 90% percentile interval.

Table 4.3: Change in log-odds (Slope) for drop-out for 1 SD-unit change in baseline characteristic (Outcome) and differences in slope between treatment groups (&Delta Treatment) at time T2 and time T4. Values are posterior means with 95% percentile intervals.

Outcome	T2		T4	
	Slope	Δ Treatment	Slope	Δ Treatment
Adiponectin (mg L-1)	0.01 [−0.50, 0.53]	0.12 [−0.62, 0.84]	0.07 [−0.37, 0.51]	0.19 [−0.50, 0.88]
HDL Cholesterol (mmol L-1)	0.05 [−0.34, 0.43]	−0.15 [−0.79, 0.51]	−0.04 [−0.37, 0.30]	−0.04 [−0.62, 0.52]
Insulin resistance index (log)	−0.14 [−0.69, 0.38]	−0.00 [−0.83, 0.78]	−0.10 [−0.59, 0.35]	−0.05 [−0.81, 0.70]
Glycated hemoglobin	−0.00 [−0.41, 0.37]	0.21 [−0.47, 0.88]	0.27 [−0.10, 0.65]	0.21 [−0.35, 0.78]
LDL Cholesterol (mmol L-1)	−0.14 [−0.52, 0.22]	−0.04 [−0.69, 0.60]	−0.16 [−0.49, 0.15]	−0.13 [−0.68, 0.44]
Triglycerides (log mmol L-1)	−0.01 [−0.41, 0.37]	0.26 [−0.41, 0.92]	0.07 [−0.26, 0.38]	0.22 [−0.33, 0.77]
Age (years)	−0.41 [−0.78, −0.05]	0.31 [−0.31, 0.95]	−0.08 [−0.42, 0.26]	0.45 [−0.09, 0.98]
Body mass index (kg m-2)	0.08 [−0.29, 0.44]	0.20 [−0.39, 0.79]	0.10 [−0.21, 0.42]	−0.08 [−0.63, 0.47]
Diastolic blood pressure (mmHg)	−0.22 [−0.62, 0.14]	0.25 [−0.40, 0.92]	−0.35 [−0.70, −0.01]	0.22 [−0.38, 0.82]
Relative fat mass (%)	0.24 [−0.15, 0.62]	0.33 [−0.26, 0.98]	0.12 [−0.24, 0.48]	0.12 [−0.45, 0.70]
Fasting glucose (mmol L-1)	0.38 [0.00, 0.77]	0.26 [−0.38, 0.91]	−0.21 [−0.55, 0.13]	0.47 [−0.11, 1.08]
Systolic blood pressure (mmHg)	−0.15 [−0.53, 0.22]	0.26 [−0.37, 0.88]	−0.25 [−0.59, 0.09]	0.32 [−0.26, 0.91]
Waist circumference (cm)	0.01 [−0.39, 0.39]	0.19 [−0.38, 0.78]	0.03 [−0.30, 0.37]	−0.00 [−0.56, 0.55]

In conclusion, age was negatively associated with probability of drop-out at T2 without robust difference between treatment groups (Figure 4.8 A; Table 4.3). Resting glucose at baseline tended to predict drop-out to T4 with no strong difference between treatment groups (Figure 4.8 B; Table 4.3). Together this suggest that baseline covariates did not robustly interact with treatment in predicting drop-out. However, baseline covariates (age and fasting glucose) were associated with higher drop-out rates.

4.0.1 Synthetic data

Synthetic data is generated with known parameter values to validate models and to enable exploration of the effects of model misspecification and missing data. The first data simulation function uses a multivariate normal distribution to simulate conditional means for K measurement types and N participants.

$$\begin{aligned} y_{i,j} &\sim \text{MVNormal}\ (\mu_{k[i,j]}, \Sigma) \\ \mu_{k[i,j]} &= \alpha + \beta_k X \end{aligned} \tag{4.2}$$

Where $y_{i,j}$ is a vector of outcome values for participant i in time-point j, drawn from a multivariate normal distribution with conditional means ($\mu_{k[i,j]}$ and a variance-covariance matrix (Σ). The conditional means are simulated using a linear predictor with parameter vector β and design matrix X.

The covariance matrix is constructed from two separate correlation matrices and a cross-time-point vector of correlations within each measurement type. Residual standard deviations for each type (and time) are used to combine standard deviations and correlations into the variance-covariance matrix[1].

[1]In the simulation function (`./R/sim-fun/sim-type.R`) separate variance-covariance matrices are constructed and then combined. But separate correlation matrices are the function's input.

$$
\begin{aligned}
\rho_{t2} &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
\rho_{t4} &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
\rho_{\text{cross}} &= \text{diag } (p_{k,\text{ time}}) \\
R &= \begin{bmatrix} \rho_{t2} & \rho_{\text{cross}} \\ \rho_{\text{cross}}^T & \rho_{t4} \end{bmatrix} \\
S &= \text{diag } (\sigma_{k,\text{ time}}) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.3}
$$

We will be able to control the within-time *residual* correlations between measurement types and between-time *residual* correlations within measurement types. The full correlation matrix (R) will set the residual correlation between measurement types at different time points to 0. These will, however, be correlated through within-time-point and pre-intervention correlations.

The pre-intervention values are also conditionally constructed using a separate multivariate normal distribution.

$$
\begin{aligned}
y_{\text{pre }[i]} &\sim \text{MVNormal } (\mu_{k[i]}, \Sigma) \\
\mu_{k[i]} &= \alpha + \beta_k X \\
R &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
S &= \text{diag } (\sigma_k) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.4}
$$

Using this definition, we can control the correlations among measurement types (at baseline) and their variation. We can also make the means conditional on a set of predictors (design matrix X) with corresponding parameter values (β_k).

The approach detailed above is convenient as we are able to define each pairwise correlation. However, this comes with two caveats. First, for 8 variables we need to set $((8 \times 8 - 8)/2) \times 3 = 84$ correlation coefficients to capture between variable correlations at each time-point. Second, this approach does not guarantee a positive definite covariance matrix. An alternative is to use a factor loading approach[2]. For each variable we can specify a loading onto a factor and a residual variance for each variable (type). This approach is determined by λ, a $K \times F$ matrix where k_i are variables and f_j are latent factors, and ψ, a vector of length K defining residual variation.

$$
\begin{aligned}
\Sigma_{\text{raw}} &= \lambda \times \lambda^T + \text{diag } (\psi) \\
D &= \sqrt{\text{diag } (\Sigma_{\text{raw}})} \\
R &= D^{-1} \times \Sigma_{\text{raw}} \times D^{-1} \\
S &= \text{diag } (\sigma_k) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.5}
$$

This approach is used in `sim_type2` where Σ is built from the factor approach. Lambda is a matrix with K rows, representing variable types and F columns representing factors.

```
build_sigma_factor <- function(Lambda, psi, S) {
  K <- nrow(Lambda)
  R_raw <- Lambda %*% t(Lambda) + diag(psi)
  d <- sqrt(diag(R_raw))
  R <- R_raw / outer(d, d)
```

[2]See here for an introduction on the maths.

```
  diag(S) %*% R %*% diag(S)
}
```

As an example, $K = 5$ variable types are defined using $F = 3$ factors below. Approximate "variance explained" can be used to define the relationship between variables and factors where for k_1 we aim for 50%, 20% and 10% explained by f_1 to f_3 and the remaining 20% is error. The squared loadings plus error sum to 1: $0.5 + 0.2 + 0.1 + 0.2 = 1$. In code:

```
# Loadings as proportions
l1 <- c(0.5,  0.2,  0.1)
# Transforms to loadings
l1 <- sqrt(l1)
error <- 0.2

sum(l1^2) + error
```

```
[1] 1
```

The above example first defines loadings as proportions (summing to 1 with errors) and then adds a step to transform the proportions to loadings. For a full $K \times F$ λ matrix we populate the matrix and use ψ for errors.

```
# Defining a matrix of proportions explained by factors
Lambda <- matrix(
  c(0.5, 0.2, 0.1,
    0.8, 0.1, 0.05,
    0.1, 0.15, 0.1,
    0.05, 0.05, 0.2,
    0.05, 0.1, 0.01
    ),
                ncol = 3,
                byrow = TRUE)
# Defining remaining error
psi <- c(0.2, 0.05, 0.65, 0.7, 0.84)

# Transform from proportions to loadings
Lambda <- sqrt(Lambda)

# Check total proportions
rowSums(Lambda^2) + psi
```

```
[1] 1 1 1 1 1
```

Each row of the λ matrix plus errors now sum to 1. Next, we can use the λ matrix to construct Σ with unit standard deviations. The full covariance matrix (Σ) is used in calling `MASS::mvnorm` to produce the multivariate normal variable. As we are using standard deviations equal to 1, the covariance matrix equals the correlation matrix.

```
S <- rep(1, 5)
Sigma <- build_sigma_factor(Lambda, psi, S)

out <- MASS::mvrnorm(1000, rep(0, 5), Sigma = Sigma)

colnames(out) <- paste0("k", 1:5)
```

Table 4.4: Correlations used to simulate data built using a factor approach.

	k1	k2	k3	k4	k5
k1	1.000	0.845	0.497	0.400	0.331
k2	0.845	1.000	0.476	0.371	0.322
k3	0.497	0.476	1.000	0.299	0.225
k4	0.400	0.371	0.299	1.000	0.165
k5	0.331	0.322	0.225	0.165	1.000

Table 4.5: Correlations from simulated data (n=1000) built using a factor approach.

term	k1	k2	k3	k4	k5
k1	NA	0.852	0.483	0.390	0.313
k2	0.852	NA	0.479	0.364	0.291
k3	0.483	0.479	NA	0.280	0.223
k4	0.390	0.364	0.280	NA	0.148
k5	0.313	0.291	0.223	0.148	NA

In the loadings we defined k_1 and k_2 to be highly correlated through being influenced to a large degree by factors f_1 and f_2. In contrast, k_5 was given a lot of variation not explained by the factors which resulted in lower correlations with the other variables.

4.0.1.1 Heteroscedasticity and non-normal noise

The synthetic data-generating models described above produces homoscedastic data, i.e., equal residual variation between treatment groups. To create synthetic data under heteroscedastic conditions (unequal variance between treatment groups) an additional data generating model is created to enable setting treatment-specific variation (`sim_type3` function in `R/sim-fun.R`). The `sim_type3` function also includes two arguments to set additional noise in post-treatment measurements `prop_noise` controls the fraction of measurements that gets extra noise added and `sd_noise` controls the zero-centered normal distribution from which additional noise is drawn.

4.0.1.2 Missing data

Two mechanisms for including missing values are used in the simulation functions. First, a proportion of pre- and post-intervention values are set to missing, affecting single observations (e.g., a single measurement within a single participant). Second, the probability of missing observations from a whole participant is simulated using a Bernoulli model. This mechanism can connect baseline predictors to missing values. A Bernoulli simulation is performed, with p_i modeled linearly by α, which controls the average probability of complete missing observations, and the additional parameters β linked to predictors in a predictor data frame.

$$\begin{aligned} y_{\text{miss }[i]} &\sim \text{Bernoulli }(p_i) \\ \text{logit }(p_i) &= \alpha + \beta X \end{aligned} \tag{4.6}$$

Simulating a full (K = 8) data set of synthetic data, with missing values, results in observed patterns similar to the observed data (Figure 4.6, Figure 4.9 and Figure 4.10).

```
# A tibble: 3,552 × 8
   participant  type group   sex   age time       y    ystd
         <int> <int> <int> <int> <dbl> <chr>  <dbl>   <dbl>
```

```
 1           1     1     1     1  1.23 T1    -1.46  -1.23
 2           1     1     1     1  1.23 T2    -0.537 -0.568
 3           1     1     1     1  1.23 T3    -0.364 -0.444
 4           1     2     1     1  1.23 T1     0.762  0.524
 5           1     2     1     1  1.23 T2     0.114 -0.0497
 6           1     2     1     1  1.23 T3     1.51   1.19
 7           1     3     1     1  1.23 T1    -1.39  -1.17
 8           1     3     1     1  1.23 T2    -0.647 -0.547
 9           1     3     1     1  1.23 T3    -1.03  -0.869
10           1     4     1     1  1.23 T1     0.400  0.178
# ℹ 3,542 more rows
```

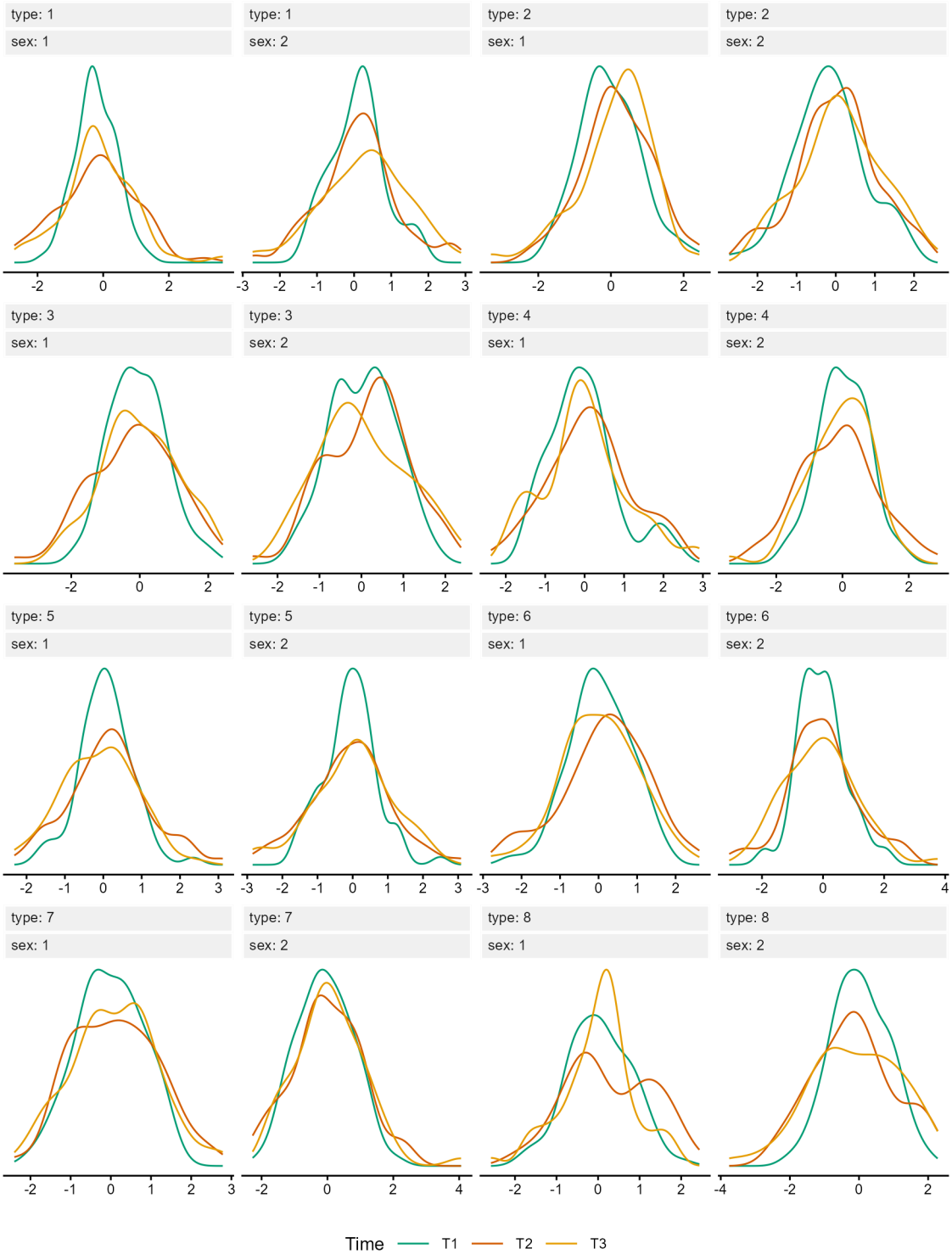

Figure 4.9: Distribution of (standardized) synthetic data across sex and measurement types.

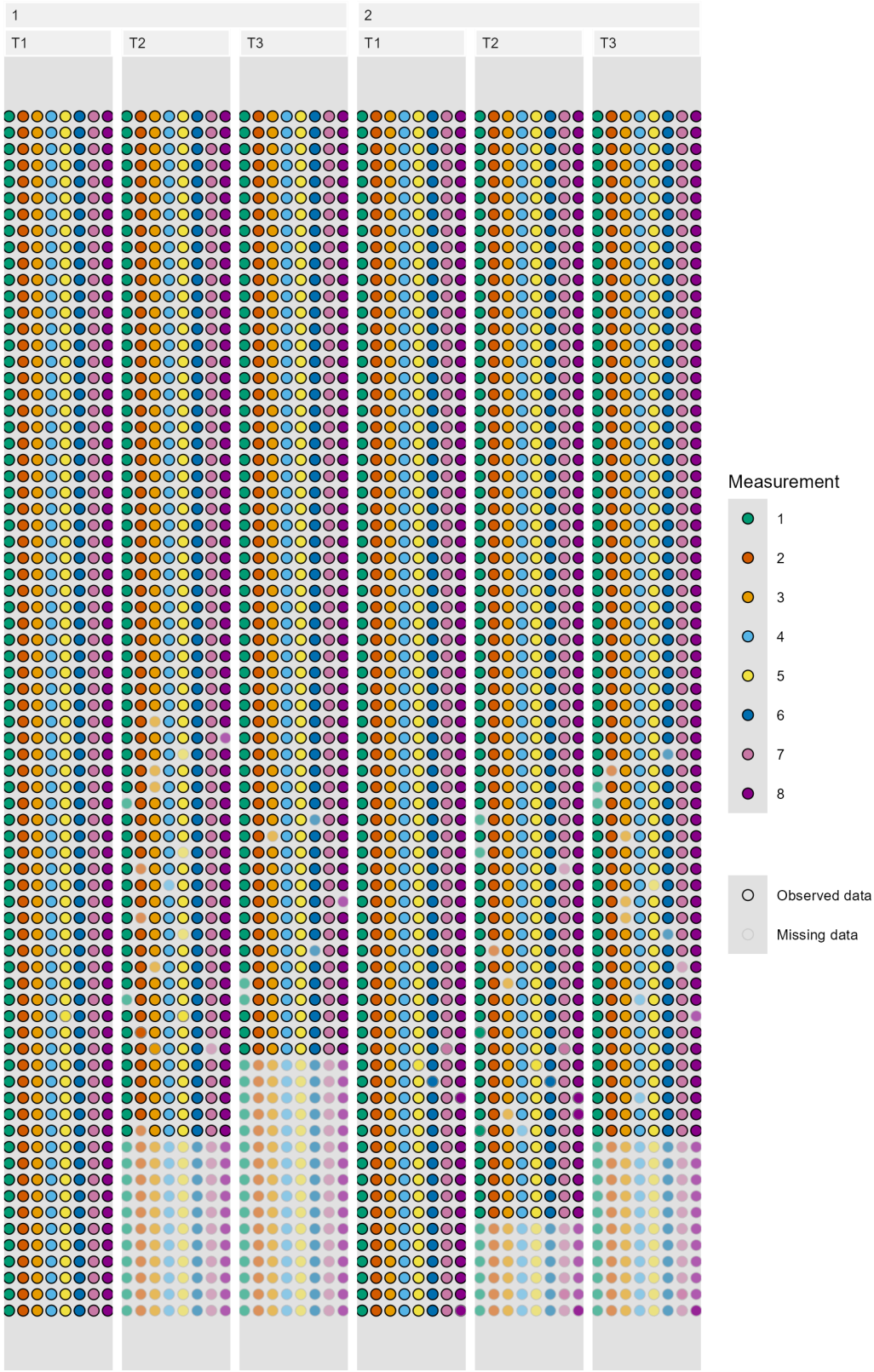

Figure 4.10: Missing data patterns from the data simulation. Missing values are found in specific measurements and in all measurements for a participant at specific time points.

5. Effects of krill oil supplementation on indications of metabolic syndrome - Models

Here models are incrementally developed by incorporating and testing features using prior predictive and posterior predictive checks, using both synthetic and observed data. Model performance (e.g., chain convergence and divergent transitions) is evaluated at relevant steps. The following chapter describes incremental model expansions towards a set of models that will be used to draw inference about the effect of krill oil supplementation on metabolic health.

The first model (model s1, `models/s1.stan`) is an ANCOVA-type model with pre-intervention values as covariates for estimating the post-intervention treatment- and measurement-type-specific means. The model is defined as:

$$
\begin{aligned}
y_i &\sim \text{Normal } (\mu_i, \sigma_i) \\
\mu_i &= \alpha_{\text{type,[TX]}} + \beta_{\text{type}} \times \text{pre}_i \\
\alpha_{\text{type,[TX]}} &\sim \text{Normal } (0, 1) \\
\beta_{\text{type}} &\sim \text{Normal } (1, 0.4) \\
\sigma_{\text{type}} &\sim \text{Exponential } (1)
\end{aligned}
\tag{5.1}
$$

Where $\alpha_{\text{type,TX}}$ is the measurement-type and treatment (TX) -specific intercept, and pre_i is the row-specific pre-treatment value.

Prior to modeling, observed values are placed on a standardized scale (mean-centered and scaled to the standard deviation). Since the model specification gives the same Normal $(0, 1)$ prior for both treatment groups, the implied prior for the differences between groups is Normal $(0, \sqrt{2})$. 95% of the prior probability mass therefore captures differences between groups from about −2.8 to 2.8. This prior is considered weakly regularizing. Similarly, the prior for the type-specific parameter for the pre-value covariates expects a 1-to-1 relationship between pre- and post-intervention values but with a wide range of possible relationships (95% prior probability mass in the range 0.2 to 1.8). The prior for the σ is also considered weakly regularizing allowing for a wide range of possible values but with more prior weight below the average (1).

The second model (model S2, `models/s2.stan`) for the main outcomes uses a multivariate normal likelihood. Here, we model outcomes as sampled from a multivariate normal distribution with a covariance matrix that connects the outcome types. This model extends model S1 by modelling the correlation structure among measurement types, enabling partial pooling across them. In further extensions of this model, the correlation structure will provide precision for imputing missing values, as observed scores will inform them through the structure. The same ANCOVA-type model is used here, in which pre-intervention scores serve as a covariate for the post-intervention group and measurement-type-level averages.

The model is defined as:

$$
\begin{aligned}
y_p &\sim \text{MVNormal}\ (\mu_p, \Sigma) \\
\mu_{p,\ \text{type}} &= \alpha_{\text{type,TX}} + \beta_{\text{type}} \times \text{pre}_{p,\ \text{type}} \\
\alpha_{\text{type,TX}} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{type}} &\sim \text{Normal}\ (1, 0.5) \\
\Sigma &= \text{S} \times \text{R} \times \text{S} \\
S &= \begin{bmatrix} \sigma_{\text{type}\ [1]} & 0 & .. \\ 0 & \sigma_{\text{type}\ [2]} & .. \\ .. & .. & .. \end{bmatrix} \\
R &= \begin{bmatrix} 1 & \rho_{12} & .. \\ \rho_{21} & 1 & .. \\ .. & .. & .. \end{bmatrix} \\
\sigma_{\text{type}} &\sim \text{Exponential}\ (1) \\
\text{R} &\sim \text{LKJcorr}\ (4)
\end{aligned}
\tag{5.2}
$$

Where y_p is a participant-level vector of outcome scores. Σ is the covariance matrix with standard deviations S and the correlation matrix R. The prior on R is defined using the LKJ distribution which provides regularization away from extreme values (-1, 1).

The third (model S3, `models/s3.stan`) includes all available data for one time-point, and imputes missing data. Imputation is done for both missing pre- and post-intervention observations. The pre-intervention imputation is done using a separate multivariate-normal model, defined as:

$$
\begin{aligned}
y^{\text{all}}_{\text{pre},p} &\sim \text{MVNormal}\ (\mu_{\text{pre},p}, \Sigma_{\text{pre}}) \\
y^{\text{all}}_{\text{pre},\ \text{type},p} &= \begin{cases} y^{\text{obs}}_{\text{pre},\ \text{type},p} \text{ if observed} \\ y^{\text{miss}}_{\text{pre},\ \text{type},p} \text{ if missing} \end{cases} \\
y^{\text{miss}}_{\text{pre},\ \text{type},p} &\sim \text{Normal}\ (0, 2)
\end{aligned}
\tag{5.3}
$$

Where missing values are treated as parameters and given a vague prior (Normal(0,2)), this makes sampling under prior-only more efficient.

The main model uses the $y^{\text{all}}_{\text{pre},p}$ as a covariate in the linear predictor. Again, missing values in the outcome are treated as parameters. The prior for missing observation is only included in prior predictive simulations.

$$
\begin{aligned}
y^{\text{all}}_{p} &\sim \text{MVNormal}\ (\mu_p, \Sigma) \\
y^{\text{all}}_{p,\ \text{type}} &= \begin{cases} y^{\text{obs}}_{p,\ \text{type}} \text{ if observed} \\ y^{\text{mis}}_{p,\ \text{type}} \text{ if missing} \end{cases} \\
y^{\text{miss}}_{\text{type},p} &\sim \text{Normal}\ (0, 2) \\
\text{pre}_{p,\ \text{type}} &= y^{\text{all}}_{\text{pre},p,\ \text{type}} \\
\mu_{p,\ \text{type}} &= \alpha_{\text{type,TX}} + \beta_{\text{type}} \times \text{pre}_{p,\ \text{type}} \\
\alpha_{\text{type,TX}} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{type}} &\sim \text{Normal}\ (1, 0.5) \\
\Sigma &= \text{S} \times \text{R} \times \text{S} \\
S &= \begin{bmatrix} \sigma_{\text{type}\ [1]} & 0 & .. \\ 0 & \sigma_{\text{type}\ [2]} & .. \\ .. & .. & .. \end{bmatrix} \\
R &= \begin{bmatrix} 1 & \rho_{12} & .. \\ \rho_{21} & 1 & .. \\ .. & .. & .. \end{bmatrix} \\
\sigma_{\text{type}} &\sim \text{Exponential}\ (1) \\
\text{R} &\sim \text{LKJcorr}\ (4)
\end{aligned}
\tag{5.4}
$$

Model S4 (`models/s4.stan`) adds sex as a covariate with the intention to improve predictions. The relevant addition to model s3 is in the linear predictor:

$$\begin{aligned} \mu_{p,\text{ type}} &= \alpha_{\text{type,TX}} + \beta_{\text{type}} \times \text{pre}_{p,\text{ type}} + \beta_{\text{sex}} \times \text{sex}_p \\ \beta_{\text{type,sex}} &\sim \text{Normal }(0, 0.5) \end{aligned} \tag{5.5}$$

The sex_p variable is created as male $= -0.5$ and female $= 0.5$ which gives that the prior for $\beta_{\text{type,sex}}$ expects that most of the difference between males and females in post-intervention scores is within ± 1 SD.

Model S5 (`models/s5.stan`) and S6 (`models/s6.stan`) further extends previous models by including both post-intervention time-points. In model S5 residuals are not correlated between time-points. Instead, separate covariance matrices are built for each time-point. The covariance matrices are therefore

$$\begin{aligned} \Sigma_{\text{time}} &= \text{S}_{\text{time}} \times \text{R}_{\text{time}} \times \text{S}_{\text{time}} \\ S_{\text{time}} &= \begin{bmatrix} \sigma_{1\text{ [time]}} & 0 & .. \\ 0 & \sigma_{2\text{ [time]}} & .. \\ .. & .. & .. \end{bmatrix} \\ R_{\text{time}} &= \begin{bmatrix} 1 & \rho_{12\text{ [time]}} & .. \\ \rho_{21\text{ [time]}} & 1 & .. \\ .. & .. & .. \end{bmatrix} \\ \sigma_{\text{type [time]}} &\sim \text{Exponential }(1) \\ \text{R}_{\text{time}} &\sim \text{LKJcorr }(4) \end{aligned} \tag{5.6}$$

The total number of parameters in each Σ is 36 for 8 measurement types.

Model 6 includes a full covariance matrix for the multivariate normal model. The total number of parameters in the matrix is 136 for 8 measurements with σ indexed 1 to 16. For S:

$$S = \begin{bmatrix} \sigma_1 & 0 & \cdots \\ 0 & \cdots & \cdots \\ \cdots & \cdots & \sigma_{2K} \end{bmatrix} \tag{5.7}$$

Finally, model S7 (`models/s7.stan`) extends model S1 by including both post-intervention time-points and missing data imputation. No correlation structures are included in the model and parameters (σ, β α are time- and type-specific).

5.0.1 Prior predictive checks, preliminary models

Prior predictive checks are performed for each model to confirm expected relationships between the pre-intervention covariate (Figure 5.1 and Figure 5.2), and prior expectations of treatment means (Figure 5.3).

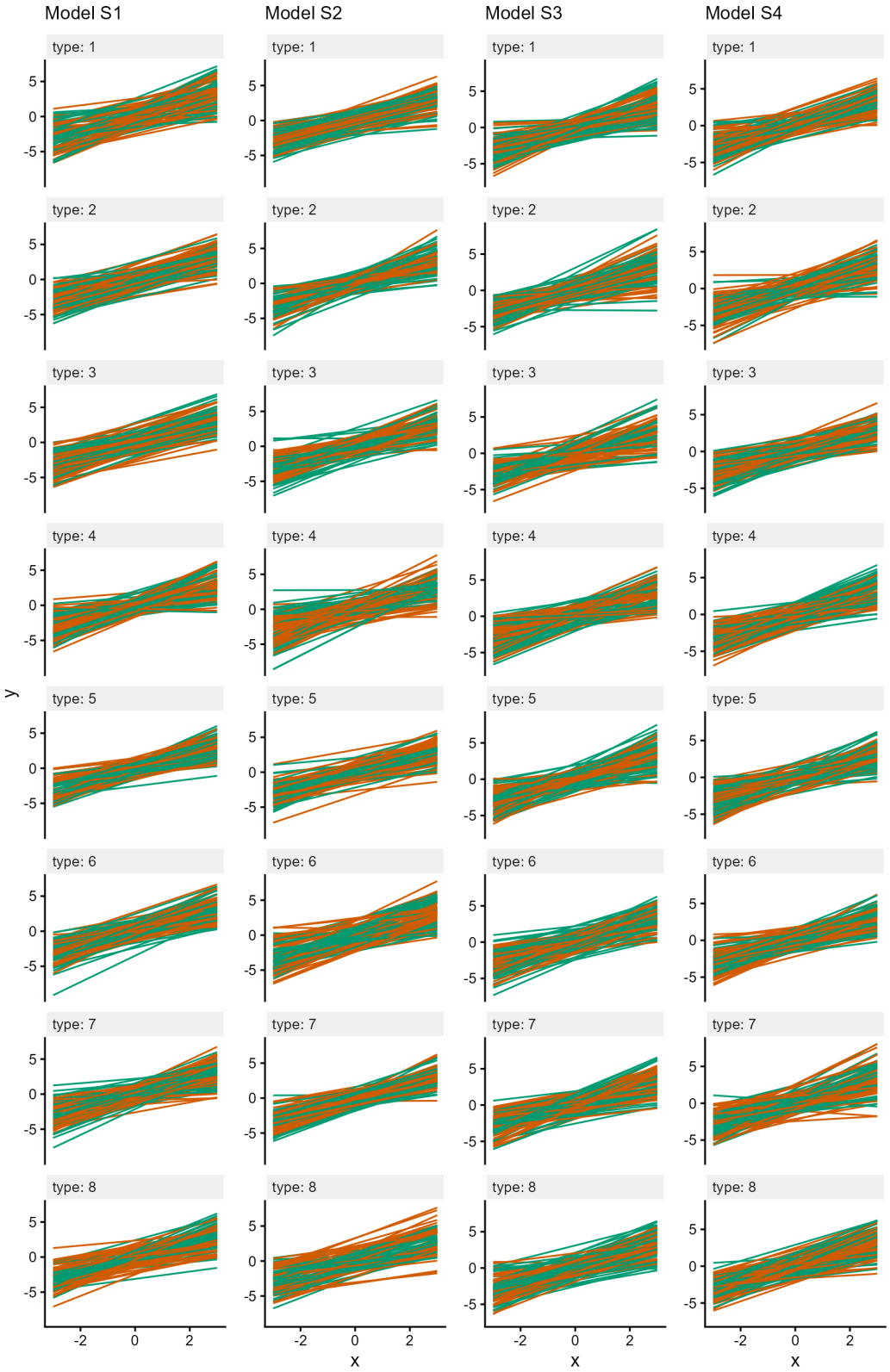

Figure 5.1: Prior predictive check of pre- to post-intervention associations in model S1-S4.

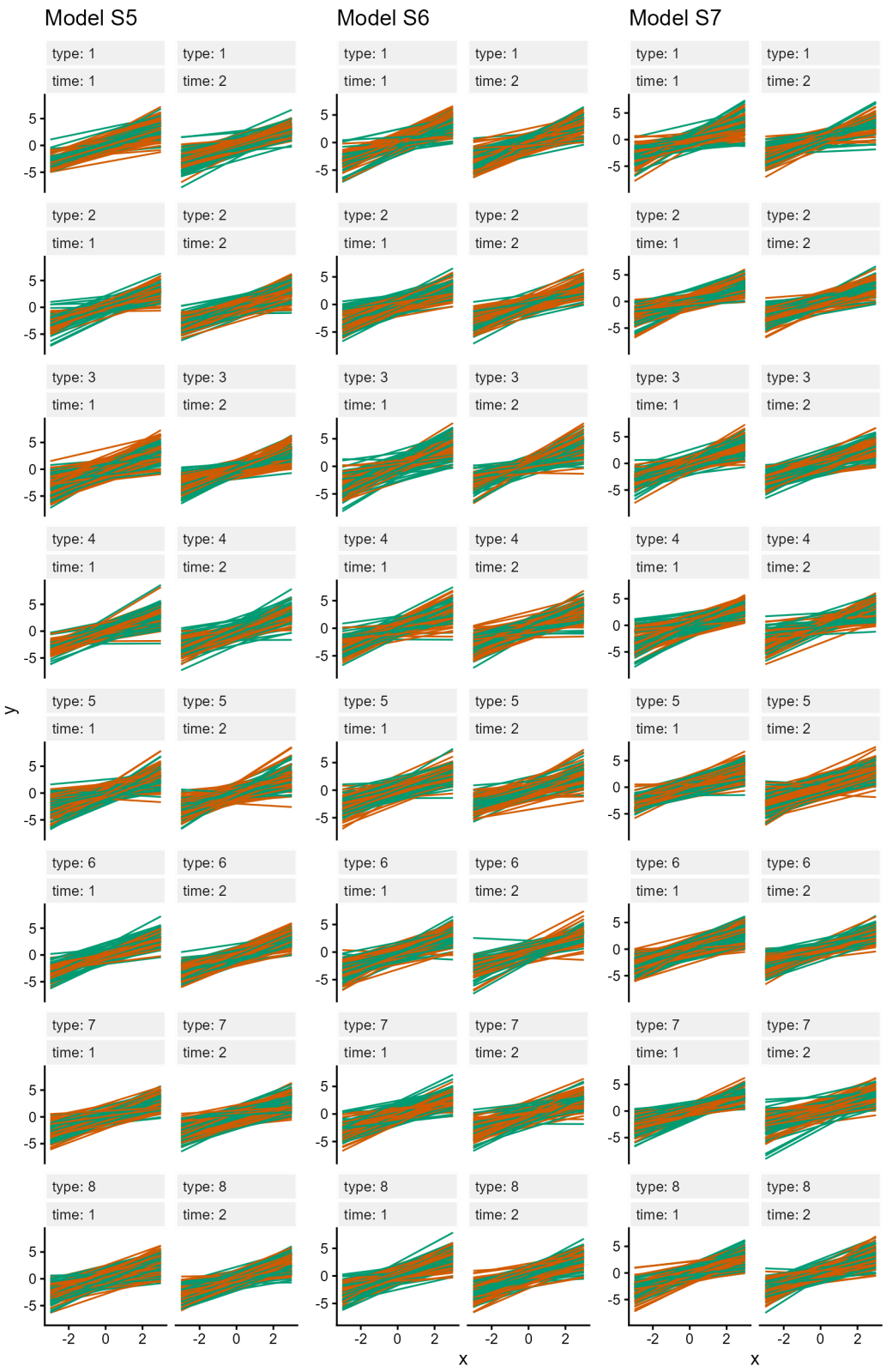

Figure 5.2: Prior predictive check of pre- to post-intervention associations in model S5-S7.

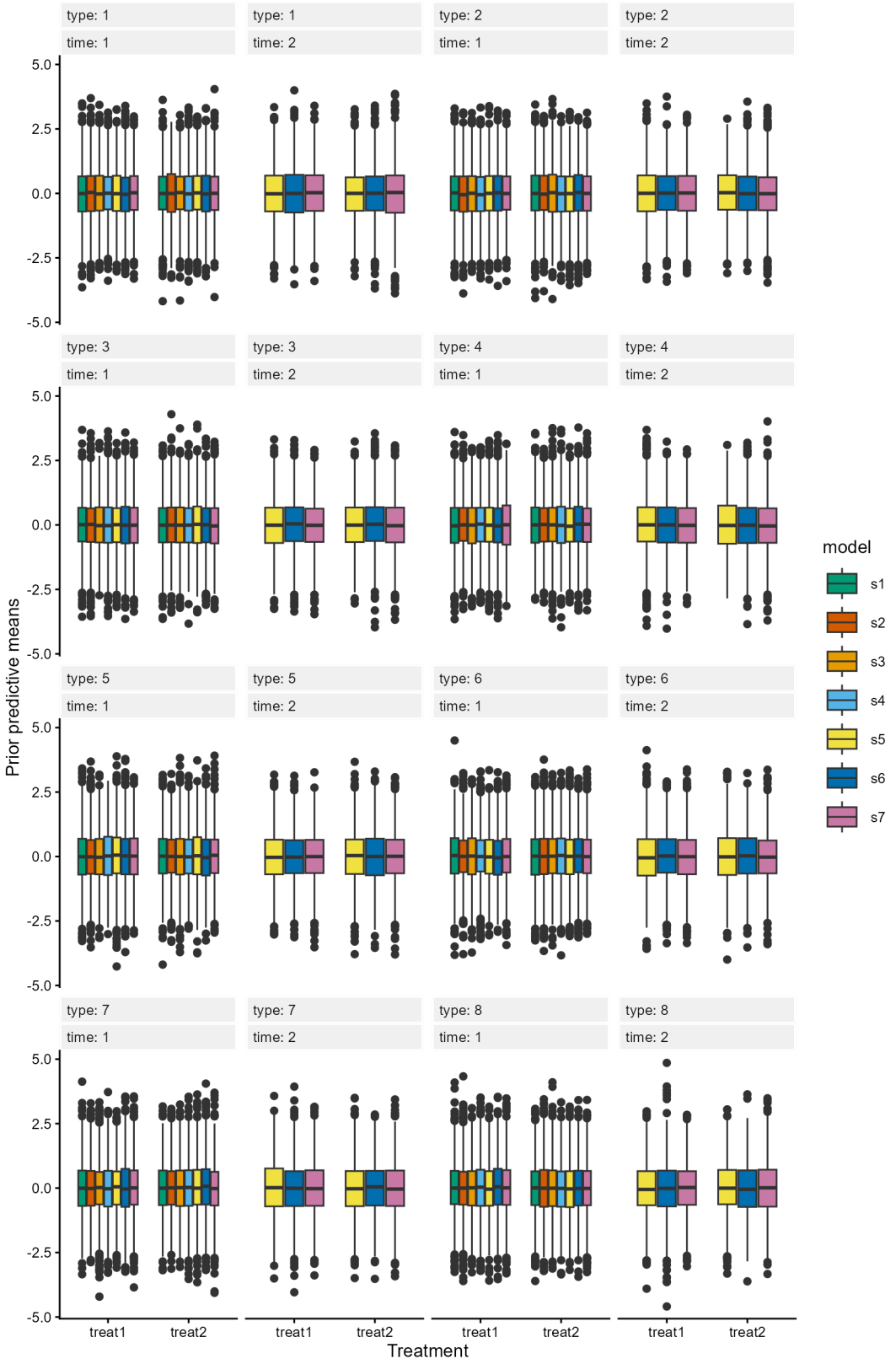

Figure 5.3: Prior predictive distributions of treatment means in models S1-S7.

The prior predictive checks confirm working and comparable models. A further check of model performance is done using synthetic data. We assess if we can recover simulated effects in the model estimates,

we also assess the models effects of the prior on “reversed” effects, i.e. effect of opposite direction in the two groups.

Here, a simulated data set is constructed where we fix the relationship between pre- and post-intervention values ($\beta_{pre} = 0.95$), residual variation is set to 0.2 and effect sizes of treatment effects are set to −0.5 in type 1 and 7, 0.5 in type 2 and 8, and 0 in all other types. The model capture these effects and this preliminary check does not suggest preference for one treatment over the other (Figure 5.4).

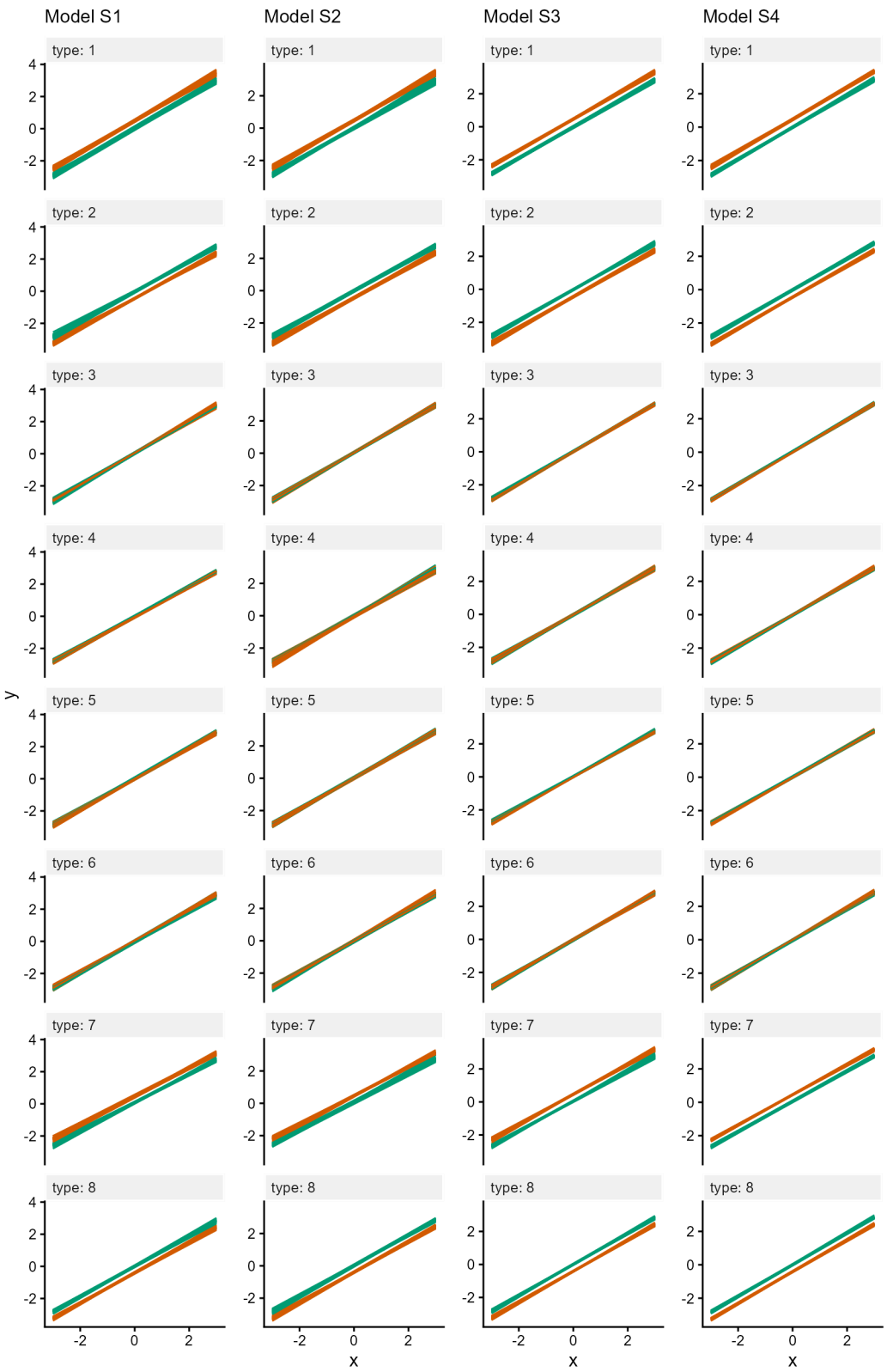

Figure 5.4: Posterior predictive check of pre- to post-intervention associations in model S1-S4 on synthetic data.

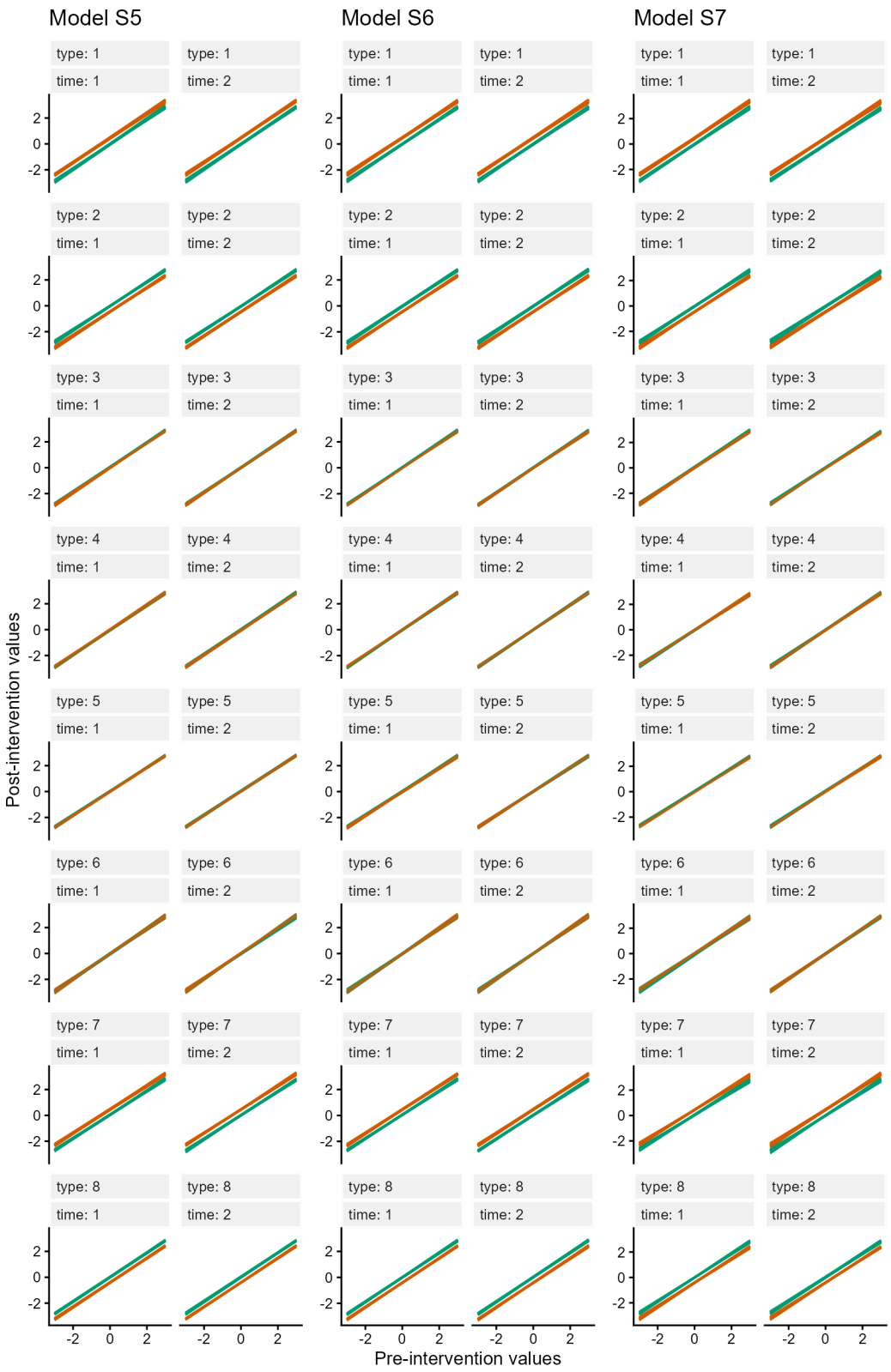

Figure 5.5: Prior predictive check of pre- to post-intervention associations in model S5-S7.

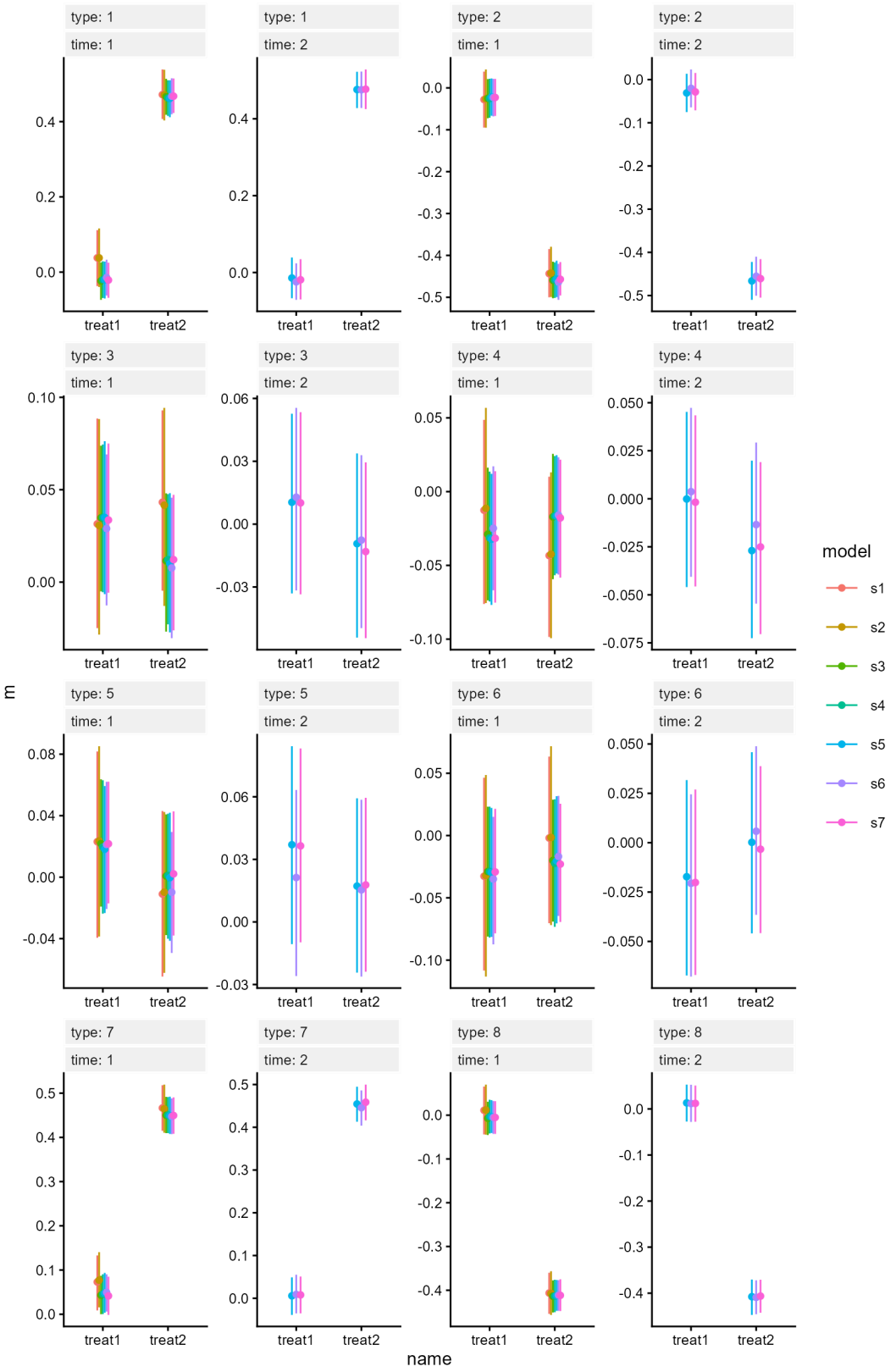

Figure 5.6: Posterior predictive distributions of treatment means in models S1-S7.

Using the synthetic data we confirm that the incrementally developed models captures the same patterns when they use the same data. In the above test, synthetic data was generated with large effects and negli-

gible correlation structures across measurement types. We should therefore not expect large differences between models that could potentially capture different between-measurement type correlations patterns.

Based on these preliminary models, model S6 and S7 are kept for further development and evaluation. Model S6 is replaced by S8 that will also generate diagnostic criteria thresholds. Additionally, to allow for treatment-specific variation we including treatment-specific σ by updating model S7 and S8 to S9 and S10, respectively. The relevant update in S7 to S9 is an type-, time- and treatment-specific σ ($\sigma_{\text{type, time, treat}} \sim \text{Exponential}(1)$). Similarly, model s10 adds type-, time- and treatment-specific σ to the model by incorporating it in the multivariate-normal model.

6. Effects of krill oil supplementation on indications of metabolic syndrome - Posterior predictive simulations and model expansions

Models S7 to S10 (Table 6.1) were next fitted with the observed data. Diagnostic statistics indicated that the HMC algorithm converges with satisfactory results (Table 6.2). Figure 6.1 shows per-parameter summaries of $\hat{R}$ and effective sample size indicating that parameters have $\hat{R} << 1.01$ and effective sample sizes are generally > 1000 per chain.

Table 6.1: Summary of model configurations

Model	Likelihood	Variance structure
s7|Univariate normal | Time- and type-specific σ| |s8	Multivariate normal	Time- and type-specific σ
s9|Univariate normal | Time-, type-, and treat-specific σ| |s10	Multivariate normal	Time-, type-, and treat-specific σ

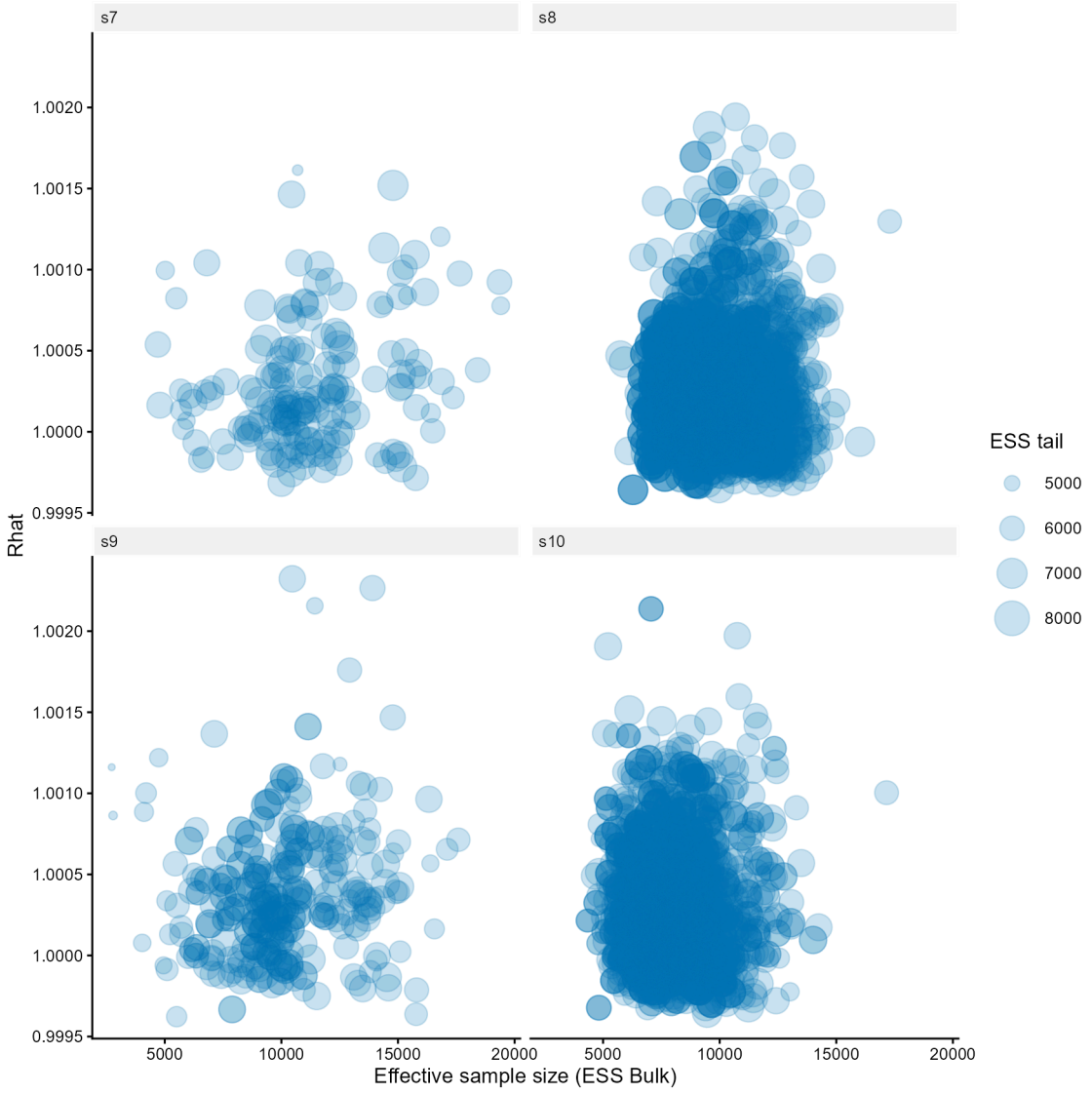

Figure 6.1: Summary chain statistics indicate satisfactory results in models s7-s10.

Table 6.2: Diagnostic summaries of models s8 to s10. BFMI, Bayesian Fraction of Missing Information.

Chain	n divergent trans.	n maximum treedepth	BFMI
		s7	
1	0	0	0.83
2	0	0	0.83
3	0	0	0.88
4	0	0	0.90
		s8	
1	0	0	0.74
2	0	0	0.74
3	0	0	0.85
4	0	0	0.80
		s9	
1	0	0	0.84
2	0	0	0.89
3	0	0	0.84
4	0	0	0.78
		s10	
1	0	0	0.87
2	0	0	0.84
3	0	0	0.77
4	0	0	0.75

Using the observed data set we are next assessing the models tendencies to produce replicated data similar to the observed data. All models (s7-s10) are still considered in these posterior predictive checks. The overall distributions of replicated data (y^{rep}) tend to look very similar to the observed data for each measurement type, treatment, and time-point combinations in all considered models (Figure 6.2, Figure 6.3, Figure 6.4, Figure 6.5).

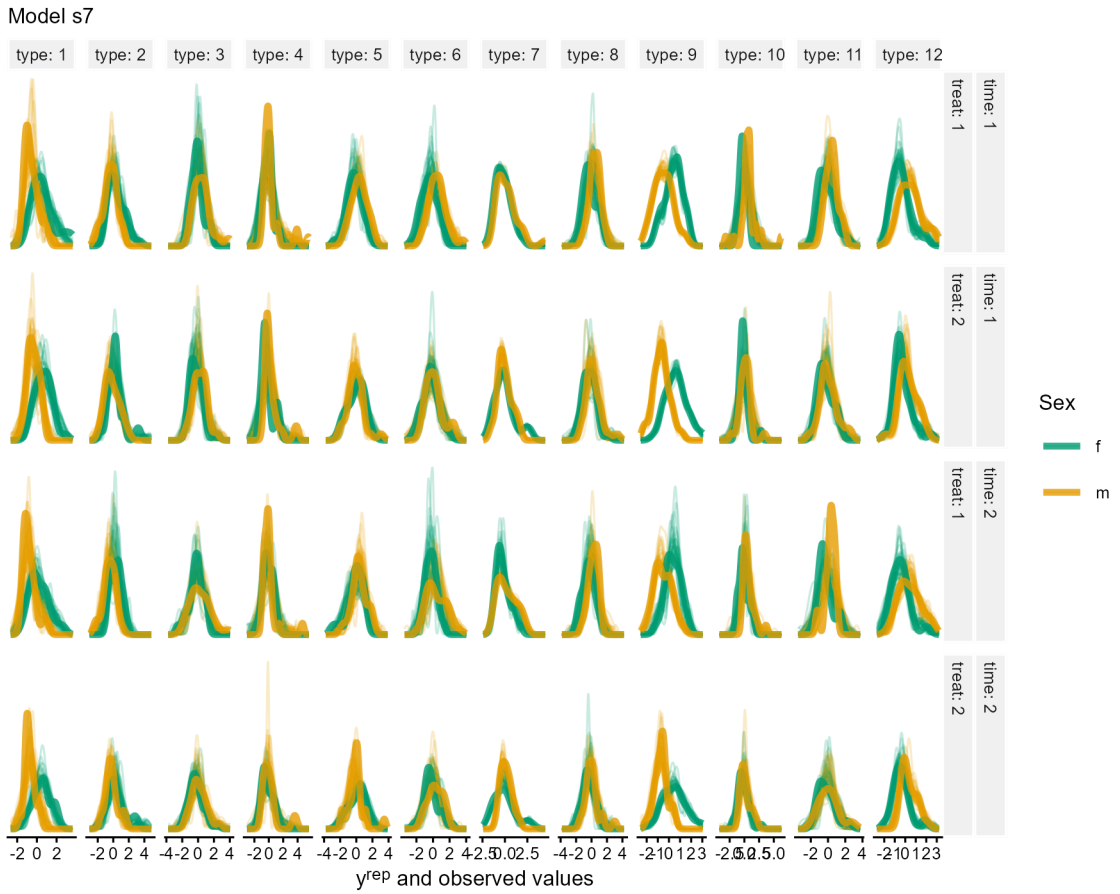

Figure 6.2: Posterior predictive check of models s7. y^{rep} (40 draws from the posterior) and observed data (thick line) indicate that the model replicate the observed data distributions.

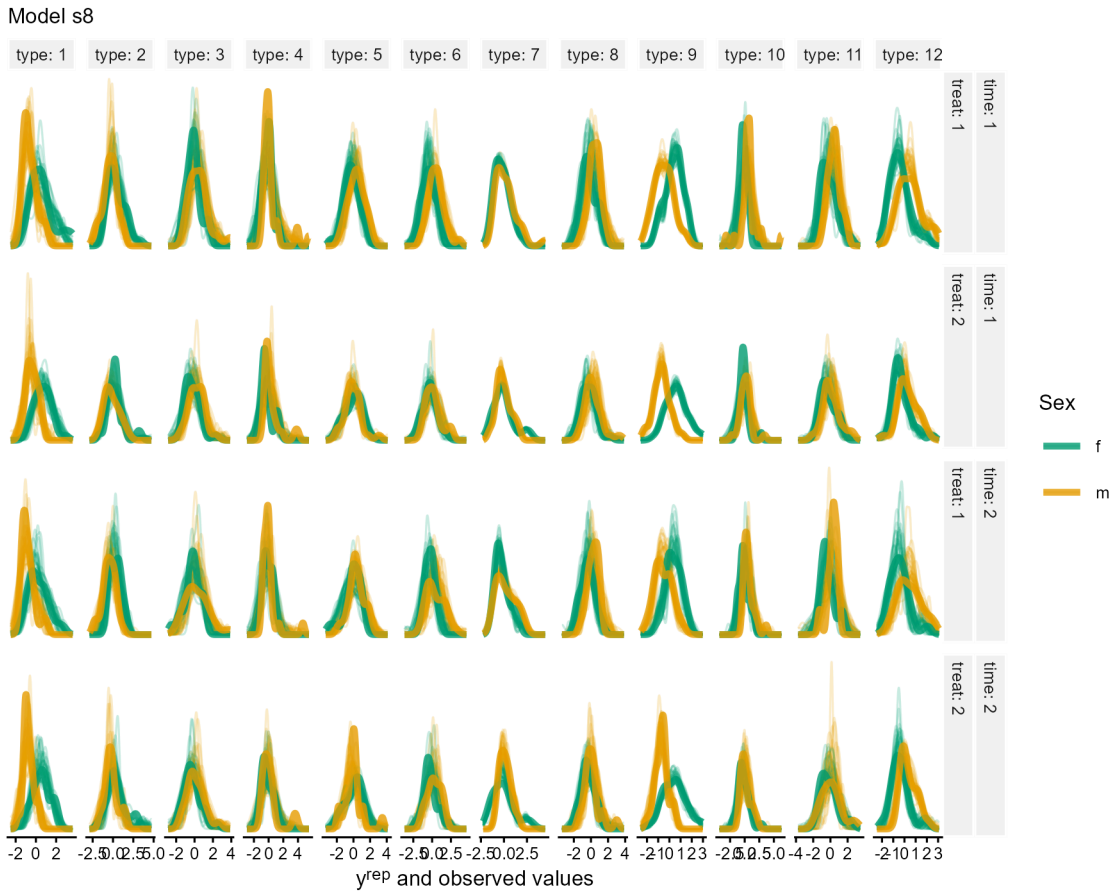

Figure 6.3: Posterior predictive check of models s8. y^{rep} (40 draws from the posterior) and observed data (thick line) indicate that the model replicate the observed data distributions.

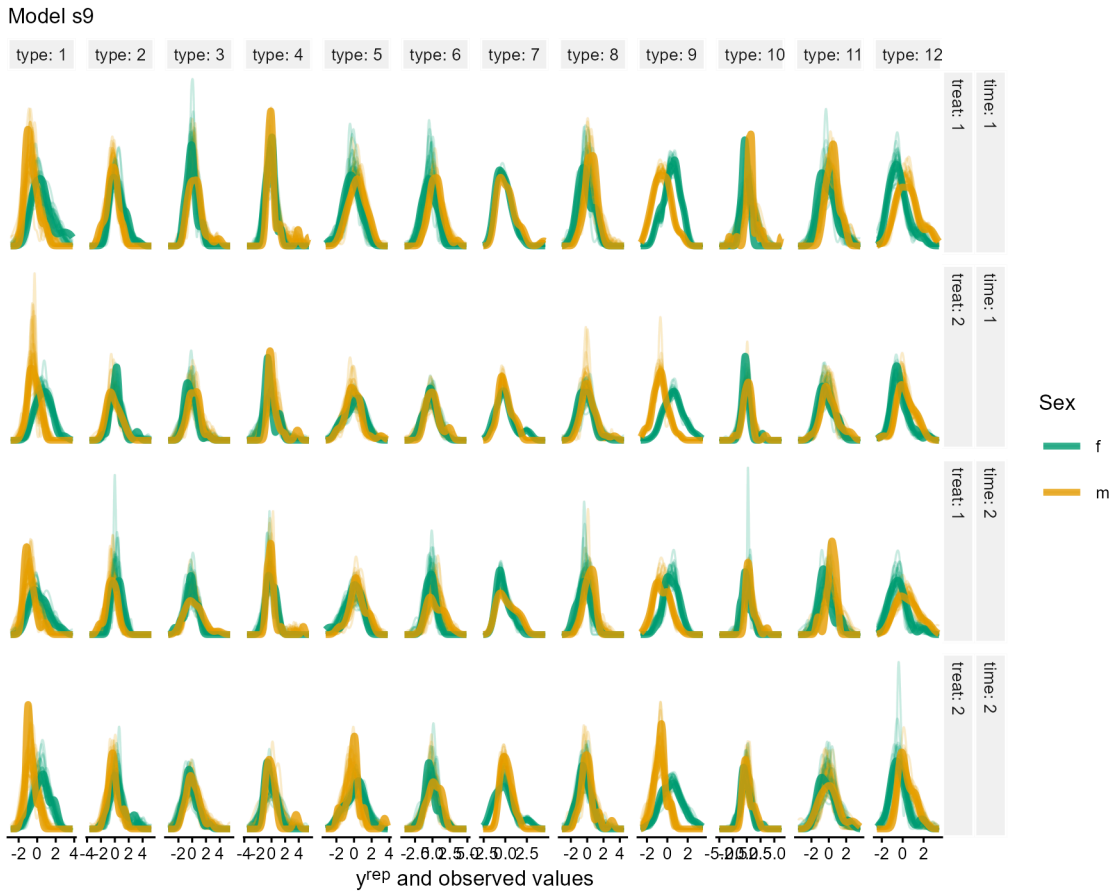

Figure 6.4: Posterior predictive check of models s9. y^{rep} (40 draws from the posterior) and observed data (thick line) indicate that the model replicate the observed data distributions.

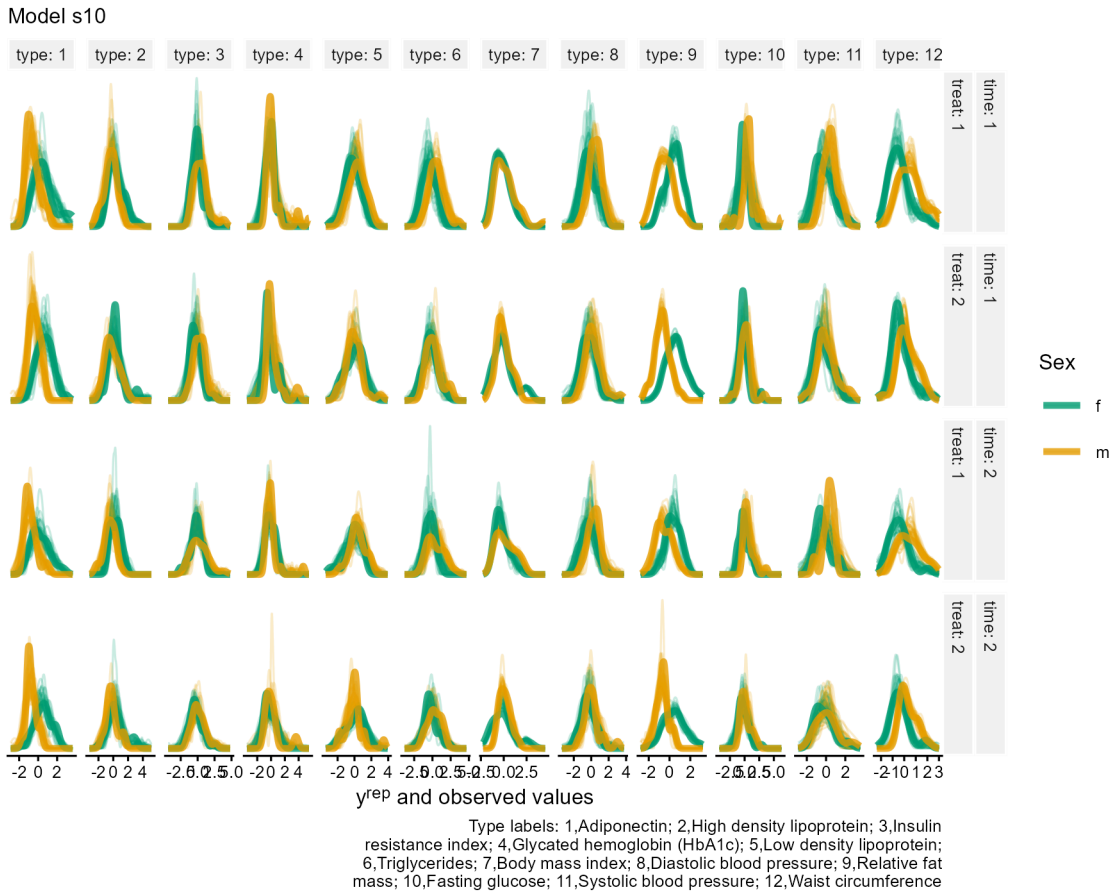

Figure 6.5: Posterior predictive check of models s10. y^{rep} (40 draws from the posterior) and observed data (thick line) indicate that the model replicate the observed data distributions.

To further explore model implications we calculate the observed standard deviation per measurement type and treatment and extract distributions implied by the model through y^{rep}. In Figure 6.6 and Figure 6.7 most distributions of y^{rep} standard deviations contain the empirical SD (shown as contrasts between y^{rep} and y^{obs} where 0 indicate agreement), however, for both model S7 and S8, some of the standard deviations observed in the data do not correspond to those implied by the model.

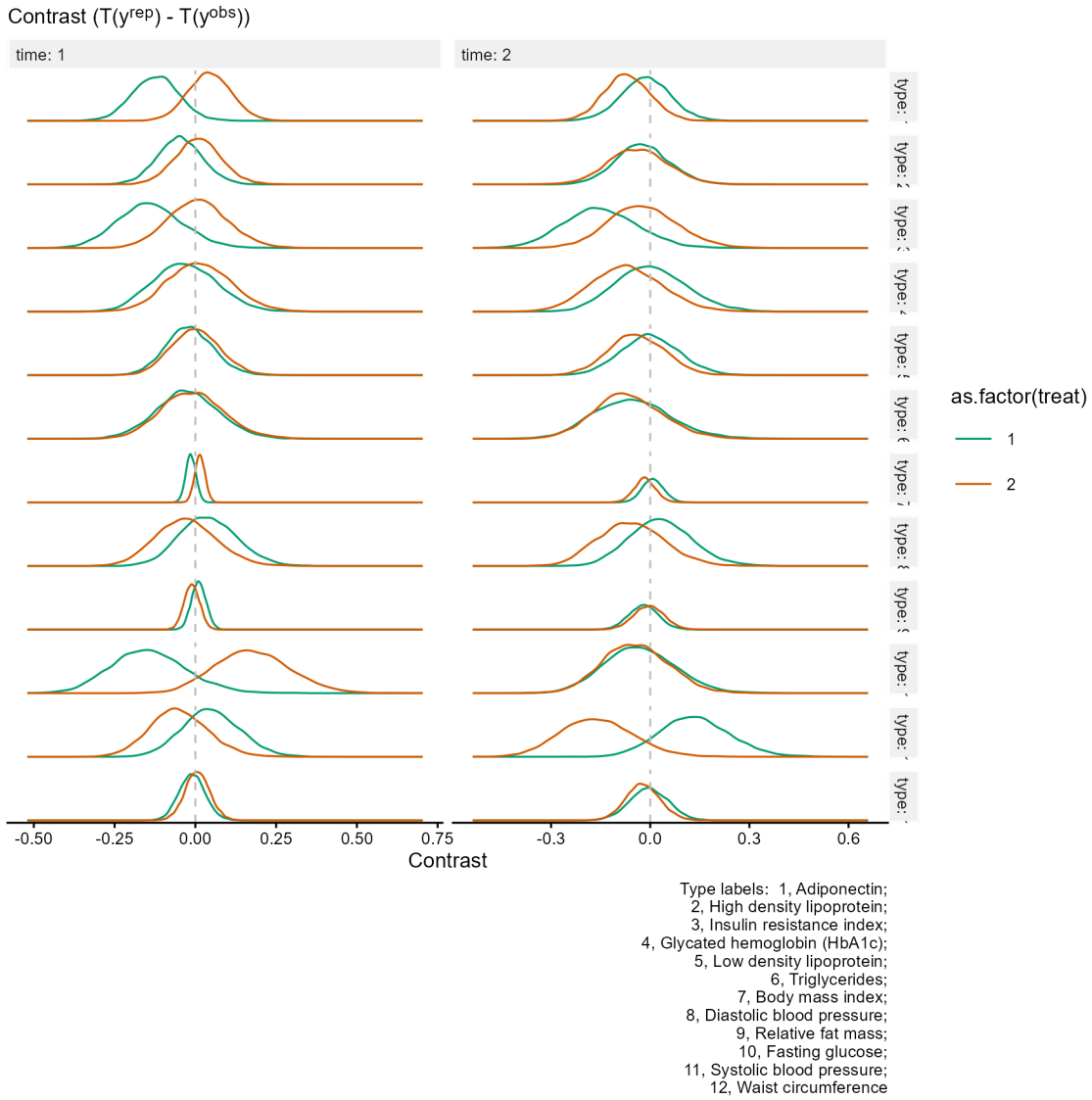

Figure 6.6: Posterior predictive check of model s7. The distributions of standard deviations calculated from y^{rep}

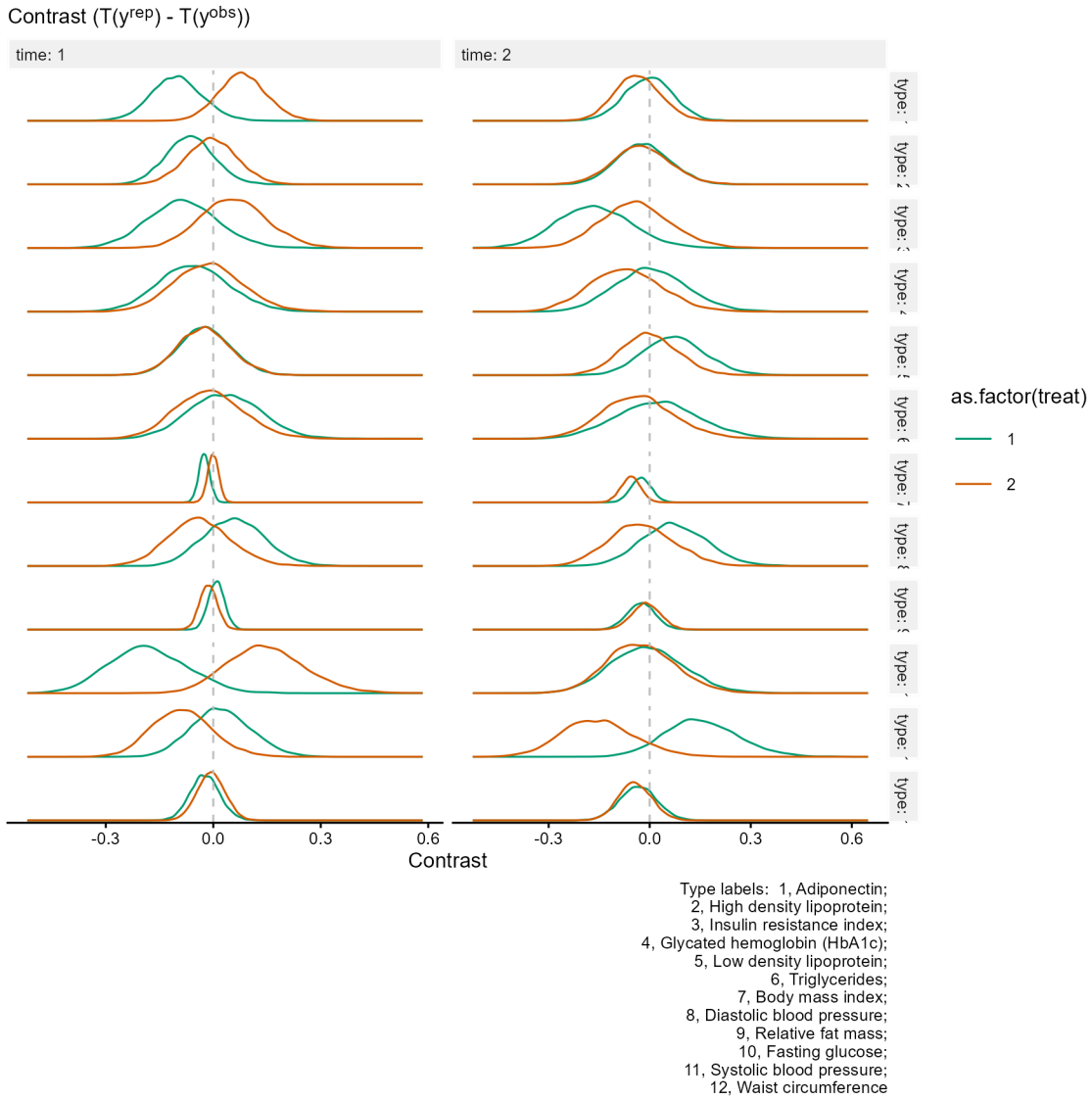

Figure 6.7: Posterior predictive check of model s8. The distributions of standard deviations calculated from y^{rep}

Allowing for the models to capture more heterogeneity in the data could improve the replicate data. Models s9 and s10 are displayed in Figure 6.8 and Figure 6.9

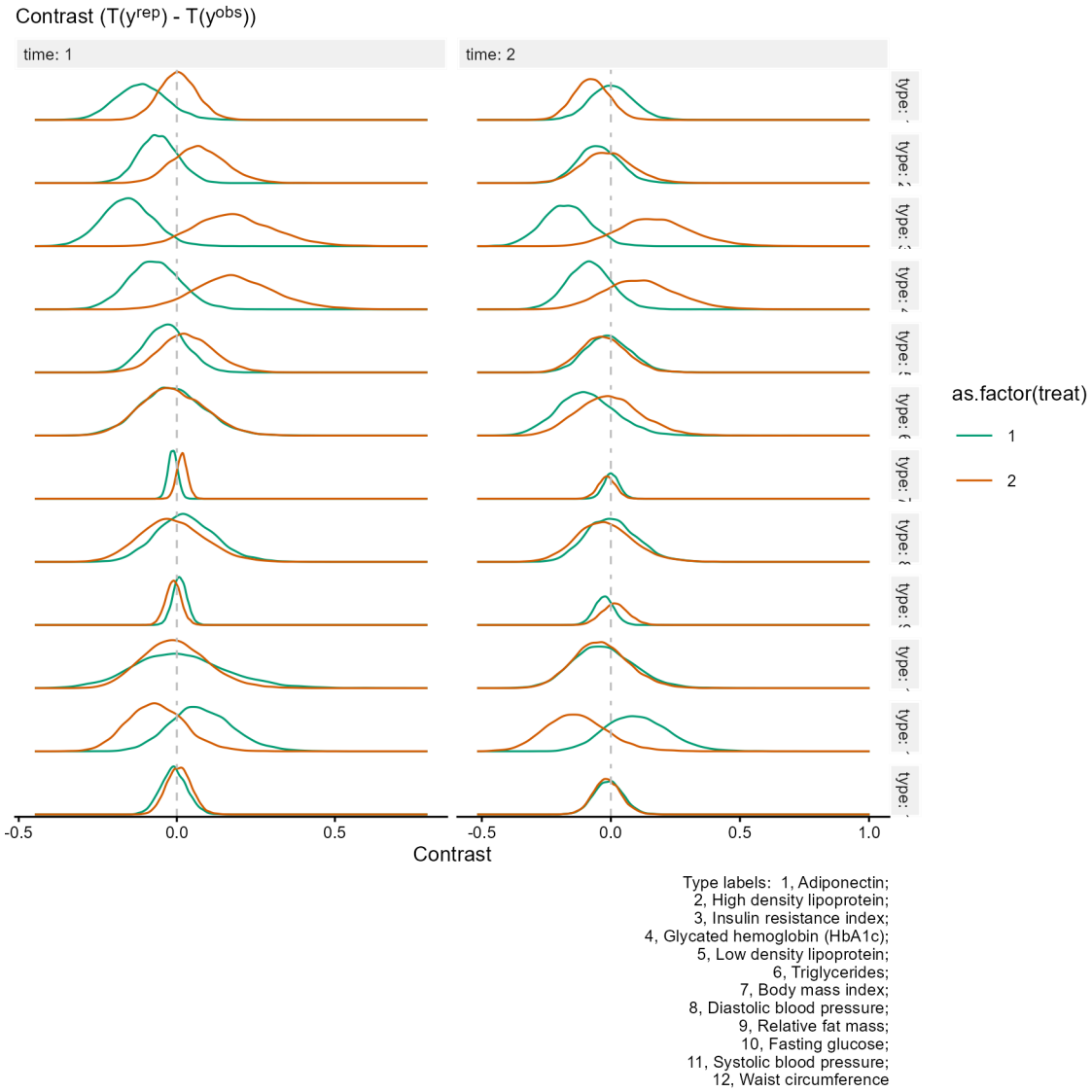

Figure 6.8: Posterior predictive check of model s9. The distributions of standard deviations calculated from y^{rep}

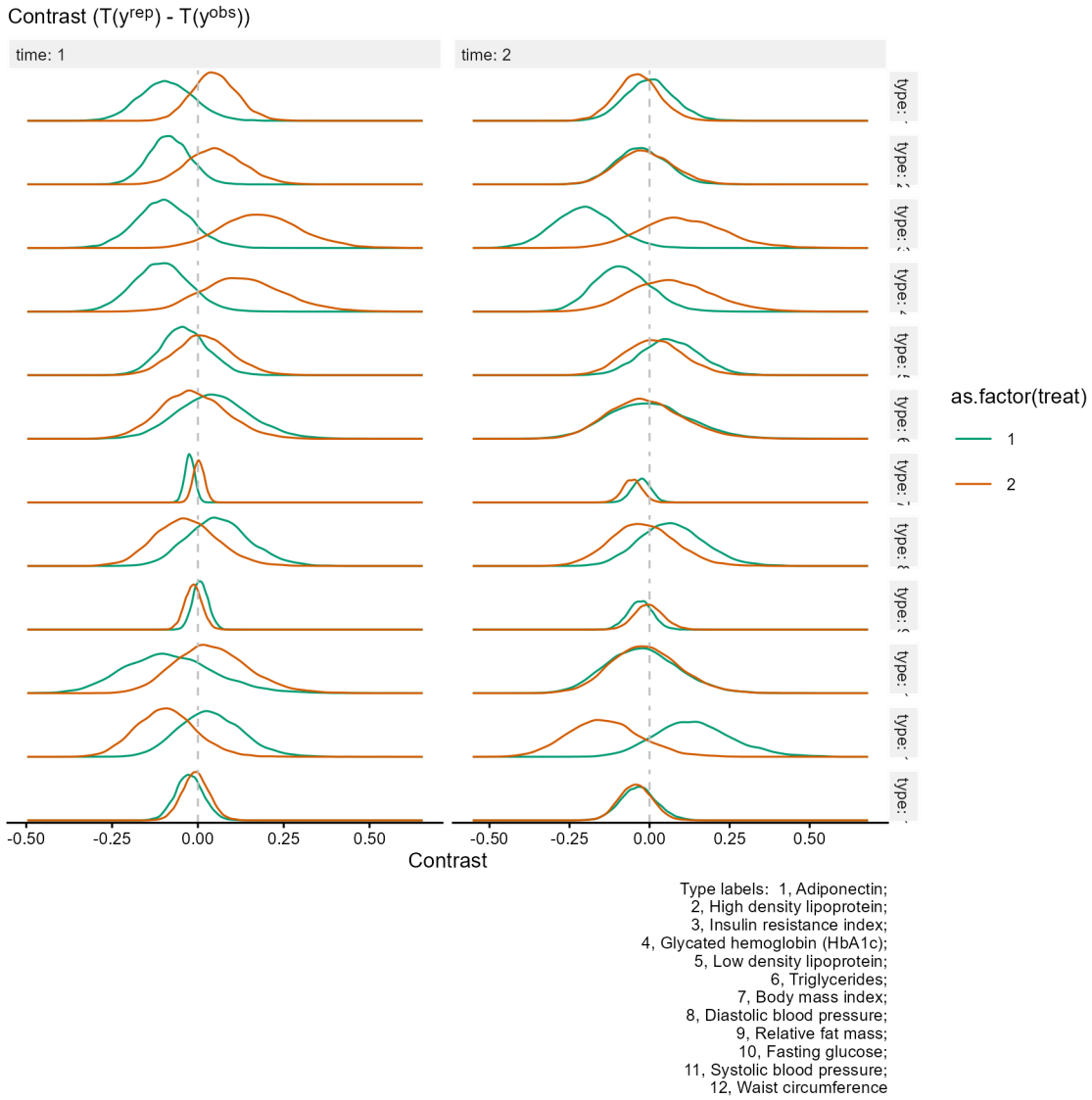

Figure 6.9: Posterior predictive check of model s10. The distributions of standard deviations calculated from y^{rep}

Combining all comparisons between replicated and observed data as p-values indicate that models s8 and s10 tend to have more examples of "extreme" observed data compared to model implications (p-values closer to 0 in Figure 6.10). No systematic under or over-estimation are evident in either model.

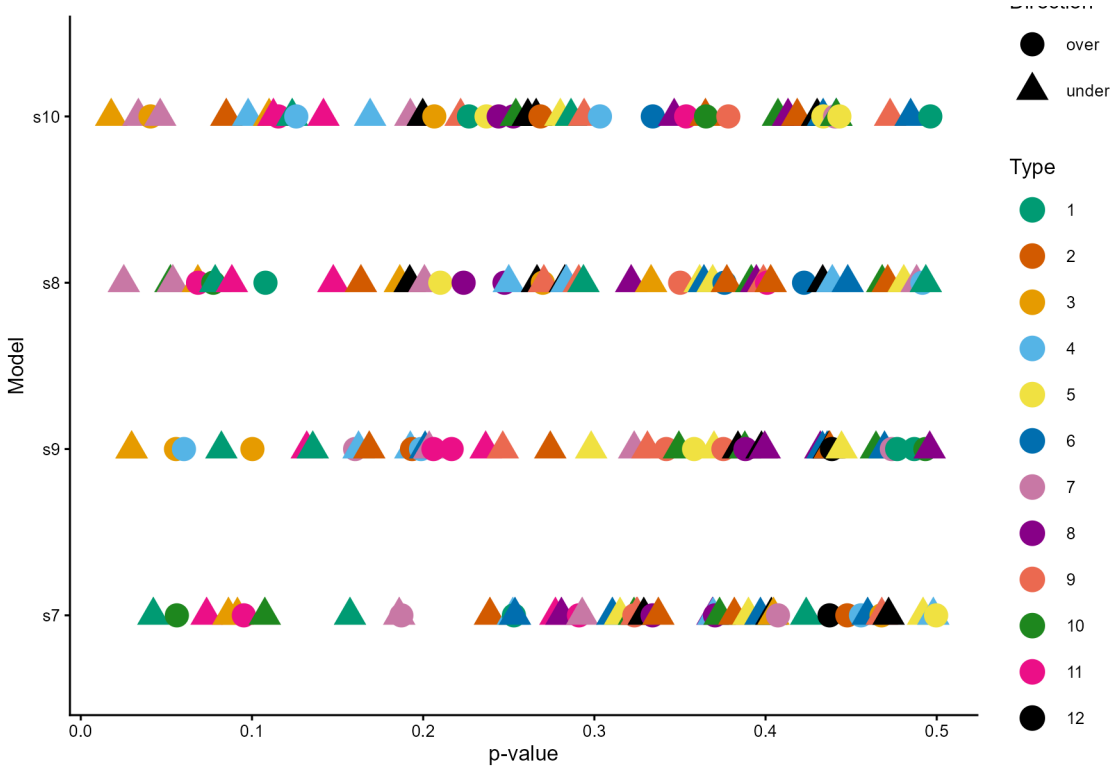

Figure 6.10: Comparing p-values from posterior-predictive checks with SD calculated as test statistics on replicated data compared to observed data.

Next we compute the approximate leave-one-out cross-validation statistics from the models using Pareto smoothed importance sampling (11). As a first step we only consider the Pareto k diagnostic values as these indirectly indicates observations that are influential for the models. Here, one observation corresponds to a participant, i.e., all values corresponding to a participant contribute to the log-likelihood. The multivariate models have more unstable estimate of the Pareto k statistic indicating that these models are "more surprised" by single observations (Figure 6.11).

Together with the indications from above that suggest that the multivariate models (and more so the heteroscedastic variant, s10) tend to under or overestimate group-specific variation in replicate simulations we will consider a more robust definition of the two model classes (uni- and multivariate models).

Although high Pareto k values indicate unreliable estimates of the expected log-pointwise predictive density we finally compare models for their out-of-sample predictive performance (Table 6.3). These estimate indicate that multivariate would predict better on new data with no differences between models with different residual variance structures.

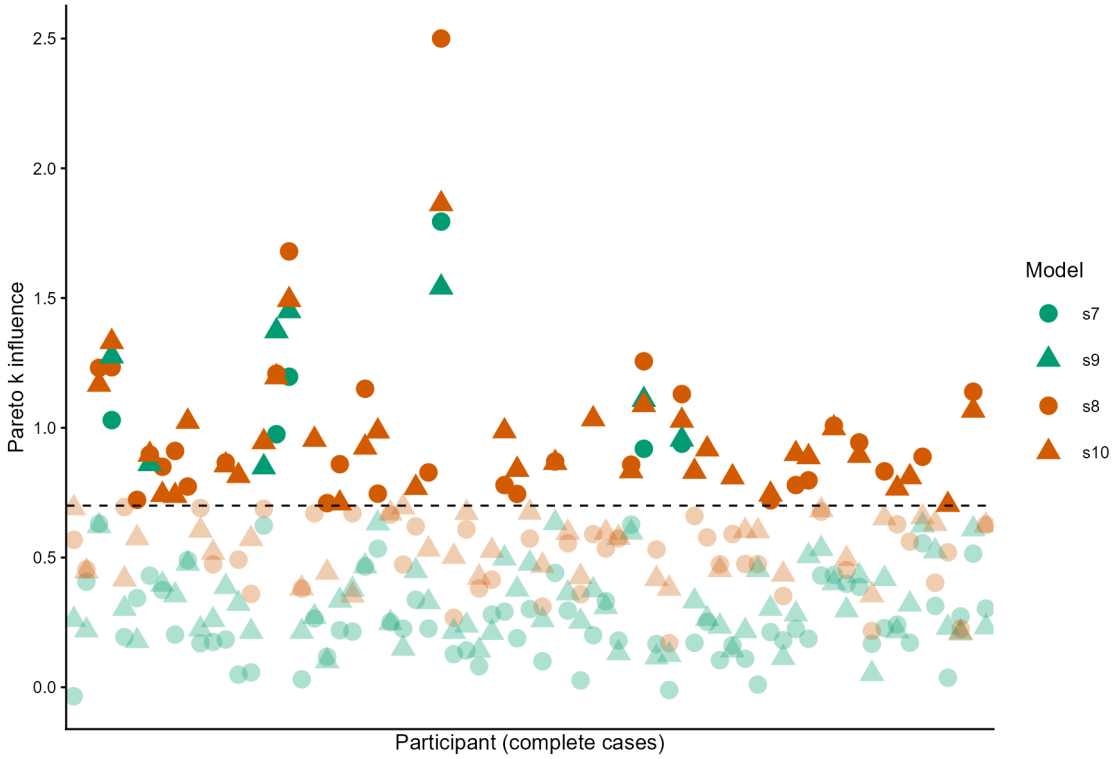

Figure 6.11: Pareto k statistic for participants with complete data in each model.

Table 6.3: ELPD estimate differences from models s7-s10.

Model	ELPD diff	SE
s8	0	0
s10	−6.563	9.177
s9	−284.724	28.589
s7	−298.182	32.969

6.0.1 Model expansion: Robust likelihood

As our initial models might suggest that they are "surprised" by specific data point we will consider a heavy-tailed likelihood to allow for inference robust to extreme observations. The Student's t-distribution is available both as a uni- and multivariate definition. Models s11 and s12 are updated from s9 and s10 to the Student-t versions of these model specifications. In model s11, the relevant change is in the definition of the likelihood ($y_i \sim \text{Student-t}\ (\mu_i, \sigma_i, \nu)$. The ν parameter is set to 4 (as pragmatic starting point, separating it from the normal likelihood (7)). In model s12 the likelihood is updated to a multivariate Student's t-distribution $y_i \sim \text{MVT}\ (\mu_i, \Sigma, \nu)$, also with ν set to 4.

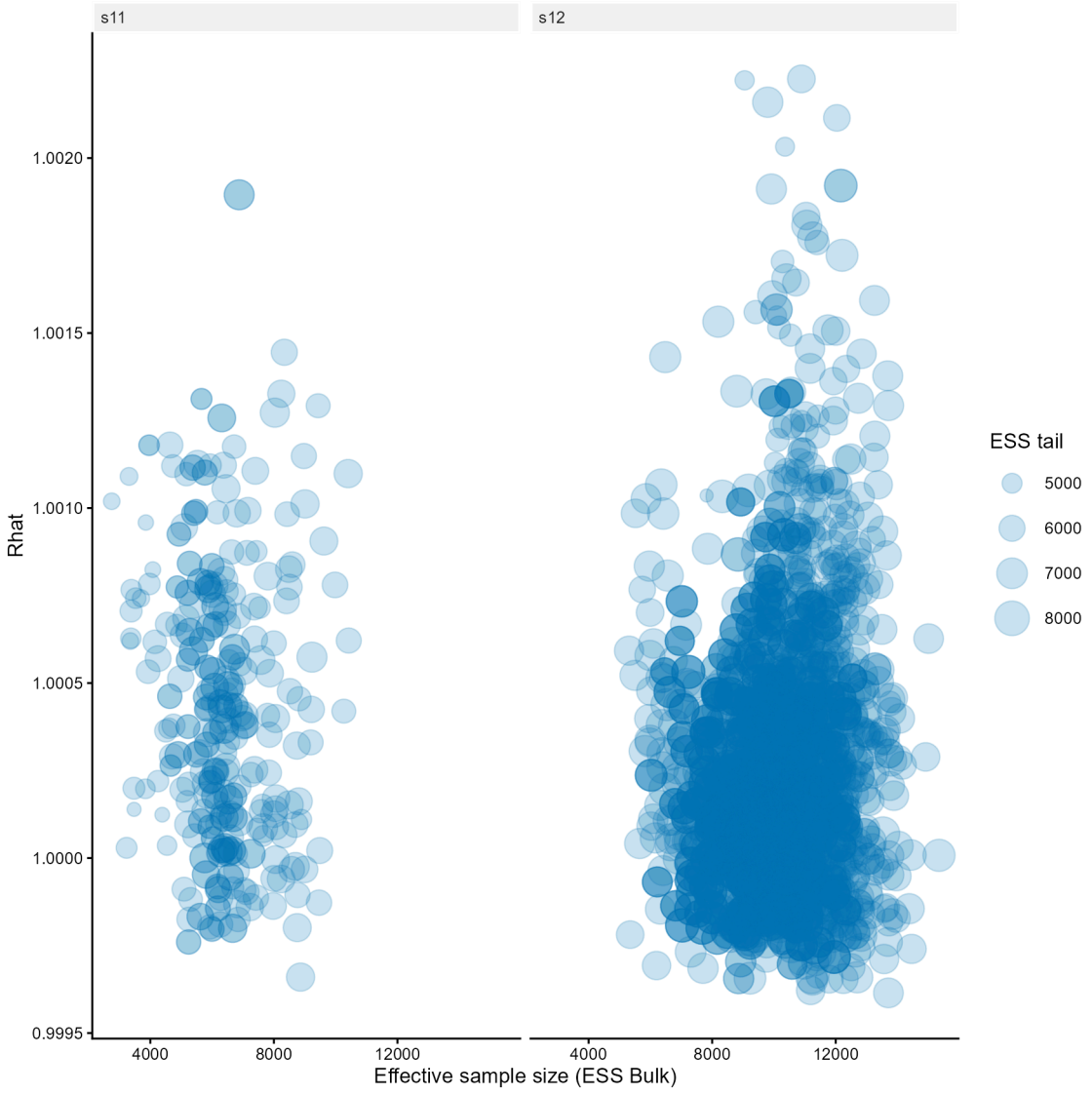

Figure 6.12: Summary chain statistics indicate satisfactory results in models s11 and s12.

Table 6.4: Diagnostic summaries of models s11 and s12. BFMI, Bayesian Fraction of Missing Information.

Chain	n divergent trans.	n maximum treedepth	BFMI
	s11		
1	0	0	0.64
2	0	0	0.60
3	0	0	0.64
4	0	0	0.64
	s12		
1	0	0	0.48
2	0	0	0.45
3	0	0	0.51
4	0	0	0.43

The chain diagnostics indicate that the HMC algorithm converges (Figure 6.12) and that no divergent transitions were evident along with healthy BFMI (Table 6.4; See here for light background on diagnostics). The plots of y^{rep} distributions against observed data distributions indicate that the model replicate the observed values (Figure 6.13, and Figure 6.14).

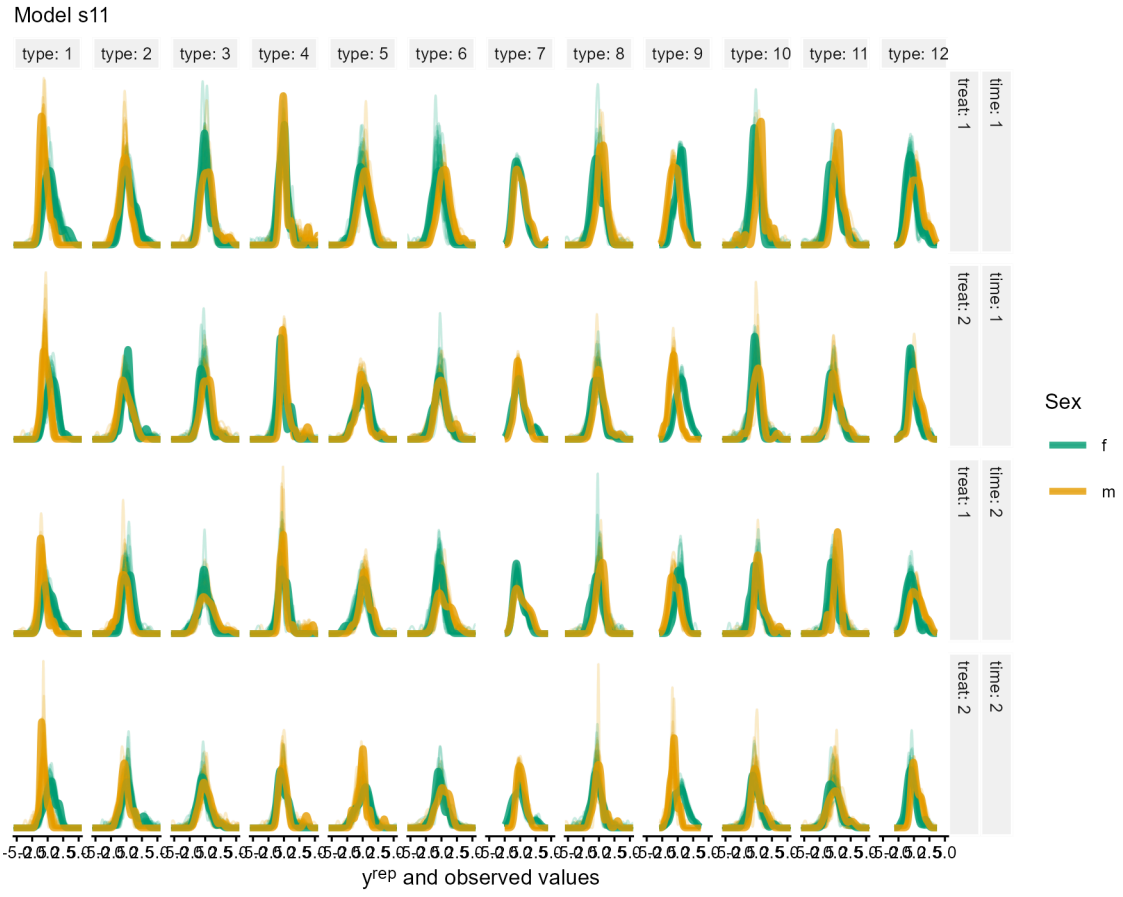

Figure 6.13: Posterior predictive check of models s11. y^{rep} (40 draws from the posterior) and observed data (thick line) indicate that the model replicate the observed data distributions.

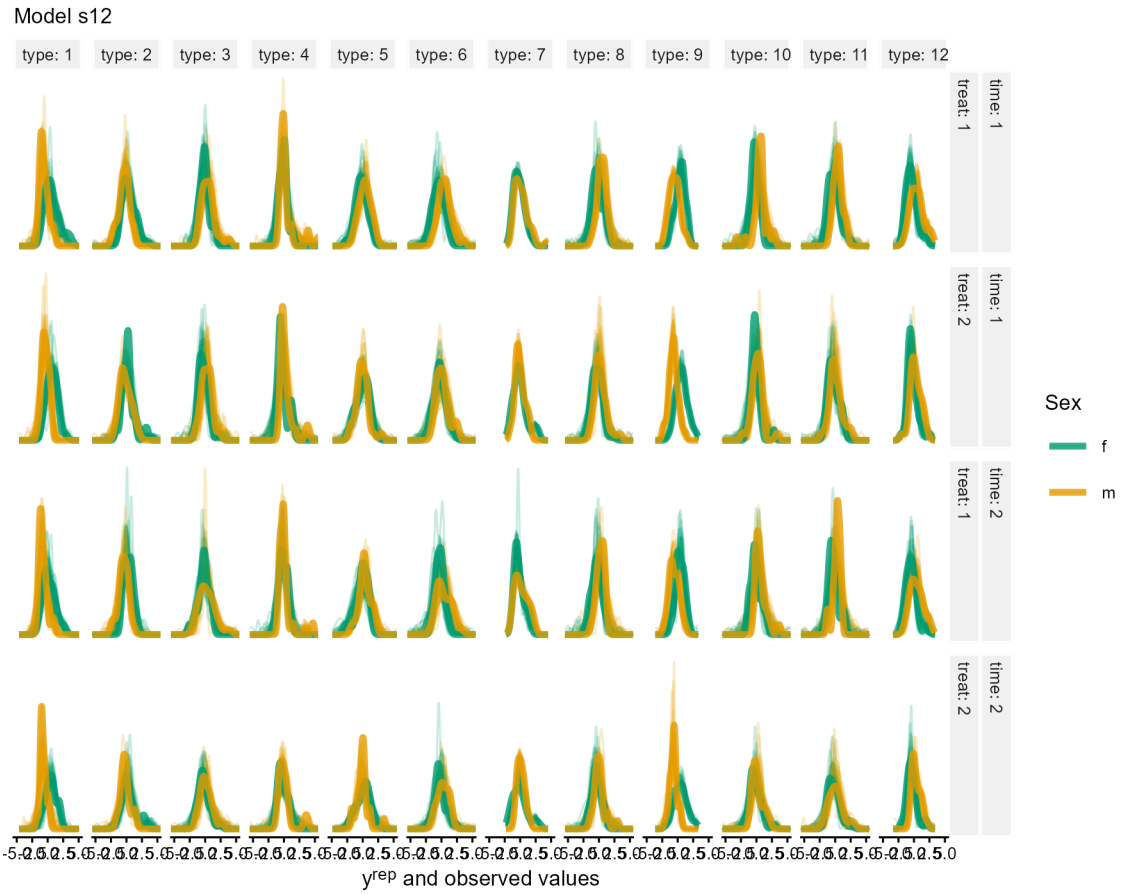

Figure 6.14: Posterior predictive check of models s12. y^{rep} (40 draws from the posterior) and observed data (thick line) indicate that the model replicate the observed data distributions.

No extreme deviations were seen from posterior predictive checks on the SD of replicated and observed data for either model (Figure 6.15 and Figure 6.16)

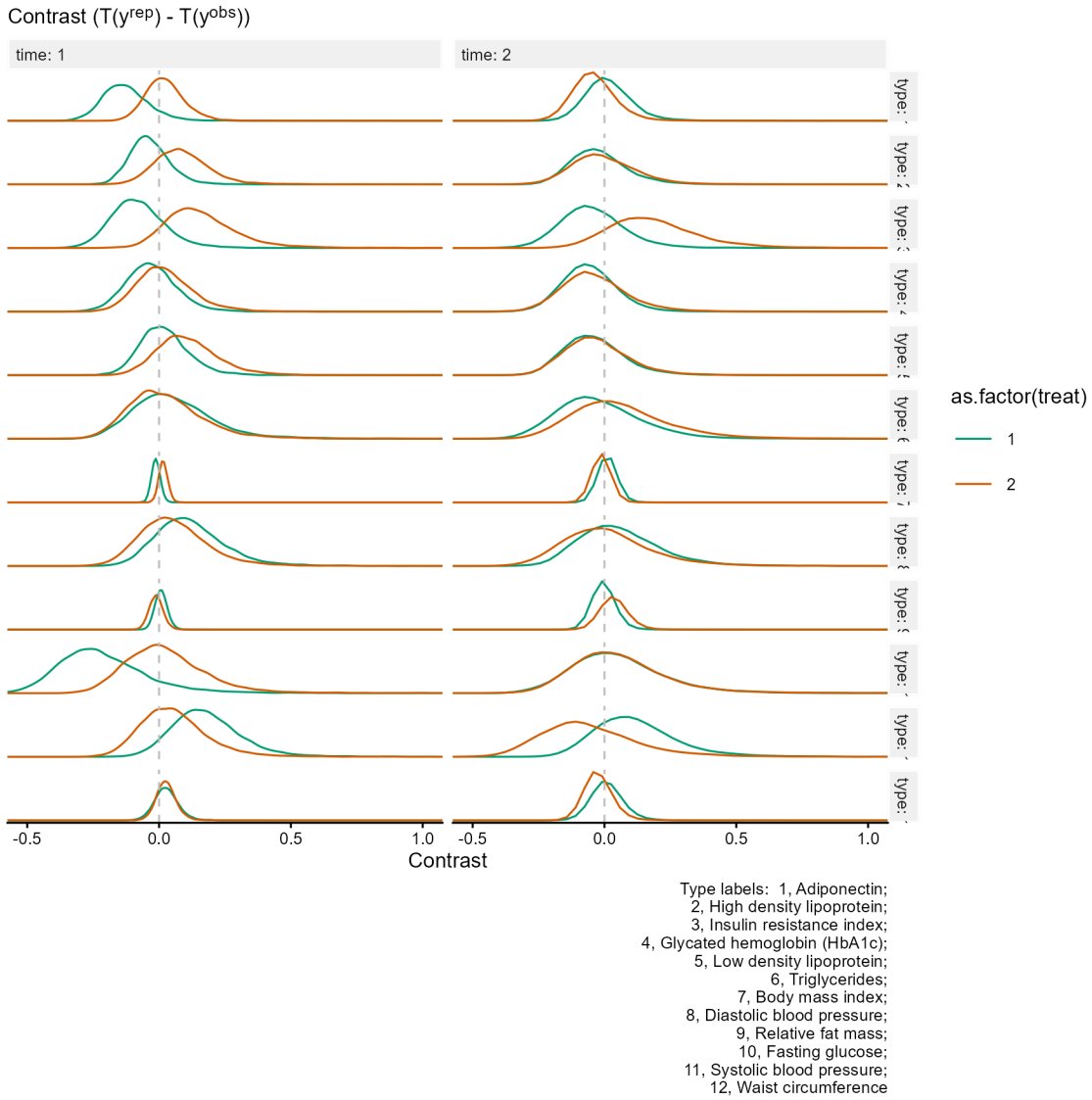

Figure 6.15: Posterior predictive check, SD from replicated data (y^{rep}) compared to observed from SD in the raw data for model s11.

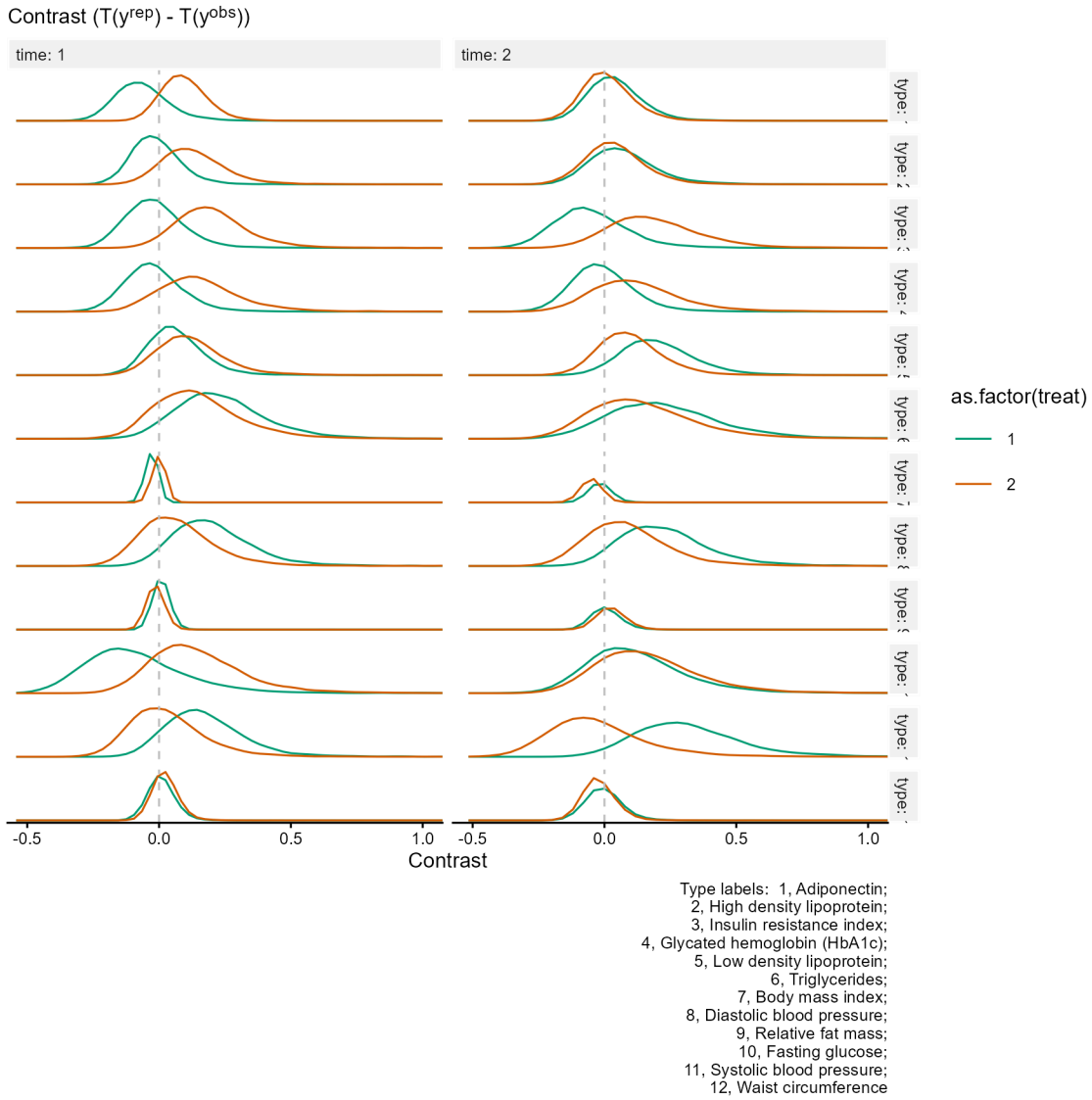

Figure 6.16: Posterior predictive check, SD from replicated data (y^{rep}) compared to observed from SD in the raw data for model s12.

Next we compare the approximate leave-one-out statistics. The univariate robust model had no unreliable Pareto K values but the multivariate variant did. However, although uncertain, the loo-comparison indicate a large advantage for the s12 model over s11.

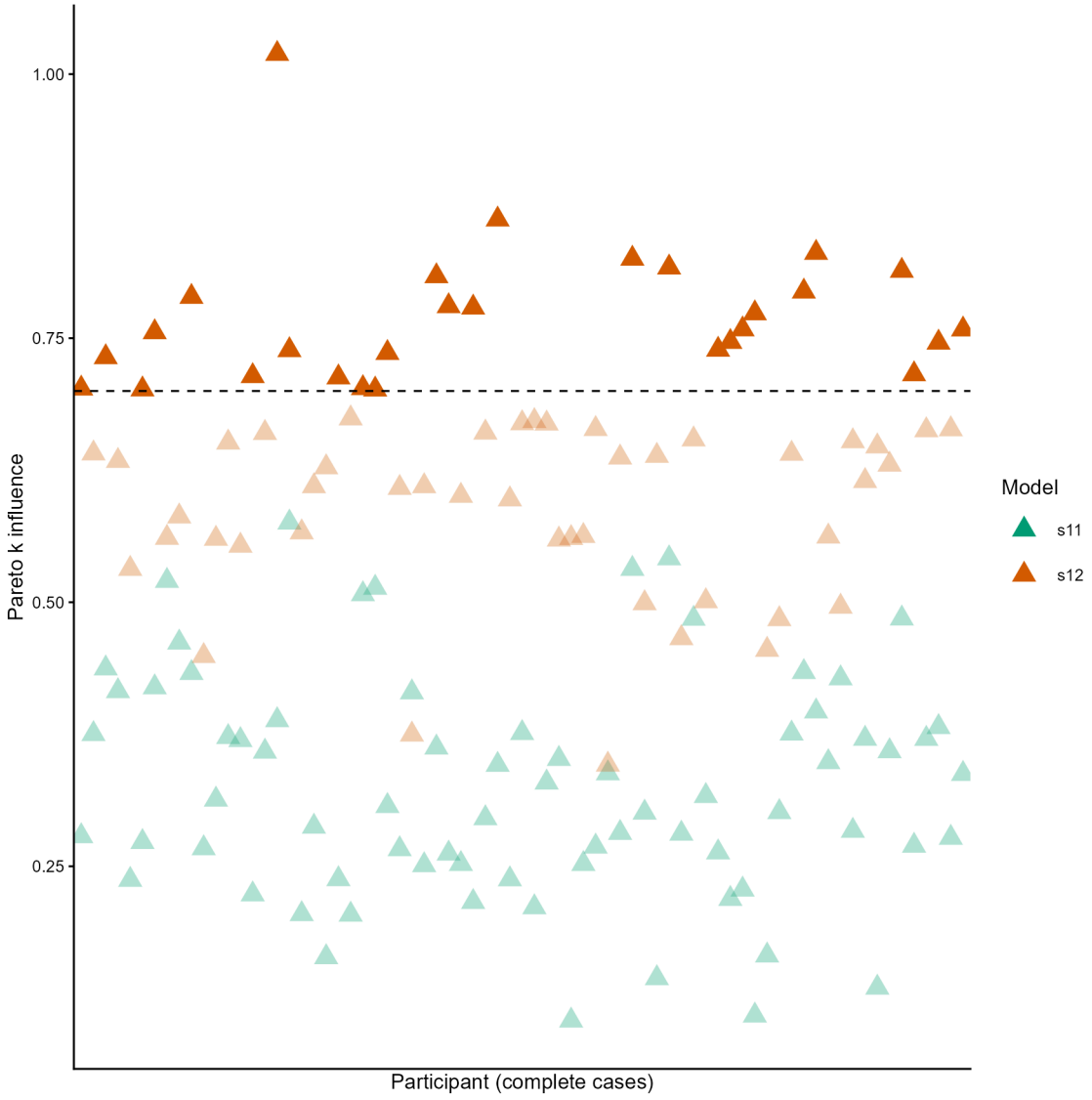

Figure 6.17: Pareto K values for complete-case participants in model s11 and s12.

Table 6.5: Pareto K statistics comparing model s11 and model s12.

model	Likelihood	σ structure	ELPD diff	SE	Pareto k > 0.7	Pareto k > 1	Max Pareto K
s12	MVT, full correlation matrix	time × type × treat	0	0	28	1	1.019
s8	MVN, full correlation matrix	time × type	−13.97	16.187	30	10	2.5
s10	MVN, full correlation matrix	time × type × treat	−20.534	13.454	36	10	1.862
s11	Univariate t-distribution	time × type × treat	−170.72	29.131	0	0	0.575

As the simple version of the multivariate normal model (s8) performed better than the more heavily parameterized s10, we will add a simplification of the s12 model. s13 uses the multivariate t-distribution but restricts the structure of σ to time × type. To isolate the contribution of the correlation structure under the t-distributed likelihood, model s18 is the univariate counterpart of s13: a univariate Student-t likelihood with $\nu = 4$ and the same time × type structure of σ, but no correlation among measurement types. s18 thus relates to s13 as s7 relates to s8, and to s11 as s7 relates to s9.

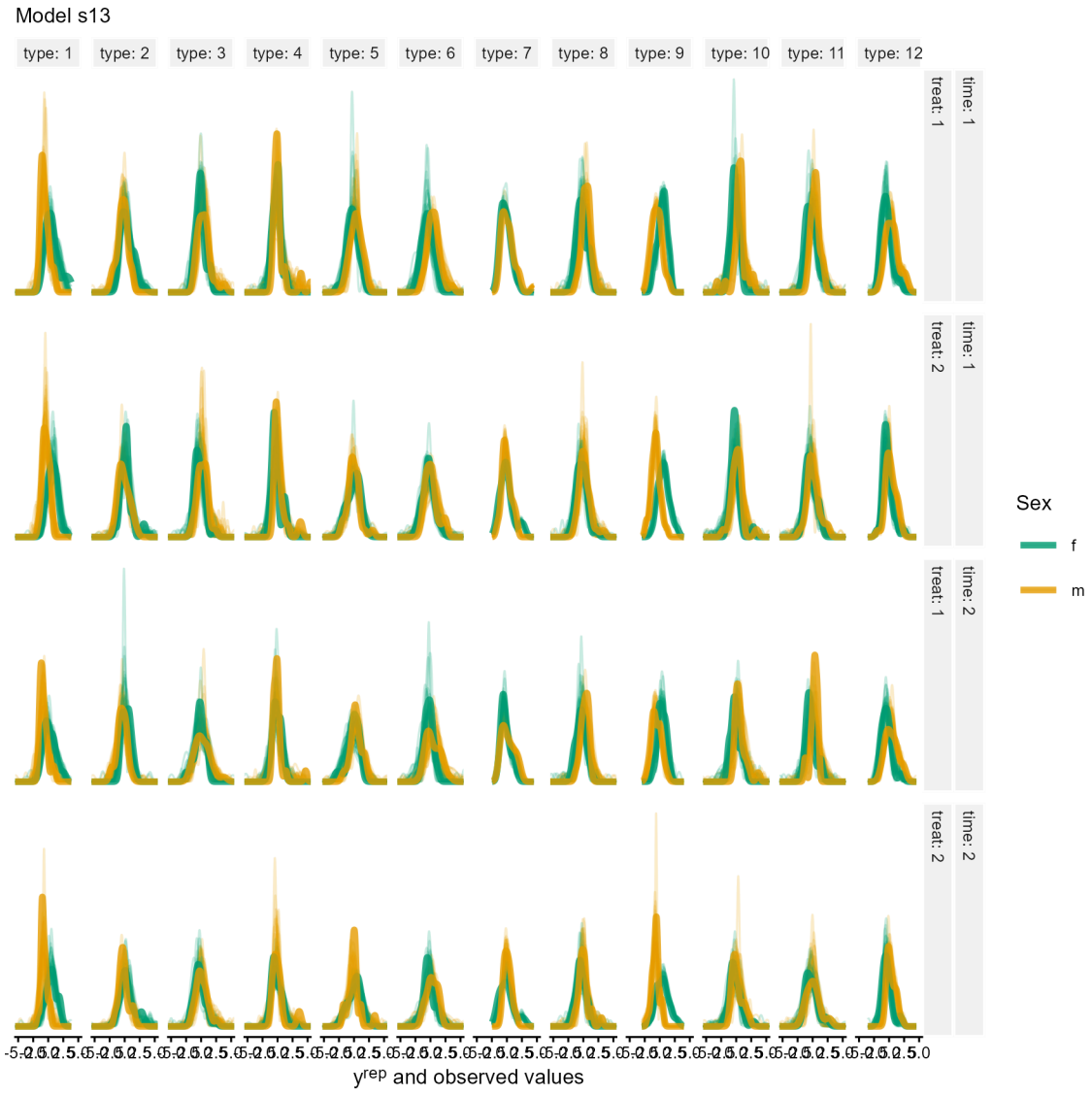

Figure 6.18: Posterior predictive check model s13

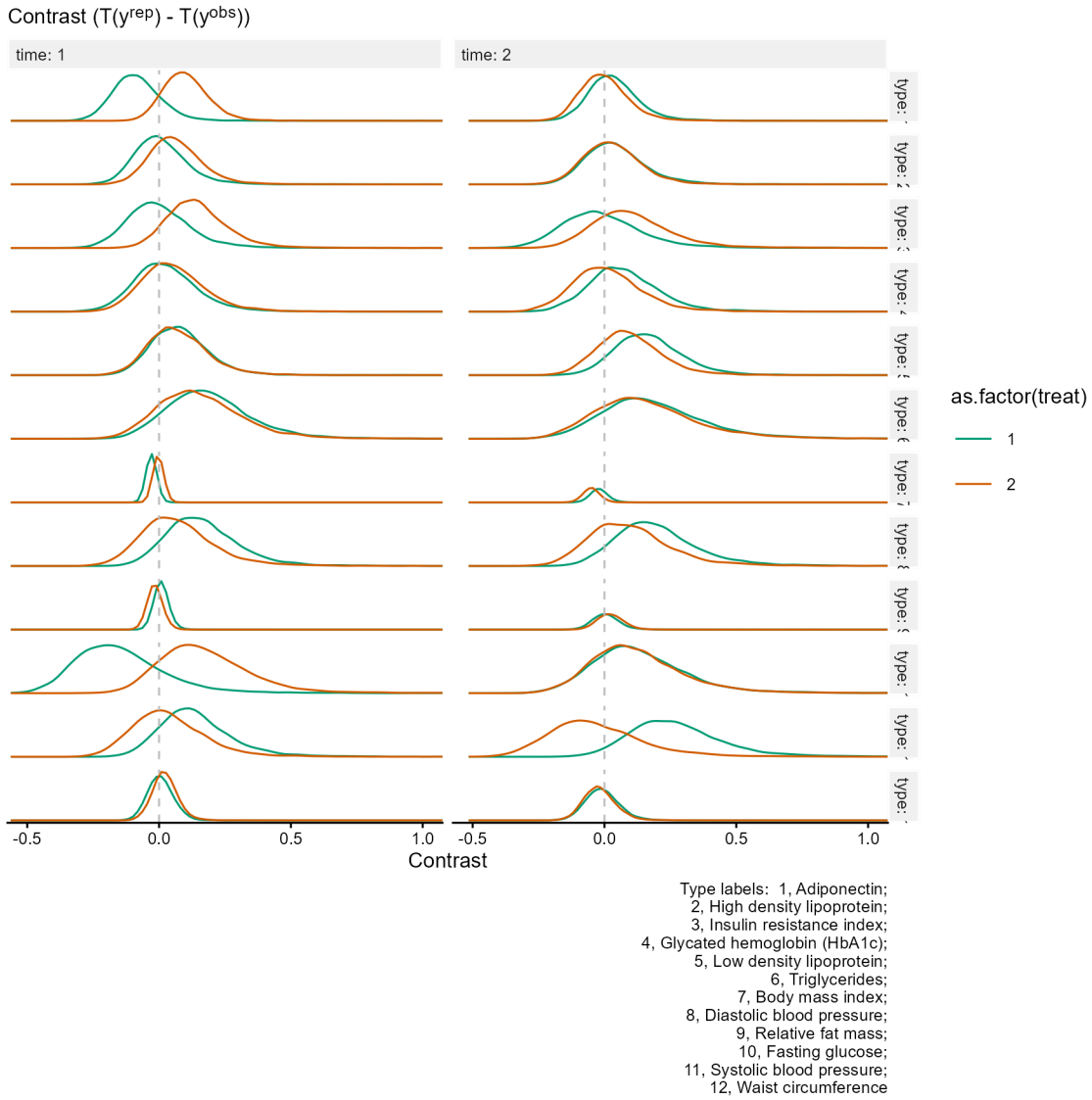

Figure 6.19: Posterior predictive check model s13

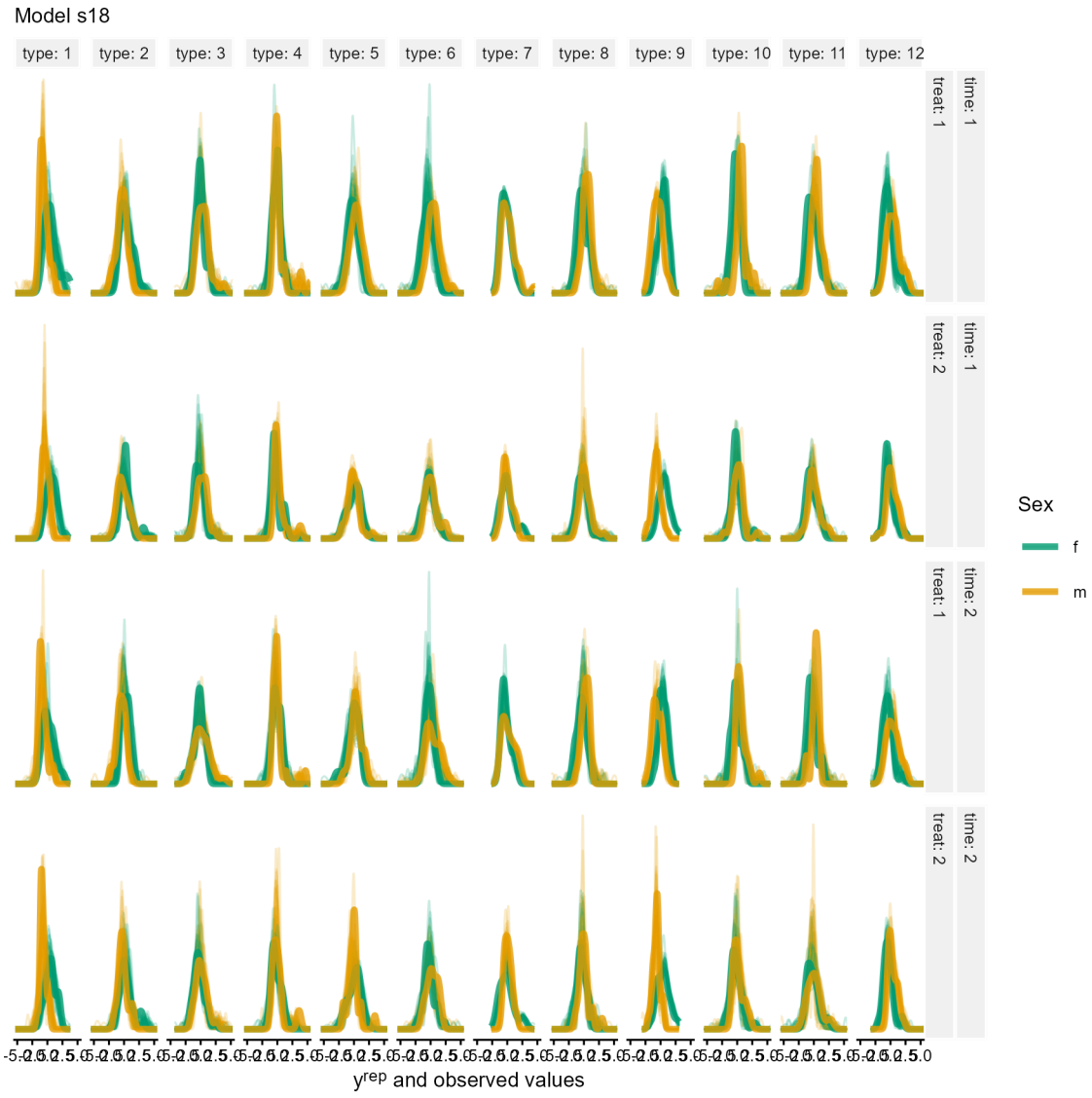

Figure 6.20: Posterior predictive check model s18

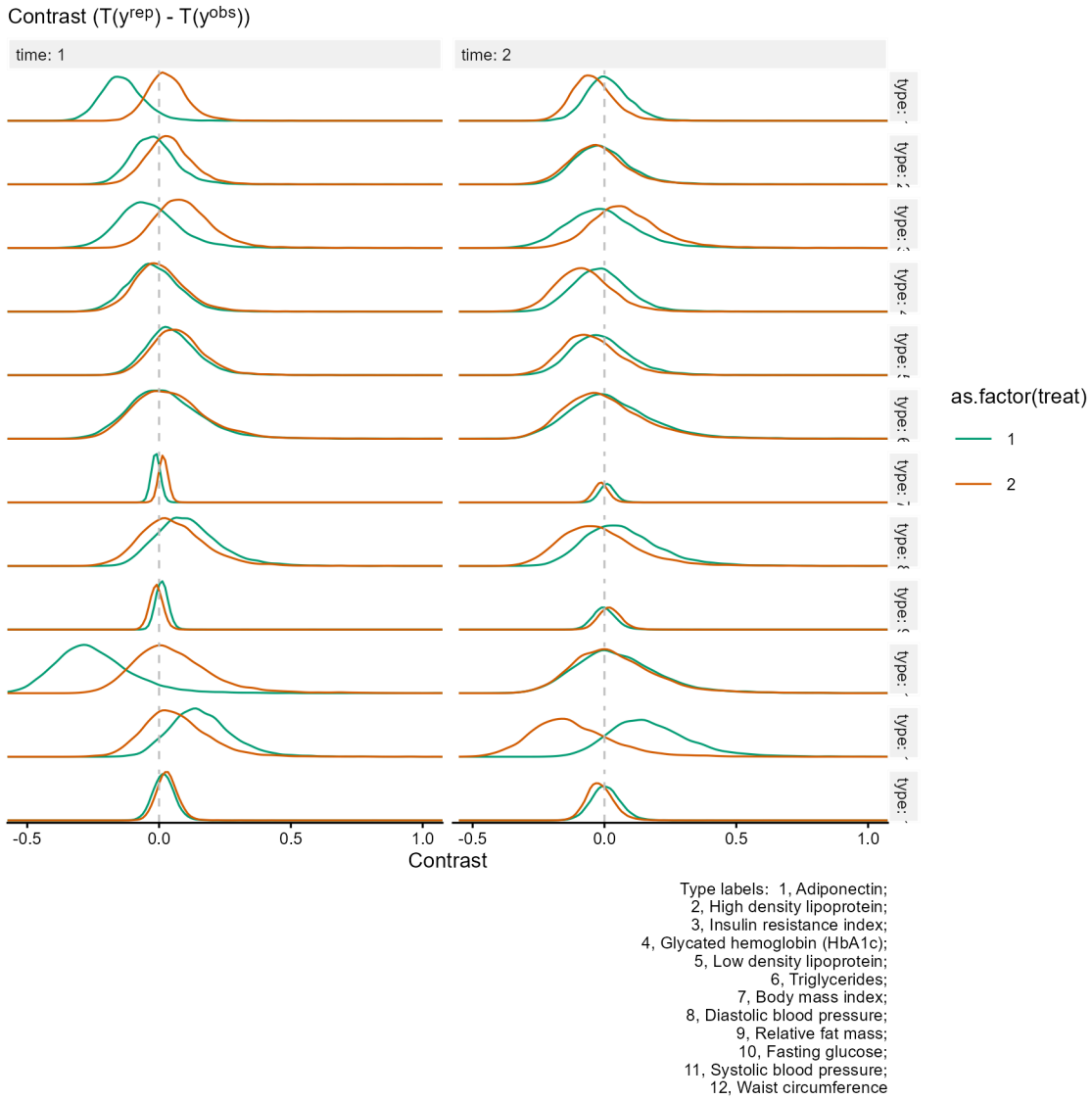

Figure 6.21: Posterior predictive check model s18

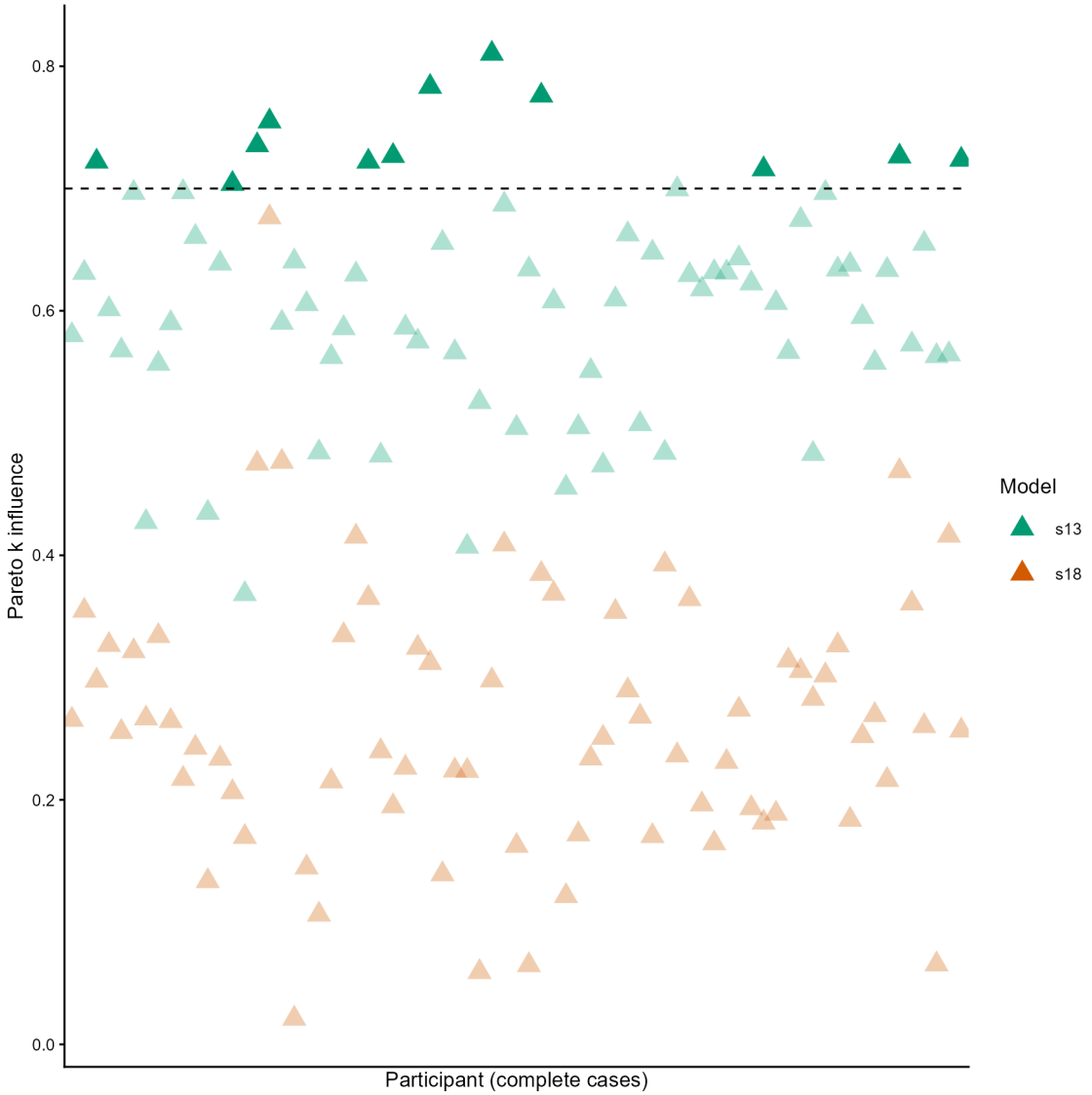

Figure 6.22: Posterior predictive check model s13

Adding s13 to the model comparison table indicate that this slightly simplified model performs better than s12. The difference is robust with regards to the ELPD difference but still uncertain because of unreliable Pareto k values (Table 6.6).

Table 6.6: Model structures, Pareto K statistics and ELPD for models s8-s13 .

model	Likelihood	σ structure	ELPD diff	SE	Pareto k > 0.7	Pareto k > 1	Max Pareto K
s13	MVT, full correlation matrix	time × type	0	0	12	0	0.81
s12	MVT, full correlation matrix	time × type × treat	−21.605	6.249	28	1	1.019
s8	MVN, full correlation matrix	time × type	−35.575	13.824	30	10	2.5
s10	MVN, full correlation matrix	time × type × treat	−42.138	14.043	36	10	1.862
s18	Univariate t-distribution	time × type	−179.79	29.522	0	0	0.676
s11	Univariate t-distribution	time × type × treat	−192.325	29.541	0	0	0.575

6.0.2 Simplification of the correlation structure

A new set of models simplifies the correlation structure. Instead of estimating the full $M \times M$ correlation matrix with an LKJ prior (as in models s8-s13, where $M = 2K$ is the number of measurement types stacked over the two post-intervention time points), this set of models relates each outcome to a small number of latent variables, following the factor approach used in the simulations (Section 4.0.1; see also the Stan User Guide). Each measurement type at each time point is given a loading onto D latent factors, collected in a $M \times D$ loading matrix Λ, together with a vector ψ of length M containing residual standard deviations (time × treat × type, or time × type). Rows 1 to K of Λ hold the loadings for the first post-intervention time point and rows $K + 1$ to M the loadings for the second. Because the two blocks of rows are estimated separately, the within-time correlation structures are free to differ between time points, whereas the between-time correlations are restricted to correlations induced by factor loadings. The implied correlation matrix is obtained by normalising the factor covariance to unit diagonal, and the covariance matrix is then built from the Cholesky factor of this correlation matrix and the residual scales σ:

$$\begin{aligned} \Sigma_{\text{raw}} &= \Lambda\Lambda^T + \text{diag}\ (\psi^2) \\ R &= \text{diag}\ (\Sigma_{\text{raw}})^{-1/2}\, \Sigma_{\text{raw}}\, \text{diag}\ (\Sigma_{\text{raw}})^{-1/2} \\ L_R &= \text{chol}\ (R) \\ L_\Sigma &= \text{diag}\ (\sigma) L_R \\ \Sigma &= L_\Sigma L_\Sigma^T \end{aligned} \tag{6.1}$$

Since R has unit diagonal, σ keeps its interpretation as the marginal residual standard deviation for each time × type (s16 and s17) or time × type × treatment (s14 and s15) combination, exactly as in the models with a full correlation matrix. Only the correlation part is constrained to a low-rank structure (low-rank in the meaning that the correlation matrix is informed by a smaller number of loadings). The factor model is not identified with respect to rotations of Λ, as any orthogonal rotation of the factors leaves $\Lambda\Lambda^T$ unchanged. Following Geweke J, Zhou G. (12), we therefore fix the upper triangle of Λ to zero and constrain its diagonal to be positive, which enables identification of the loading parameters. With $D = 4$ latent factors and $K = 12$ measurement types this gives $DM - D(D+1)/2 = 86$ free

lower-triangular loadings, $D = 4$ diagonal loadings and $M = 24$ residual standard deviations, compared to $M(M-1)/2 = 276$ free correlation coefficients in the full correlation matrix. Loadings were given standard normal priors and ψ a log-normal prior centred at 0.6 to keep the residual standard deviations positive and away from zero. The models are otherwise identical to s8 (s16, multivariate normal), s10 (s14, multivariate normal), and their multivariate t-distribution counterparts (s17 and s15, respectively).

Increasing the number of loadings to $D = 6$ gives $DM - D(D+1)/2 = 123$ free parameter and potentially more flexibility. Models s14 to s17 has been run also with $D = 6$ to assess the effect of a larger set of loadings. We include this run in the full comparison table below. Together the full comparison suggest that model s13 (MVT with time × type variance structure) has the best predictive accuracy. A small number of participants have Pareto k values larger than 0.7, but none larger than 1 (Table 6.7). The multivariate models have better estimated predictive accuracy compared to the univariate counterparts and the robust versions (t-distributions) perform better than the normal likelihoods. Finally, the simplified variance structure generally performs better than the more complex structure. Together this suggest that the candidate model s13 is preferred for inference. We will use it together with the univariate version in subsequent steps.

Table 6.7: Model definitions, variance structures, ELPD and Pareto K estimates from models s8-s18

model	Likelihood	σ structure	ELPD diff	SE	Pareto k > 0.7	Pareto k > 1	Max Pareto K
s13	MVT, full correlation matrix	time × type	0	0	12	0	0.81
s12	MVT, full correlation matrix	time × type × treat	−21.605	6.249	28	1	1.019
s8	MVN, full correlation matrix	time × type	−35.575	13.824	30	10	2.5
s17-D6	MVT, 6 latent factors	time × type	−41.651	13.046	2	0	0.963
s10	MVN, full correlation matrix	time × type × treat	−42.138	14.043	36	10	1.862
s17	MVT, 4 latent factors	time × type	−58.036	14.75	1	0	0.941
s15-D6	MVT, 6 latent factors	time × type × treat	−60.997	14.125	6	1	1.045
s15	MVT, 4 latent factors	time × type × treat	−79.424	15.78	3	0	0.994
s16-D6	MVN, 6 latent factors	time × type	−80.857	17.2	16	4	2.215
s14-D6	MVN, 6 latent factors	time × type × treat	−85.705	18.01	20	6	1.965
s16	MVN, 4 latent factors	time × type	−135.83	24.135	11	5	1.916
s14	MVN, 4 latent factors	time × type × treat	−166.262	24.814	20	8	2.56
s18	Univariate t-distribution	time × type	−179.79	29.522	0	0	0.676
s11	Univariate t-distribution	time × type × treat	−192.325	29.541	0	0	0.575
s9	Univariate normal	time × type × treat	−320.299	33.869	8	5	1.541

7. Effects of krill oil supplementation on indications of metabolic syndrome - Simulations

To compare the multivariate Student's t models to the univariate counterpart (S13 vs. S18) we will use a more realistic simulation inducing correlations between variables and a larger data set (k = 12). Data are generated as having homoscedastic variance, including equal variance between groups (`sigma_ratio = 1` in `sim_wrap3`, see `R/sim-fun.R`), and added outlier noise. For comparisons, data included participant-level drop-out and values missing at random corresponding to about 10, 28 and 40% of the follow-up values with baseline values having about 8% missing values. Correlations were set to weak (independence between variables and time-points) and strong, about |r|=0.25 between measurement types and |r|=35 between time-points.

As an example, the following synthetic data set was simulated using the simulation routine used in `R/model-comp3.R` (Figure 7.1).

Simulating the full data set leads to synthetic data patterns presented in Figure 7.1.

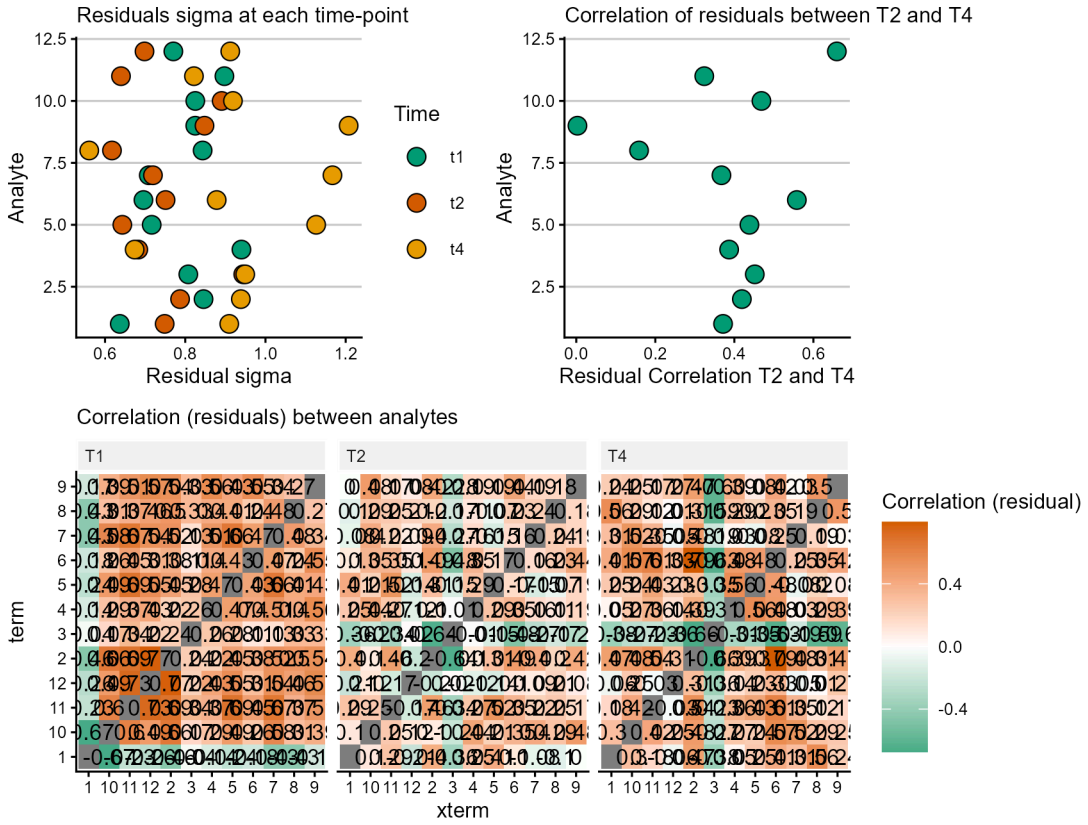

Figure 7.1: Residual variation and correlation pattern within and between variables in the synthetic data example.

The missing data patterns from the medium sized missingness scenario corresponds approximately to the observed data (Figure 7.2, see also Chapter 4.).

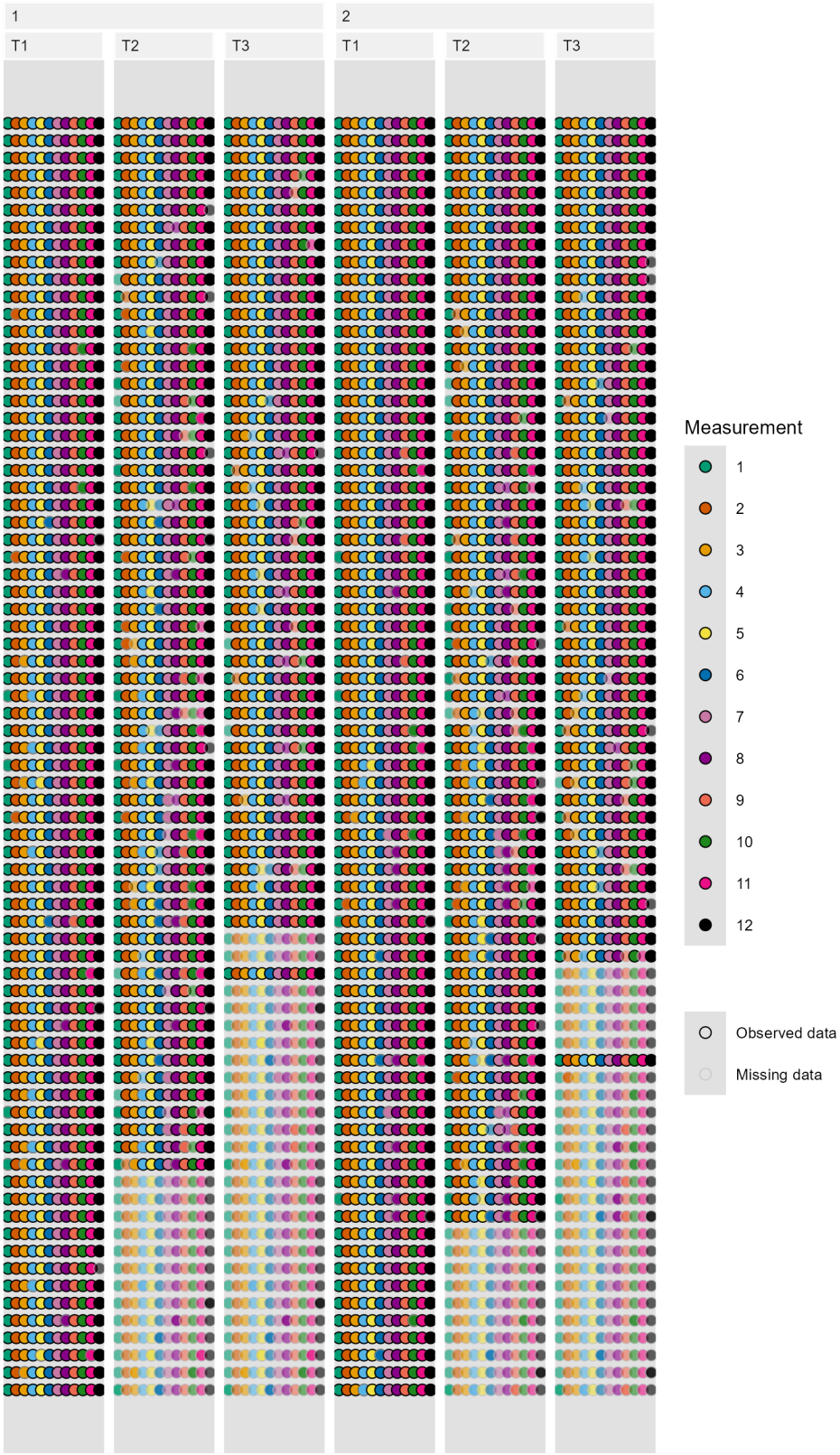

Figure 7.2: Missing data patterns in the exemple simulated data set.

Using 1200 simulated data sets we are comparing models S13 to s18 for modelling performance using two scenarios of correlations (strong vs. weak) and three levels of missingness (low, mid, high). Figure 7.3

shows chain diagnostics from the fits, no simulated data sets (n = 0) were excluded from comparisons due unsatisfactory chain convergence or effective sample sizes.

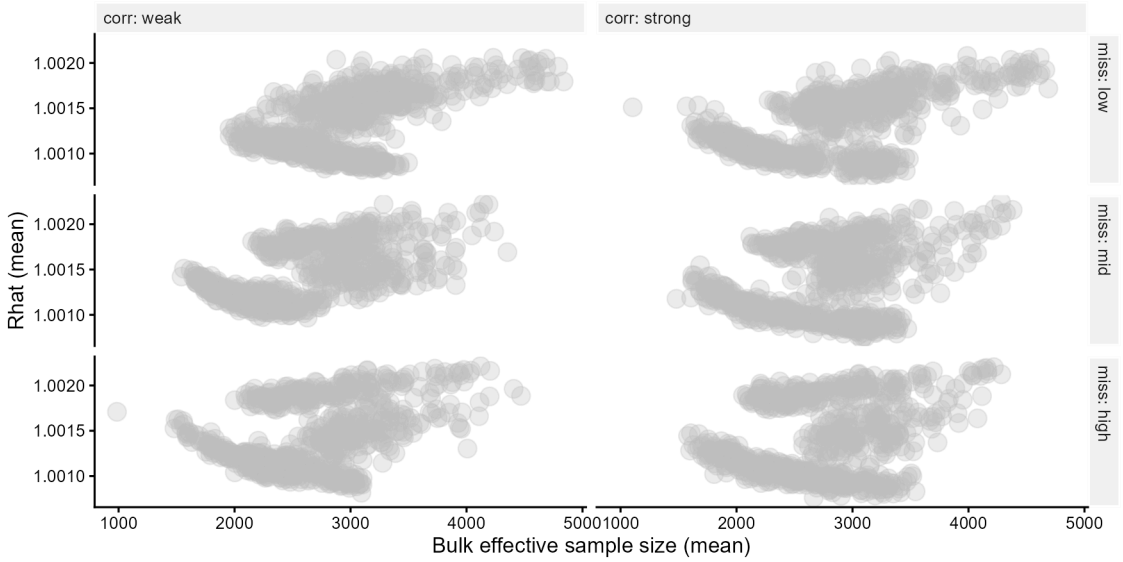

Figure 7.3: Chain dioagnostics under the full three by two simulated scenarios.

For each estimate (treatment contrasts, 2 × 12 per data set), coverage of a 90% percentile interval was determined and averaged within each model, variant (full data and missing values) and simulation cell (missingness × correlation strength). With weak correlations, coverage of S13 was on avergae closer to the nominal level while S18 had slightly lower coverage. With strong correlations, both models had similar coverage, slightly below nominal at low missingness and close to nominal at mid and high missingness (Figure 7.4 A). Credible intervals were wider in the missing data variant and widened with increasing proportions of missing values, while interval widths in the full data variant were similar across conditions (Figure 7.4 B). With weak correlations, S13 had slightly wider intervals than S18 in both variants, consistent with its higher coverage. With strong correlations, interval widths were similar in the full data variant, but S13 tended to have narrower intervals than S18 in the missing data variant, a difference that increased with the proportion of missing values (Figure 7.4 C).

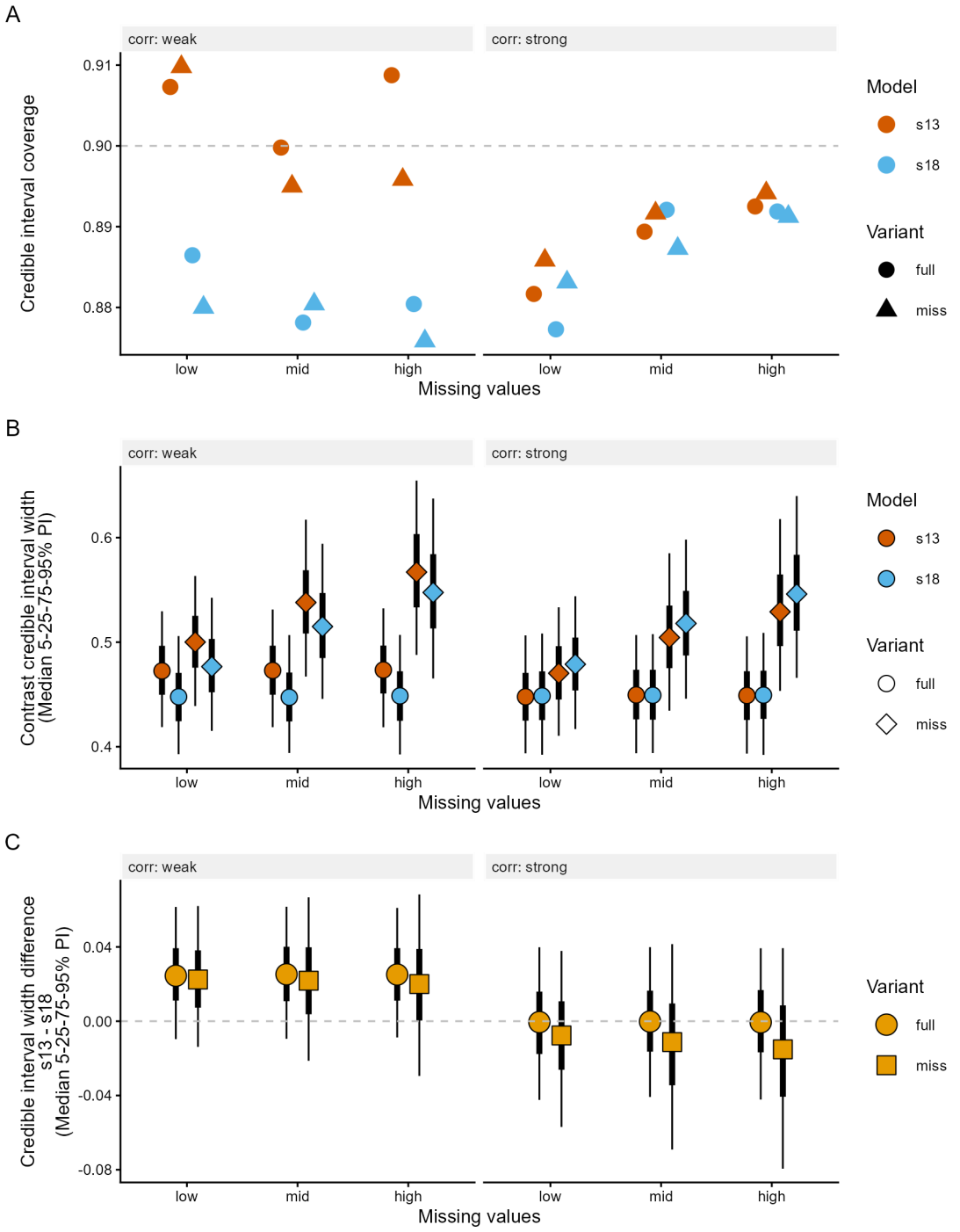

Figure 7.4: (A) Coverage of 90% credible intervals, (B) credible interval widths, and (C) differences in credible interval widths between models.

Accuracy was assessed as the error of the posterior mean of each treatment contrast (Estimate - Simulated). In the full data variant, the root mean squared error (RMSE) was practically identical between S13 and S18. Missing values increased the RMSE for both models, and the increase was smaller for S13 when correlations were strong (Figure 7.5 A). The paired differences in absolute error per estimate were centered close to zero in all comparisons and in both zero and non-zero effects, with wide distributions (Figure 7.5 B). S13 had the lower absolute error in a small majority of estimates in most comparisons (overall about 50-54%), with the largest proportions in the missing data variant with strong correlations (Figure 7.5 C).

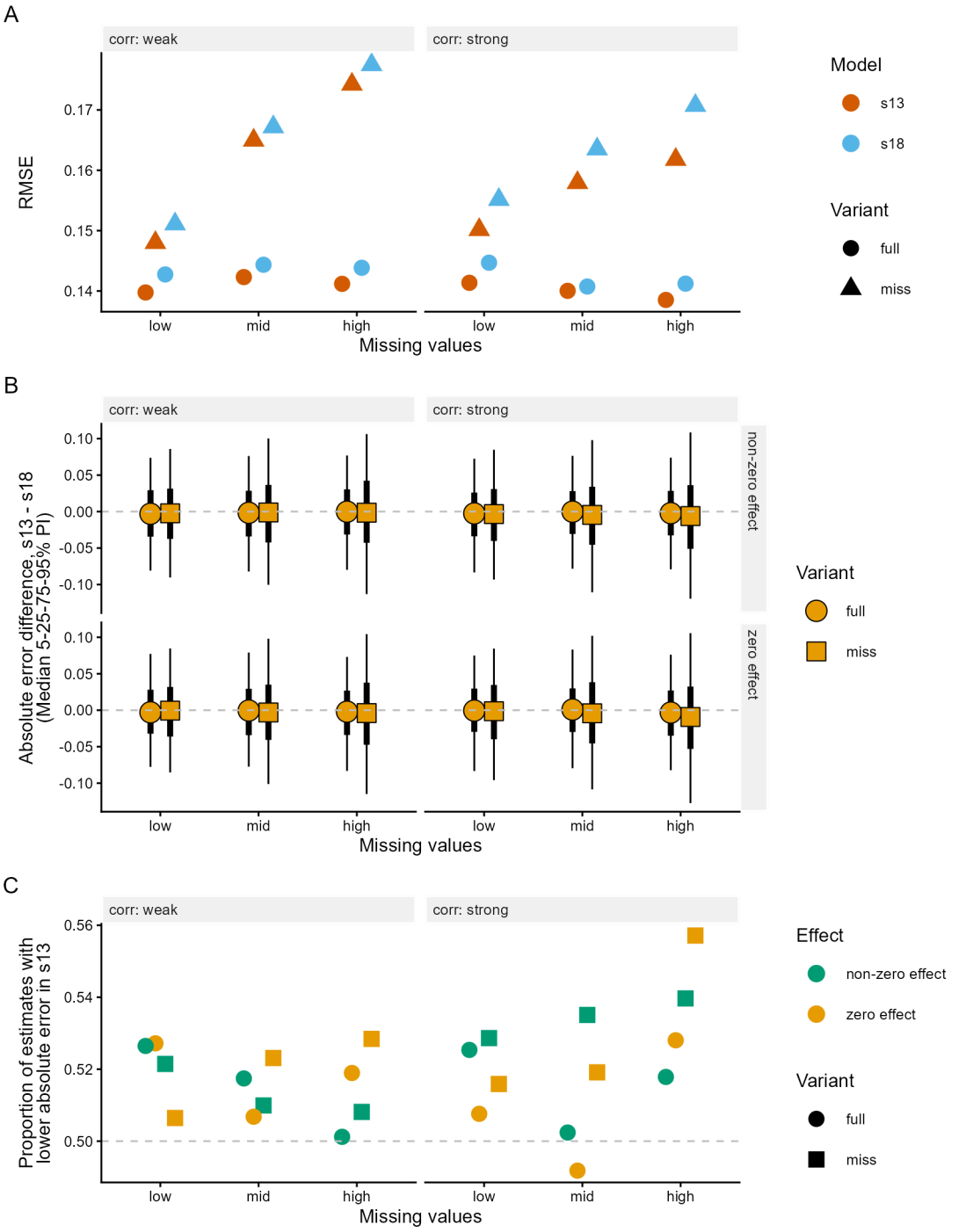

Figure 7.5: (A) Root mean squared error (RMSE) of posterior mean treatment contrasts, (B) paired differences in absolute error between models (S13 - S18) in zero and non-zero effects, and (C) proportion of estimates where S13 had lower absolute error than S18.

Finally, we compared the error associated with data imputation. Overall, the RMSE was lower in s13 compared to s18 with larger benefit of s13 in imputation of pre-values and follow-up values under stronger correlations (Figure 7.6).

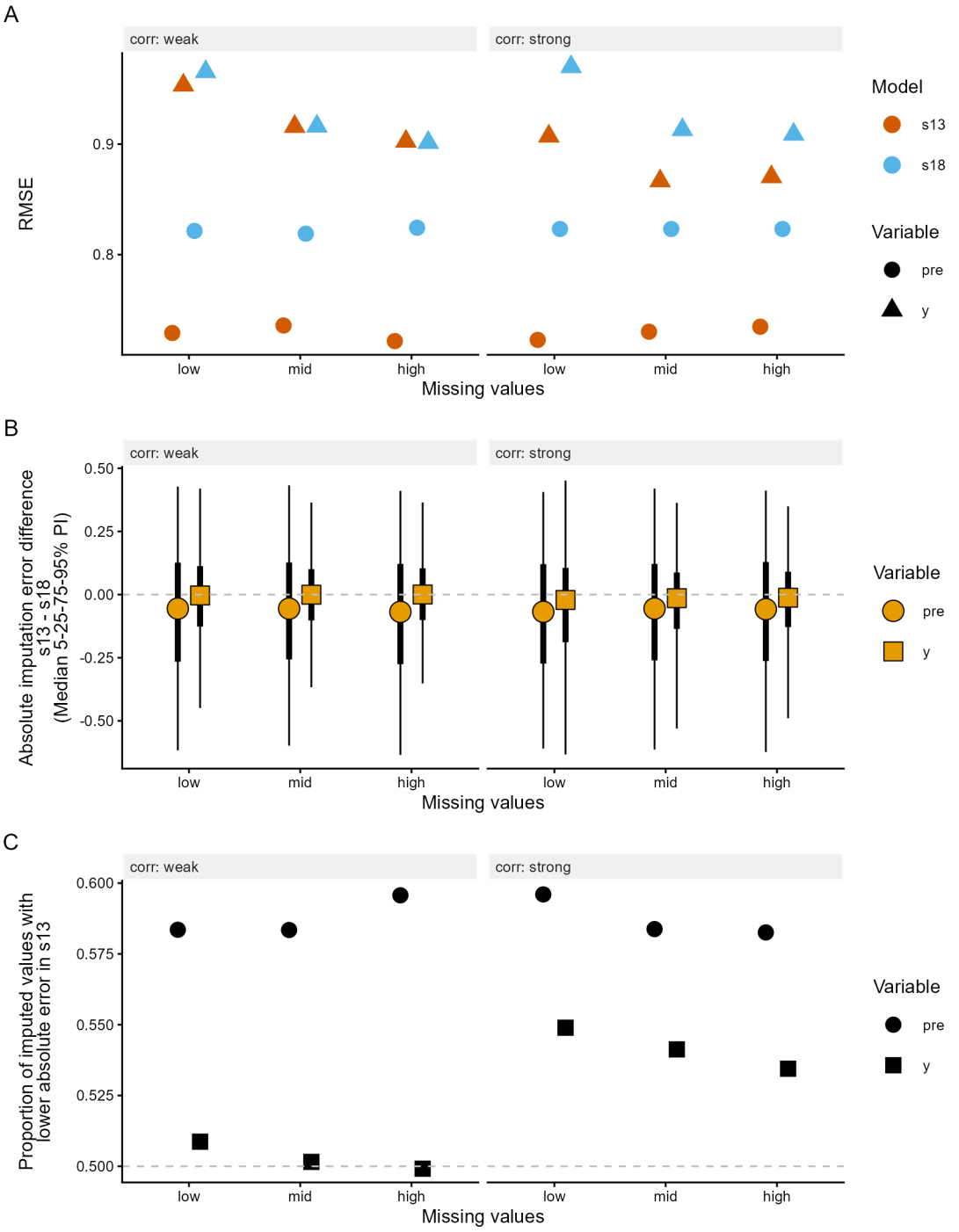

Figure 7.6: (A) Root mean squared error (RMSE) of posterior mean imputed values, (B) paired differences in absolute imputation error between models (S13 - S18) for baseline (pre) and post-intervention (y) values, and (C) proportion of imputed values where S13 had lower absolute error than S18.

Together, although differences are small, these results indicate that S13 is at least as accurate as S18, with a modest gain in accuracy when correlations between variables and time-points are strong and data are missing.

8. Effects of krill oil supplementation on indications of metabolic syndrome - Inference

Here, we draw inferences about the effect of krill oil supplementation on metabolic health markers. First, we examine the differences in outcomes between placebo and krill oil, followed by a combined analysis of the dichotomous diagnostic score (0/1) and a continuous version derived from the sum of the sub-scores, each given equal weight.

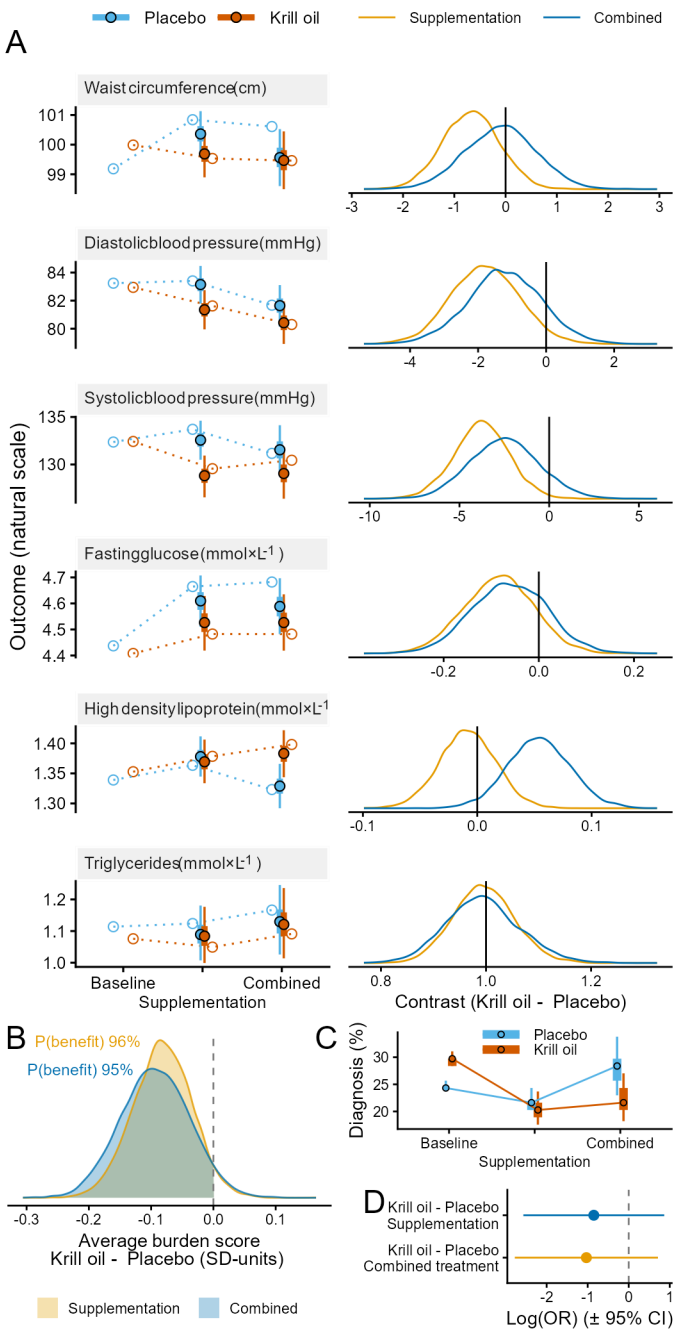

Figure 8.1: A, Observed group means (open symbols, all available observations; geometric means for triglycerides) and model-based adjusted group means (filled symbols, posterior median with 50% and 95% CI) for each outcome in the metabolic syndrome score on its natural scale (A). Adjusted means after supplementation and combined periods are model predictions conditional on the overall mean pre-intervention value and averaged over sex. The right panels show posterior distributions of treatment effects (krill oil - placebo) for each measurement. Baseline represents pre-intervention values (observed); Supplementation and Combined show estimates and observed means after the supplementation period and after the Supplementation + resistance-training period, respectively. B, posterior distribution of the average burden score contrast (krill oil - placebo) at post-supplementation and after the combined period. P(benefit) indicates the percentage of the posterior associated with any benefit for krill oil. C, the percentage of participants meeting the diagnostic criteria in each group; error bars represent the uncertainty associated with missing-value imputation (95th and 50th percentile intervals). D, treatment contrast for the occurrence of metabolic syndrome diagnosis, baseline-adjusted differences between groups at post-supplementation and after the combined treatment period.

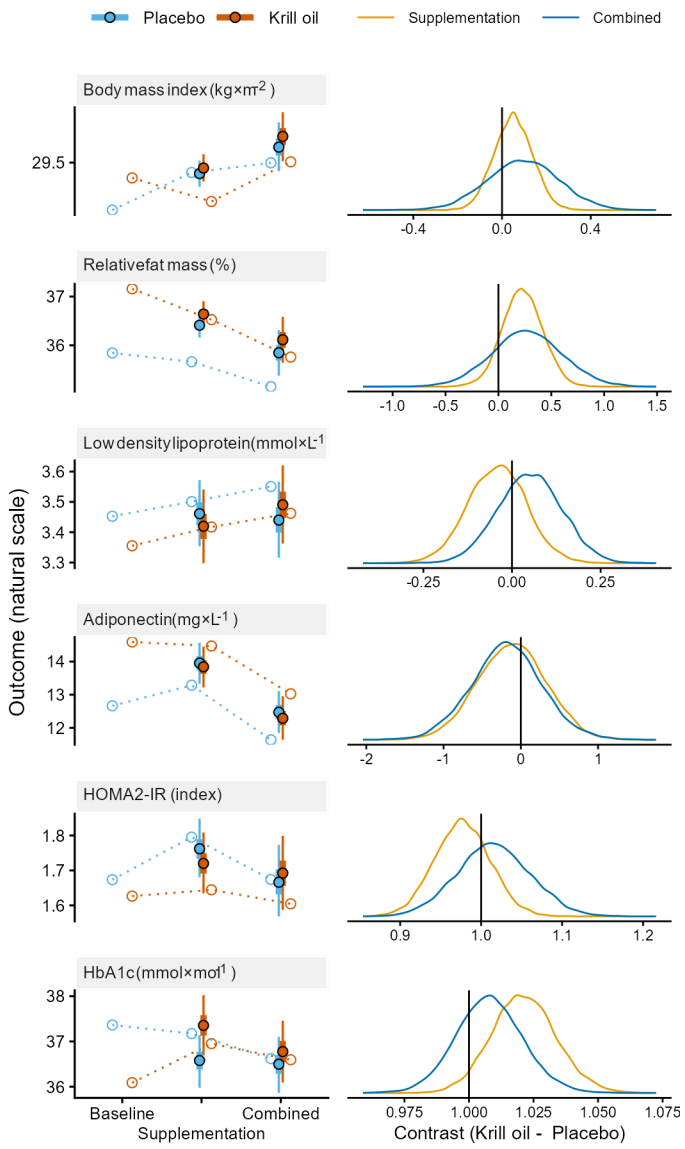

Figure 8.2: Observed group means (open symbols, all available observations; geometric means for HOMA2-IR and HbA1c) and model-based adjusted group means (filled symbols, posterior median with 50% and 95% CI) for each outcome on its natural scale. Adjusted means after supplementation and combined periods are model predictions conditional on the overall mean pre-intervention value and averaged over sex. The right panels show posterior distributions of treatment effects (krill oil - placebo) for each measurement. Baseline represents pre-intervention values (observed); Supplementation and Combined show estimates and observed means after the supplementation period and after the Supplementation + resistance-training period, respectively.

At post-supplementation (T2), baseline- and sex-adjusted differences between groups indicated a reduced waist circumference in krill oil compared to placebo −0.7 cm (95% PI: −1.8, 0.4) with the probability of any benefit (proportion of the posterior density indicating benefit for the krill oil group, P(benefit)) being 89%. After the combined period (supplementation and resistance training), the between-group contrast for waist circumference indicated no difference, −0.1 cm (95% CrI: −1.5, 1.3; P(benefit) 55%) (Figure 8.1). Both diastolic and systolic blood pressure indicated beneficial effects in krill oil compared to placebo at T2 (-1.8 mmHg (-3.7, 0.2; P(benefit) 96%) and −3.8 mmHg (-6.9, −0.7; P(benefit) >99%) for diastolic and systolic blood pressure, respectively). Although both the average diastolic and systolic blood pressure were more similar between krill oil and placebo after the second period, the probability distribution indicated a higher

probability of benefit in the krill oil group (-1.2 mmHg (-3.4, 0.9; P(benefit) 87%) and −2.5 mmHg (-6.2, 1.4; P(benefit) 90%) for diastolic and systolic blood pressure, respectively; Figure 8.1).

After the supplementation period, the krill oil group displayed similar levels of high-density lipoproteins (HDL) as compared to the placebo group (-0.01 mmol/L (-0.06, 0.04; P(benefit) 37%)). However, following the combined period, the krill oil group increased HDL levels compared to placebo, indicating a higher probability of beneficial effects of krill oil supplementation (0.05 mmol/L (-0.00, 0.11; P(benefit) 97%)). Triglyceride levels were estimated to be similar between intervention protocols, both after the post-supplementation period (-1% (-11, +11%; P(benefit) 53%)) and after the combined period (-1% (-14, +14%; P(benefit) 55%)).

A combined score was created by summing the five sub-scores of the metabolic syndrome diagnostic criteria from their continuous, standardized scales, with each score given equal weight (the average blood pressure score was one-fifth of the total score). Comparing the combined scores between treatments indicated a high probability of benefit from krill oil at both post-treatment time points (post-supplementation, −0.08 SD-units (-0.18, 0.01; P(benefit) 96%); post combined treatment, −0.09 SD-units (-0.20, 0.02; P(benefit) 95%); Figure 8.1 B). The differences in the combined score did not yield a significant (95% confidence interval not containing 0) difference in the dichotomous diagnostic criteria (harmonised definition, at least three of the five criteria passing their thresholds (10); Figure 8.1 C and D).

Comparing BMI between treatment groups indicated no clear difference between groups at follow-up time points (0.05 kg/m² (95% CrI: −0.12, 0.22; P(benefit) 28%) and 0.09 kg/m² (-0.22, 0.41; P(benefit) 28%), for after the supplementation and combined periods, respectively). However, relative fat mass indicated a higher probability of an adverse effect of krill oil compared to placebo after the supplementation period (0.2 %-points (-0.1, 0.6; P(benefit) 11%)). After the combined period, differences were smaller (0.3 %-points (-0.4, 0.9; P(benefit) 20%)). Adjusted differences between groups in low density lipoprotein (LDL) and Adiponectin indicated similarity between treatments (LDL post-supplementation, −0.04 mmol/L (-0.21, 0.12; P(benefit) 69%); LDL post-combined treatment, 0.05 mmol/L (-0.13, 0.22; P(benefit) 29%); Adiponectin post-supplementation, −0.11 mg/L (-0.98, 0.73; P(benefit) 40%); Adiponectin post-combined treatment −0.18 mg/L (-1.09, 0.74; P(benefit) 34%)). Insulin resistance, estimated using the HOMA2-IR model score, indicated a greater posterior probability of benefit at post-supplementation after krill oil compared to placebo (-2% (-9, +4%; P(benefit) 75%)), a difference that was not present after the combined time period (+1% (-7, +10%; P(benefit) 36%)). Glycated haemoglobin (HbA1c) was slightly higher in krill oil compared to placebo after the supplementation period, indicating a low probability of benefit (+2% (95% CrI: −0, +5%; P(benefit) 4%)). After the combined period, the difference was smaller and more uncertain (+1% (-2, +3%; P(benefit) 27%)).

8.1 Model expansion: Baseline-dependent treatment effects

To assess whether the effect of krill oil supplementation depends on the participants' baseline status, four expansions of model s13 are considered. All keep the full s13 structure (joint imputation of missing pre- and post-intervention values, the multivariate Student's t-likelihood with $\nu = 4$, the time × type residual scales and the LKJ correlation matrix) and change only the linear predictor for the expected post-intervention value. In s13, the expected value for participant p, measurement type k and post-intervention time-point t is

$$\mu_{p,k,t} = \alpha_{k,t,\,\mathrm{tx}\,[p]} + \beta_{k,t}\, x_{p,k} + \beta^{\mathrm{sex}}_{k,t}\, s_p \tag{8.1}$$

where $x_{p,k}$ is the standardised pre-intervention value (observed or imputed), $s_p \in \{-0.5, 0.5\}$ codes sex, and $\mathrm{tx}\,[p] \in \{1, 2\}$ indexes placebo and krill oil, respectively. The treatment effect is the difference between intercepts, $\alpha_{k,t,2} - \alpha_{k,t,1}$, and does not depend on the baseline value. Let $\omega_p = 1$ for participants in the krill oil group and $\omega_p = 0$ for placebo.

Model s19: treatment × own-baseline interaction. The pre-intervention slope is allowed to differ between treatment groups,

$$\begin{aligned} \mu_{p,k,t} &= \alpha_{k,t,\,\mathrm{tx}\,[p]} + \beta_{k,t}\, x_{p,k} + \delta_{k,t}\, \omega_p\, x_{p,k} + \beta^{\mathrm{sex}}_{k,t}\, s_p \\ \delta_{k,t} &= \tau_\delta\, z_{k,t} \\ z_{k,t} &\sim \mathrm{Normal}\ (0, 1) \\ \tau_\delta &\sim \text{Half-Normal}\ (0, 0.25) \end{aligned} \tag{8.2}$$

so that the treatment effect for a participant with standardised baseline value x is

$$\Delta_{k,t}(x) = \alpha_{k,t,2} - \alpha_{k,t,1} + \delta_{k,t}\, x. \tag{8.3}$$

Since the outcomes are standardised, the baseline mean is close to zero and the intercept difference keeps its interpretation from s13 as the effect at an average baseline value, while $\delta_{k,t}$ is the change in the treatment effect per SD of the baseline value. The $2K = 24$ interaction terms share a common scale τ_δ (non-centred parameterisation), which shrinks individual interactions towards zero unless they are jointly supported by the data. A prior scale of 0.25 places most prior mass on interactions smaller than half an SD of the outcome per SD of baseline.

Model s20: treatment × baseline metabolic burden. Rather than each outcome's own baseline, s20 uses a single moderator shared across outcomes: a continuous baseline metabolic burden score defined as the mean of the sign-aligned standardised baseline values of the five diagnostic criteria,

$$\begin{aligned} b_p &= \frac{1}{5}\Big(-x_{p,\,\mathrm{HDL}} + x_{p,\,\mathrm{TG}} + \frac{1}{2}\big(x_{p,\,\mathrm{dias}} + x_{p,\,\mathrm{sys}}\big) + x_{p,\,\mathrm{glucose}} + x_{p,\,\mathrm{waist}}\Big) \\ \tilde{b}_p &= b_p - \frac{1}{N_P}\sum_{p'} b_{p'} \end{aligned} \tag{8.4}$$

where HDL enters with a negative sign so that higher values of b_p indicate worse metabolic health, and blood pressure contributes as the mean of the diastolic and systolic values. The score is computed inside the model from the partly imputed baseline matrix, so it remains a continuous and differentiable function of the imputed values (in contrast to the number of criteria met, or the baseline diagnosis, which are step functions). The linear predictor becomes

$$\begin{aligned} \mu_{p,k,t} &= \alpha_{k,t,\,\mathrm{tx}\,[p]} + \beta_{k,t}\, x_{p,k} + \gamma_{k,t}\, \omega_p\, \tilde{b}_p + \beta^{\mathrm{sex}}_{k,t}\, s_p \\ \gamma_{k,t} &= \tau_\gamma\, z_{k,t} \\ z_{k,t} &\sim \mathrm{Normal}\ (0, 1) \\ \tau_\gamma &\sim \text{Half-Normal}\ (0, 0.25) \end{aligned} \tag{8.5}$$

with treatment effect $\Delta_{k,t}(\tilde{b}) = \alpha_{k,t,2} - \alpha_{k,t,1} + \gamma_{k,t}\,\tilde{b}$. Centring the burden score on its sample mean keeps the intercept difference interpretable as the effect at average baseline burden.

Model s21: treatment × baseline relative fat mass. s21 has the same structure as s20, but the shared moderator is the participant's baseline relative fat mass. The moderator is taken from the partly imputed baseline matrix and standardised within each posterior draw to mean 0 and SD 1 over participants,

$$\begin{aligned} f_p &= \frac{x_{p,\,\mathrm{fat.rel}} - \bar{x}_{\mathrm{fat.rel}}}{\mathrm{SD}\ (x_{\mathrm{fat.rel}})} \\ \mu_{p,k,t} &= \alpha_{k,t,\,\mathrm{tx}\,[p]} + \beta_{k,t}\, x_{p,k} + \gamma_{k,t}\, \omega_p\, f_p + \beta^{\mathrm{sex}}_{k,t}\, s_p \end{aligned} \tag{8.6}$$

with $\gamma_{k,t}$ given the same partially pooled prior as in s20. The intercept difference is the treatment effect at the average baseline relative fat mass and $\gamma_{k,t}$ is the change in the treatment effect per SD of baseline relative fat mass.

Model s22: treatment × baseline omega-3 index. In s22 the shared moderator is the baseline red blood cell omega-3 index (8), which is not one of the modelled outcomes. The index was standardised to mean 0 and SD 1 over participants with an observed baseline value (o_p). The baseline index was missing for 4 participants; these values were imputed in the model with a standard normal prior, $o_p \sim \mathrm{Normal}\ (0, 1)$, so that all randomised participants are kept. The linear predictor is

$$\mu_{p,k,t} = \alpha_{k,t,\,\mathrm{tx}\,[p]} + \beta_{k,t}\, x_{p,k} + \gamma_{k,t}\, \omega_p\, o_p + \beta^{\mathrm{sex}}_{k,t}\, s_p \tag{8.7}$$

with the same partially pooled prior on $\gamma_{k,t}$ as in s20. Here, $\gamma_{k,t}$ is the change in the treatment effect per SD of the baseline omega-3 index (1 SD = 1.35 %-points).

Treatment effects across the moderator. To show what the interactions imply, treatment effects were predicted from each model at the moderator mean (0) and at ± 1 SD of the moderator,

$$\Delta_{k,t}(m) = \alpha_{k,t,2} - \alpha_{k,t,1} + \gamma_{k,t}\, m, \quad m \in \{-\ \mathrm{SD}, 0, +\ \mathrm{SD}\}, \tag{8.8}$$

with $\delta_{k,t}$ in place of $\gamma_{k,t}$ for s19. For s21 and s22 the moderator is standardised, so SD = 1. For s19, the SD is the observed baseline SD of each standardised outcome, and for s20 it is the SD of the centered burden

score over participants, computed per posterior draw. Effects are computed per posterior draw and shown on the standardised scale (outcome SD units), where a negative value indicates a lower value with krill oil compared to placebo (Figure 8.4).

All four models had satisfactory chain convergence statistics and effective sample size enabling their use for subsequent inference (Figure 8.3).

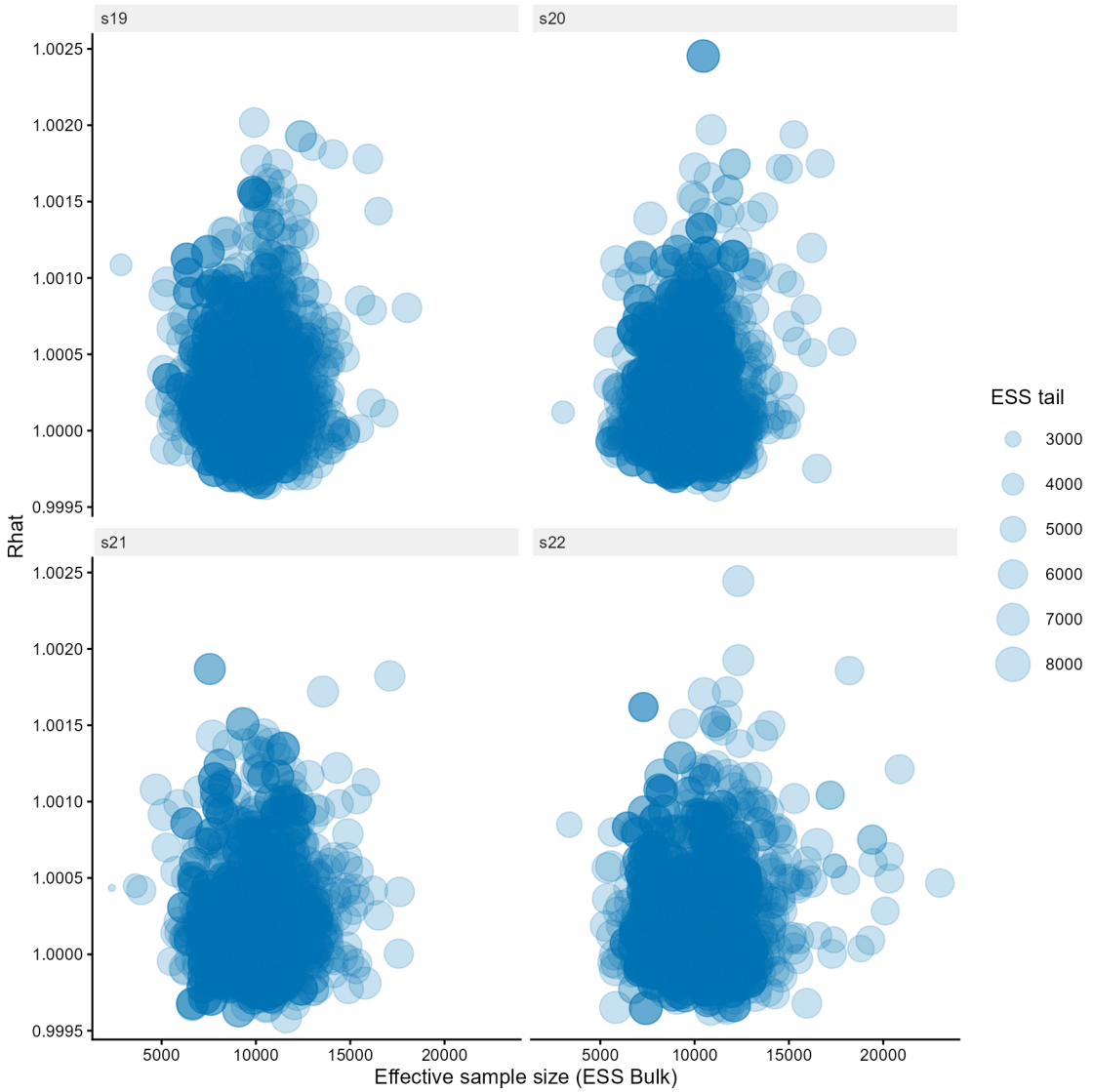

Figure 8.3: Summary chain statistics (Rhat vs. bulk and tail effective sample size) for the moderation models s19 to s22.

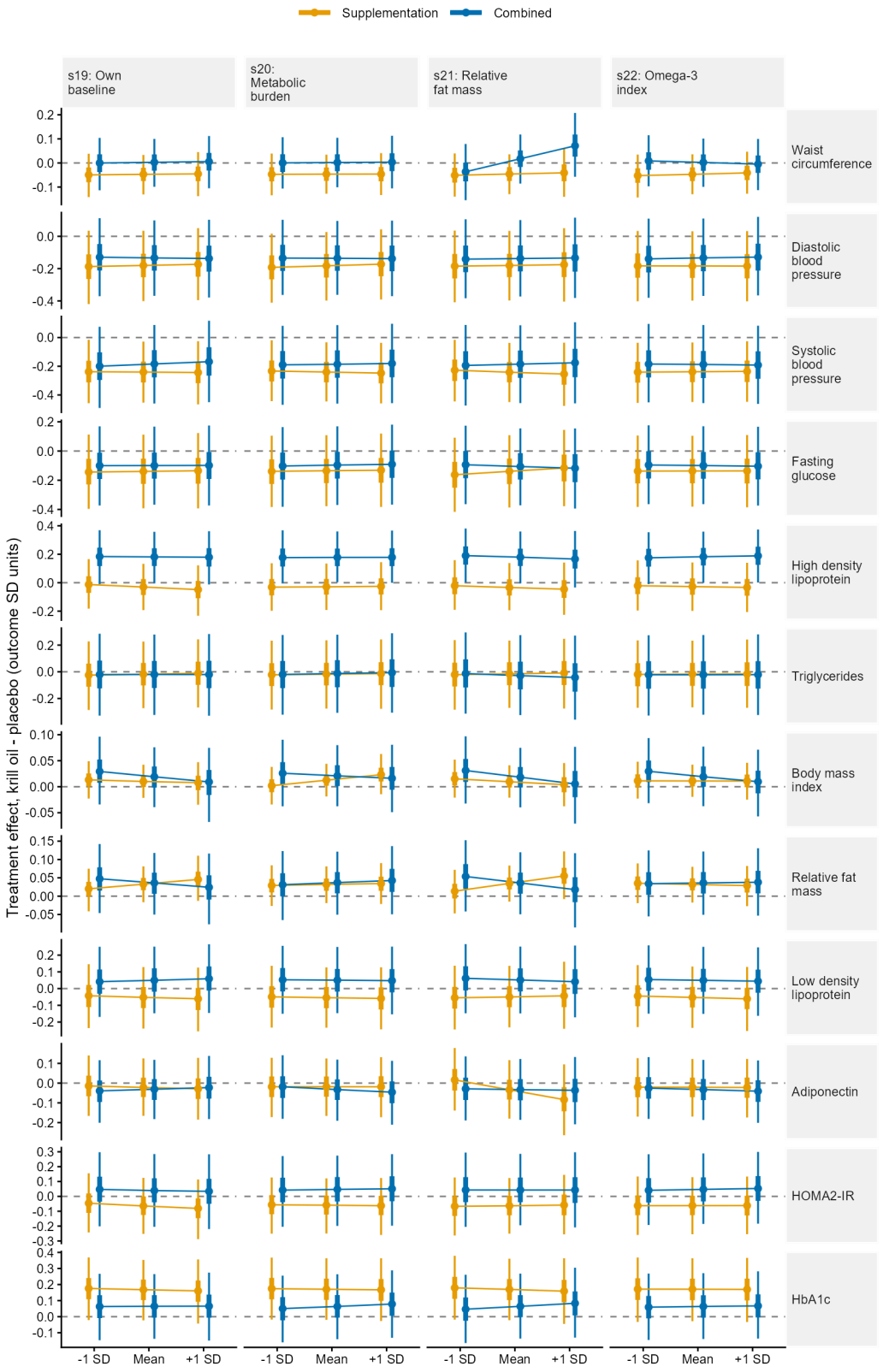

Figure 8.4: Predicted treatment effects (krill oil - placebo) at the moderator mean and at ± 1 SD of the moderator in models s19 (each outcome's own baseline value), s20 (baseline metabolic burden score), s21 (baseline relative fat mass) and s22 (baseline omega-3 index), after the supplementation and combined periods. Points and error bars show posterior medians with 50% and 95% CrI on the standardised scale (outcome SD units).

Combining the predicted effects on the diagnostic criteria per posterior draw, in the same way as for the metabolic burden score in s13, gives the predicted treatment effect on the burden score across each moderator (Figure 8.5).

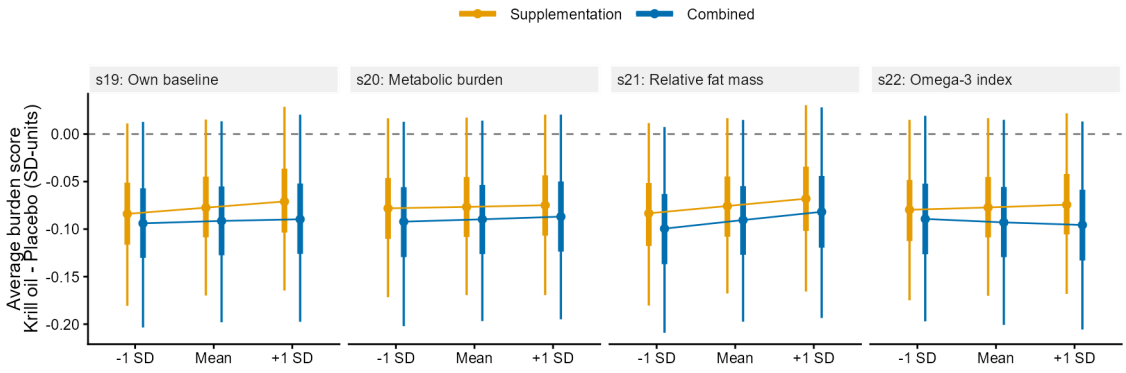

Figure 8.5: Predicted treatment effects (krill oil - placebo) on the metabolic burden score at the moderator mean and at ± 1 SD of the moderator in models s19 (each outcome's own baseline value), s20 (baseline metabolic burden score), s21 (baseline relative fat mass) and s22 (baseline omega-3 index), after the supplementation and combined periods. The burden score effect is combined per posterior draw from the effects on the diagnostic criteria. Points and error bars show posterior medians with 50% and 95% CrI on the standardised scale (SD units).

The moderation parameters of all outcomes and the burden score are summarised in Table 8.1.

Table 8.1: Moderation parameters, the change in the treatment effect (krill oil - placebo) per SD of the moderator, in models s19 (each outcome's own baseline value, $\delta_{k,t}$), s20 (baseline metabolic burden score, $\gamma_{k,t}$), s21 (baseline relative fat mass, $\gamma_{k,t}$) and s22 (baseline omega-3 index, $\gamma_{k,t}$), after the supplementation and combined periods. The burden score parameter is combined per posterior draw from the parameters of the diagnostic criteria. Values are posterior medians with 50% and 95% CrI on the standardised scale (SD units). P(benefit) is the posterior probability that the treatment effect at +1 SD of the moderator is more beneficial than at the moderator mean (higher values for HDL and adiponectin, lower values otherwise).

	Supplementation				Combined			
	Median	50% CrI	95% CrI	P(benefit)	Median	50% CrI	95% CrI	P(benefit)
s19: Own baseline								
Waist circumference	0.00	−0.01, 0.01	−0.04, 0.05	47%	0.00	−0.01, 0.01	−0.04, 0.05	44%
Diastolic blood pressure	0.00	−0.01, 0.02	−0.05, 0.08	42%	−0.00	−0.02, 0.01	−0.08, 0.05	55%
Systolic blood pressure	−0.00	−0.01, 0.01	−0.07, 0.06	53%	0.01	−0.00, 0.03	−0.03, 0.12	33%
Fasting glucose	0.00	−0.01, 0.01	−0.05, 0.07	44%	0.00	−0.01, 0.01	−0.05, 0.06	48%
High density lipoprotein	−0.01	−0.03, 0.00	−0.10, 0.02	29%	−0.00	−0.01, 0.01	−0.06, 0.05	48%
Triglycerides	0.00	−0.01, 0.02	−0.04, 0.08	41%	−0.00	−0.01, 0.01	−0.06, 0.06	51%
Body mass index	−0.00	−0.01, 0.00	−0.03, 0.02	60%	−0.01	−0.02, 0.00	−0.06, 0.02	72%
Relative fat mass	0.01	0.00, 0.02	−0.01, 0.05	20%	−0.01	−0.02, 0.00	−0.07, 0.02	70%
Low density lipoprotein	−0.00	−0.02, 0.00	−0.08, 0.04	62%	0.00	−0.00, 0.02	−0.04, 0.08	37%
Adiponectin	−0.00	−0.02, 0.00	−0.07, 0.04	40%	0.00	−0.00, 0.02	−0.04, 0.08	62%
HOMA2-IR	−0.01	−0.03, 0.00	−0.11, 0.03	70%	−0.00	−0.02, 0.01	−0.09, 0.04	57%
HbA1c	−0.00	−0.02, 0.00	−0.08, 0.04	60%	0.00	−0.01, 0.01	−0.05, 0.06	47%
Metabolic burden score	0.00	−0.00, 0.01	−0.01, 0.04	32%	0.00	−0.00, 0.01	−0.02, 0.03	42%
s20: Metabolic burden								
Waist circumference	−0.00	−0.01, 0.01	−0.03, 0.04	51%	0.00	−0.01, 0.01	−0.03, 0.04	45%
Diastolic	0.00	−0.00,	−0.02,	34%	−0.00	−0.01,	−0.05,	54%

9. Effects of krill oil supplementation on indications of metabolic syndrome - Model sensitivity

9.1

The residual correlations modeled in s13 allows us to account for associations between outcome variables when estimating treatment effects. Conceptually this could affect estimates in two ways. First, adding uncertainty associated with the between-variable correlation could increase uncertainty in the treatment effect. However, the correlations will help in estimating missing data, this would reduce uncertainty in estimation of single measurements which propagates into the treatment effect.

10. Oral glucose-tolerance test

10.1 Introduction

Here we model the oral glucose-tolerance test data. Blood glucose was measured every fifteen minute after a standard 75 g glucose drink over the course of a 2 hour period. The test was performed before the intervention period (T1), after the supplementation-only period (T2), and after the full intervention (combined period, supplementation and resistance training vs. placebo and resistance training). The goal is to evaluate the effect of the intervention (krill oil supplementation) on the glucose tolerance curve.

We aim to build a model that captures the glucose time-course after ingesting the drink. We will accomplish this by fitting a spline model to participant-level data with treatment-groups and time-points as population-level effects.

10.2 Data preparations

```
# Libraries
source("./R/fig-opt-fun.R")
source("./R/packages.R")
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
This is cmdstanr version 0.9.0

- CmdStanR documentation and vignettes: mc-stan.org/cmdstanr

- CmdStan path: C:/Users/706194/.cmdstan/cmdstan-2.39.0

- CmdStan version: 2.39.0

This is posterior version 1.7.0


Attaching package: 'posterior'


The following objects are masked from 'package:stats':

    mad, sd, var


The following objects are masked from 'package:base':

    %in%, match
```

```
This is loo version 2.9.0

- Online documentation and vignettes at mc-stan.org/loo

- As of v2.0.0 loo defaults to 1 core but we recommend using as many as possible. Use
the 'cores' argument or set options(mc.cores = NUM_CORES) for an entire session.

- Windows 10 users: loo may be very slow if 'mc.cores' is set in your .Rprofile file
(see https://github.com/stan-dev/loo/issues/94).


Attaching package: 'loo'


The following object is masked from 'package:posterior':

    gpdfit


Loading required package: htmlTable


Attaching package: 'Gmisc'


The following object is masked from 'package:tidyr':

    spread


The following object is masked from 'package:base':

    append
```

```
source("./R/helpers.R")
source("./R/data-prep.R")
```

```
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant)`
Joining with `by = join_by(participant, sex)`
```

```
# The spline package is needed for basis spline construction
library(splines)
library(ggdist)
```

```
Warning: package 'ggdist' was built under R version 4.6.1
```

```
# Settings
refit.models <- FALSE

# Get the full index of participants
```

```
part_full <- criteria_dat |>
  distinct(participant) |>
  pull(participant)


# Glucose data
glucose_dat <- AOData::glucose |>
  select(participant:glucose)  |>
  # Treatment
  inner_join(
    AOData::idkeys |>
  select(participant, treat))
```

```
Joining with `by = join_by(participant)`
```

10.3 Preliminary models

10.3.1 Baseline model with group-level effects (participants)

```
# Use a complete data set (no missing data) at time 1 to set up models
pre_dat <- glucose_dat |>
  filter(time == "T1") |>
  filter(!(is.na(glucose))) |>
  mutate(stglucose = (glucose - mean(glucose)) / sd(glucose),
         participant = as.factor(participant)) |>
  # To avoid problems, dropping participants with less than 7
  # measurements
  mutate(.by = participant,
         n = n()) |>
  filter(n == 7)


npart <- pre_dat |>
  distinct(participant) |>
  nrow()
```

In a preliminary model we fit complete data (seven measurements per participant) from the pre-intervention data (n = 146). The basis splines are visualized in Figure 10.1.

```
# Spline basis: bs(min, df = 4) WITHOUT an intercept column, so the random
# intercept is not confounded with the spline weights. This is similar to
# what brms do, see archive/ogtt-analysis.qmd for a preliminary analysis
# using brms.

B <- bs(pre_dat$min,
        df = 5)

# Do a "prediction" using the Basis splines. This will show the smooth not only
# at break-points.
min_pred <- seq(from = 0, to = 120, by = 1)

B_pred <- predict(B,  min_pred)



p1 <- data.frame(B_pred) |>
  cbind(min = min_pred) |>
  pivot_longer(starts_with("X")) |>
  ggplot(aes(min, value, group = name, color = name)) +
```

```
  geom_line() +
  alpha_theme() +
  scale_y_continuous(breaks = c(0, 1)) +
  labs(x = "Minute",
       y = "Basis value")  +
  theme(legend.position = "none")


p1
```

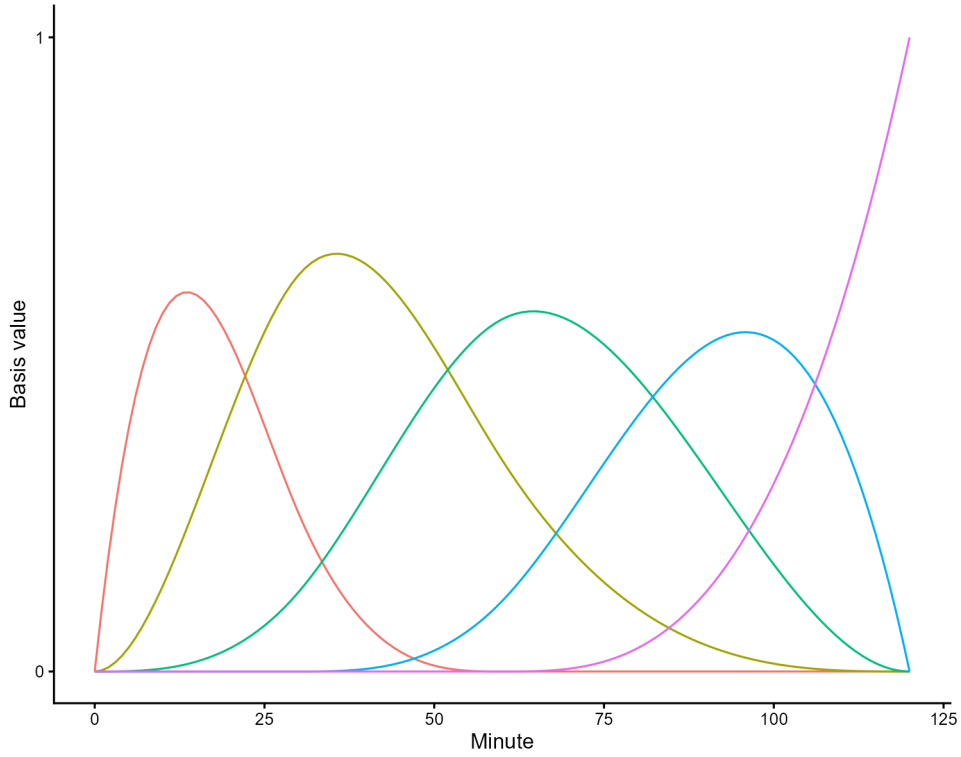

Figure 10.1: Basis splines across the glucose tolerance test.

The pre-intervention multilevel model (`mo1.stan`) for blood glucose (y) gives each participant p their own curve via a random intercept a_p and random spline coefficients $b_{k,p}$, drawn jointly (intercept + spline) from a correlated multivariate normal (this is equivalent to the brms `m5` parameterization `archive/ogtt-analysis.qmd`):

$$
\begin{aligned}
y_i &\sim \text{Normal}\ (\mu_i, \sigma) \\
\mu_i &= \alpha + a_{p[i]} + \sum_{k=1}^{K} (\beta_k + b_{k,p[i]})\, B_{k,i} \\
(a_p, b_{1,p}, \ldots, b_{K,p})^\top &\sim \text{MVNormal}\ (0, \mathbf{S}\,\mathbf{R}\,\mathbf{S}) \\
\mathbf{S} &= \text{diag}\ (\sigma_{\text{sd}}) \\
\sigma_{\text{sd}} &\sim \text{Exponential}\ (2) \\
\mathbf{R} &\sim \text{LKJ}\ (1) \\
\alpha &\sim \text{Normal}\ (0, 2) \\
\beta_k &\sim \text{Normal}\ (0, 2) \\
\sigma &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{10.1}
$$

The basis B is `bs(min, df = 5)` (no intercept column), so the random intercept a_p is identified separately from the spline weights. The choice of 5 spline components was done after preliminary prior- and posterior simulations indicating that this sufficiently regularize individual curves. The correlation matrix $\mathbf{R}$ is sampled via a non-centered Cholesky factor.

In preliminary versions of this model (`archive/ogtt-analysis.qmd`), specific priors were used to regularize the population-level spline towards increased glucose in the first half of the test followed by declining values in the second half. Since the data will likely wash away such prior information we will simplify this part of the model to avoid prior-likelihood conflicts.

Below we run a prior predictive check of the pre-intervention model. The Prior-predictive check (Figure 10.2) indicates that model does not expect systematic changes over the course of the test but allow for variation across participants.

```
# Integer participant id (1..N_P)
pre_dat <- pre_dat |>
  mutate(P = dense_rank(participant))

# Save the mean and sd from the subset of data
mglucose <- mean(pre_dat$glucose)
sdglucose <- sd(pre_dat$glucose)


# Create the data list (now with participant index)
dat <- list(
  y   = pre_dat$stglucose,
  N   = nrow(pre_dat),
  K   = ncol(B),
  B   = B,
  N_P = max(pre_dat$P),
  P   = pre_dat$P,
  prior_only = 1
)


mo1 <- cmdstan_model(stan_file = "models/mo1.stan",
                     exe_file = "models/mo1.exe")


fit_wrap(mo1, fit = "fit.mo1.pp", DATA = dat)


# Predict from the basis spline
min_seq <- seq(from = 0, to = 120, by = 1)

B_pred_long <- predict(B, min_seq) |>
  data.frame() |>
  cbind(min_seq) |>
  pivot_longer(starts_with("X"),
               values_to = "B")


draws <- fit.mo1.pp$draws(format = "df")

# Population level model (mean) using the global intercept and
# splines
b_long <- draws |>
  dplyr::select(.draw, a, starts_with("b[")) |>
  pivot_longer(starts_with("b["), values_to = "b") |>
  mutate(name = gsub("b\\[", "X", name),
         name = gsub("\\]", "", name))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(b_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, min_seq),
            mu = (first(a) + sum(B * b) )  * sdglucose + mglucose) |>
  summarise(.by = min_seq,
            m   = mean(mu) ,
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975))

# Participant-level curves (posterior-mean random effects r_p[p, m])
# r_p[p, 1] = random intercept, r_p[p, 2:(K+1)] = spline deviations
rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_p = val)
rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(name = paste0("X", m - 1)) |>
  dplyr::select(p, name, dev = val)

# Posterior-mean population coefficients
a_mean <- mean(draws$a)
b_mean_long <- draws |>
  dplyr::select(starts_with("b[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "name", values_to = "b") |>
  mutate(name = gsub("b\\[", "X", name),
         name = gsub("\\]", "", name))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
part_curves <- B_pred_long |>
  inner_join(b_mean_long, by = "name") |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, min_seq),
            mu = sum(B * (b + dev))) |>
  inner_join(rp_int, by = "p") |>
  mutate(mu = (a_mean + a_p + mu) * sdglucose + mglucose)

# Population ribbon + per-participant spaghetti

ggplot(pop_curve, aes(min_seq, m)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = p),
            alpha = 0.3, linewidth = 0.3) +

  geom_line(linewidth = 1) +
  alpha_theme() +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>") +
  theme(axis.title.y = ggtext::element_markdown())
```

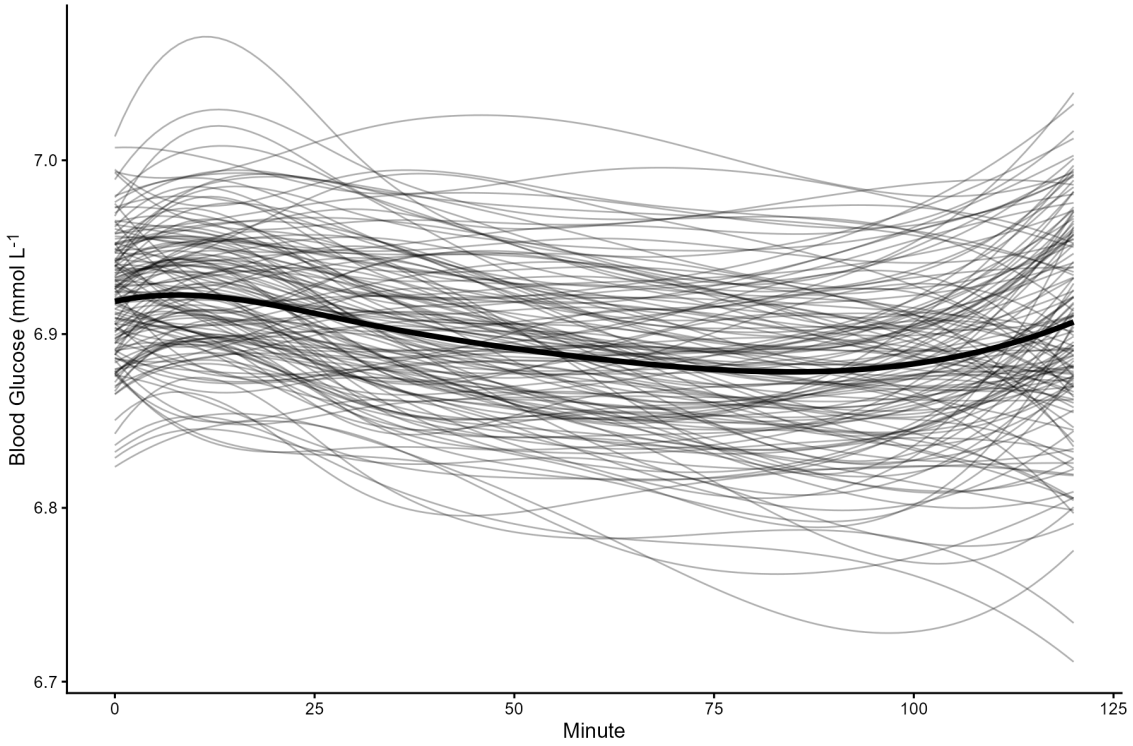

Figure 10.2: Prior predictive simulation of population average and participant-level glucose curves.

Running the model on the actual data let's us explore the model fit to data. The overall model fit gives a population mean curve and allows for large variations between individuals (Figure 10.3 A). Inspecting three individuals indicate that individual curves are shrunken from very wiggly shapes (Figure 10.3 B). Analysis of residuals indicate that there might be a tendency towards larger positive errors at higher fitted values (Figure 10.3 C), the overall data patterns indicate that the observed data can be replicated using the model. Overall these checks validate the model for further development.

```
# Set prior_only to 0 to sample from the likelihood
dat$prior_only <- 0


fit_wrap(mo1, fit = "fit.mo1", DATA = dat)


# Predict from the basis spline
min_seq <- seq(from = 0, to = 120, by = 1)

B_pred_long <- predict(B, min_seq) |>
  data.frame() |>
  cbind(min_seq) |>
  pivot_longer(starts_with("X"),
               values_to = "B")


draws <- fit.mo1$draws(format = "df")

# Population level model (mean) using the global intercept and
# splines
b_long <- draws |>
  dplyr::select(.draw, a, starts_with("b[")) |>
  pivot_longer(starts_with("b["), values_to = "b") |>
```

```
  mutate(name = gsub("b\\[", "X", name),
         name = gsub("\\]", "", name))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(b_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, min_seq),
            mu = (first(a) + sum(B * b) )  * sdglucose + mglucose) |>
  summarise(.by = min_seq,
            m   = mean(mu) ,
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975))

# Participant-level curves (posterior-mean random effects r_p[p, m])
# r_p[p, 1] = random intercept, r_p[p, 2:(K+1)] = spline deviations
rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_p = val)
rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(name = paste0("X", m - 1)) |>
  dplyr::select(p, name, dev = val)

# Posterior-mean population coefficients
a_mean <- mean(draws$a)
b_mean_long <- draws |>
  dplyr::select(starts_with("b[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "name", values_to = "b") |>
  mutate(name = gsub("b\\[", "X", name),
         name = gsub("\\]", "", name))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
part_curves <- B_pred_long |>
  inner_join(b_mean_long, by = "name") |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, min_seq),
            mu = sum(B * (b + dev))) |>
  inner_join(rp_int, by = "p") |>
  mutate(mu = (a_mean + a_p + mu) * sdglucose + mglucose)

# Population ribbon + per-participant spaghetti

p1 <- ggplot(pop_curve, aes(min_seq, m)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = p),
            alpha = 0.3, linewidth = 0.3) +
  geom_ribbon(aes(ymin = lwr, ymax = upr),
              alpha = 0.2,
              fill = colscale[3]) +
```

```
  geom_line(linewidth = 1) +
  alpha_theme() +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>") +
  theme(axis.title.y = ggtext::element_markdown())


# Individual curves examples
p2 <- ggplot(pop_curve, aes(min_seq, m)) +
  geom_line(data = filter(part_curves, p %in% c(1,2,3)),
            aes(min_seq, mu, group = p, color = as.factor(p)),
            alpha = 0.3, linewidth = 1) +
  geom_ribbon(aes(ymin = lwr, ymax = upr),
              alpha = 0.2) +

  geom_point(data = filter(pre_dat, P %in% c(1,2,3)),
             aes(min, glucose, color = as.factor(P)),
             size = 3) +

  alpha_theme() +
  geom_line(linewidth = 1) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>",
       color = "Participant ID") +
  theme(axis.title.y = ggtext::element_markdown()) +
  scale_color_manual(values = colscale)


p3 <- draws |>
  dplyr::select(.draw, starts_with("y_pred")) |>
  pivot_longer(cols = starts_with("y_pred")) |>
  mutate(obs = rep(pre_dat$stglucose, 2000*4),
         resid = obs - value) |>
  summarise(.by = name,
            m_fitted = mean(value),
            s_fitted = sd(value),
            m_resid = mean(resid),
            s_resid = sd(resid)) |>
  ggplot(aes(m_fitted, m_resid)) +

  geom_errorbar(aes(ymin = m_resid - s_resid,
                    ymax = m_resid + s_resid),
                alpha = 0.3) +
    geom_errorbar(aes(xmin = m_fitted - s_fitted,
                      xmax = m_fitted + s_fitted),
                  orientation = "y",
                alpha = 0.3) +

  geom_point(alpha = 0.3,
             size = 3) +



  geom_smooth(color = colscale[3]) +
  geom_hline(yintercept = 0, lty = 2, alpha = 0.4) +
  alpha_theme()  +
  labs(x = "Predicted values (standardized glucose)",
       y = "Observed - Predicted values") +
  theme()
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
p4 <- draws |>
  dplyr::select(.draw, starts_with("y_rep")) |>
  pivot_longer(cols = starts_with("y_rep")) |>
  filter(.draw %in% 1:40) |>
  ggplot(aes(value, group = .draw))  +
  geom_line(stat = "density", alpha = 0.3) +
  geom_line(data = pre_dat,
            aes(stglucose, group = NULL),
            color = colscale[3],
            linewidth = 1,
            stat = "density") +
  alpha_theme(yaxis = FALSE)   +
  labs(x = "Standardized glucose (simulated and observed)",
       y = "")
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
cowplot::plot_grid(p1, p2, p3, p4,
                   ncol = 2)
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

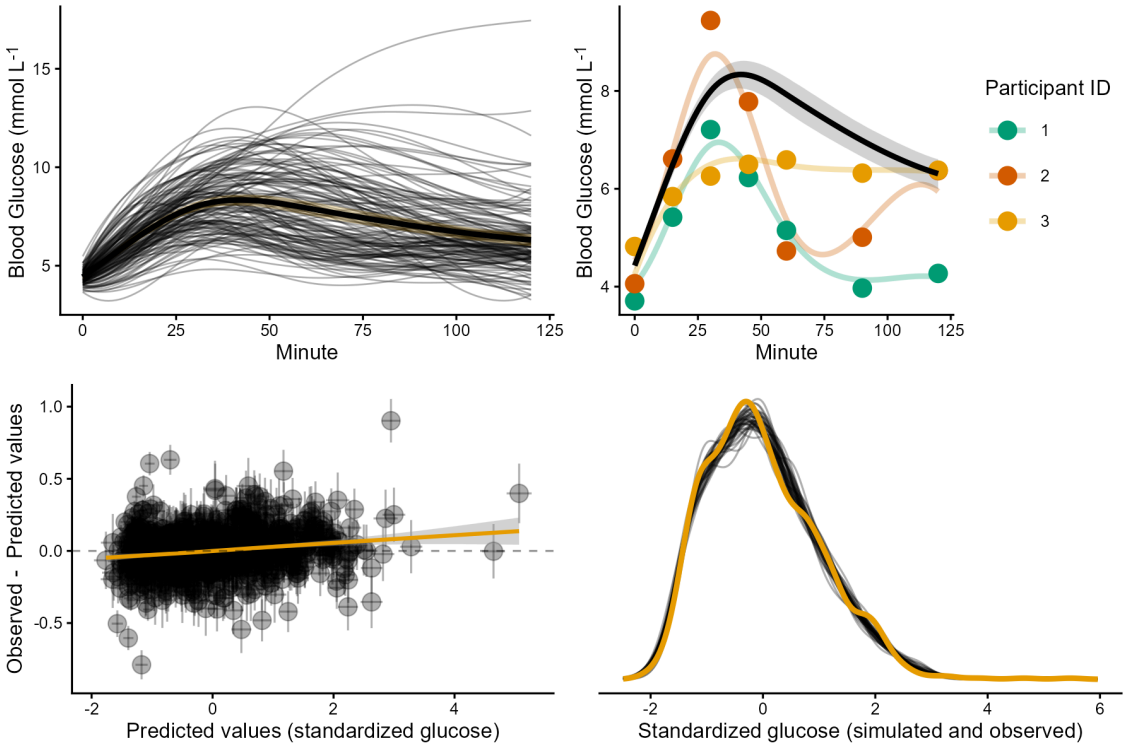

Figure 10.3: Posterior predictive simulation of population average and participant-level glucose curves (A and B). Residuals vs. predicted values (C) and distributions of simulated (yrep) data sets and the observed data.

10.3.2 Three-timepoint multilevel model

The model (`mo2.stan`) extends the pre-intervention model to include three timepoints (T1, T2, T4) and the treatment groups, following the brms `m5` parameterization (`archive/ogtt-analysis.qmd`). The population mean is a cell mean plus a spline curve per cell g (here g = time x treatment). Each participant gets a random intercept and random spline coefficients that may differ by time t, drawn jointly from a correlated multivariate normal:

$$
\begin{aligned}
y_i &\sim \text{Normal}\ (\mu_i, \sigma) \\
\mu_i &= \left(a_{g[i]} + \sum_k b_{k,g[i]}\, B_{k,i} \right) + \left(a_{p[i]} + \sum_k \beta_{k,t[i],p[i]}\, B_{k,i} \right) \\
(a_p, \beta_p)^\top &\sim \text{MVNormal}\ (\mathbf{0}, \mathbf{S}\,\mathbf{R}\,\mathbf{S}) \\
\mathbf{S} &= \text{diag}\ (\sigma_{\text{sd}}) \\
\sigma_{\text{sd}} &\sim \text{Exponential}\ (1) \\
\mathbf{R} &\sim \text{LKJ}\ (1) \\
a_g &\sim \text{Normal}\ (0, 2) \\
b_{k,g} &\sim \text{Normal}\ (0, 2) \\
\sigma &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{10.2}
$$

The participant block has $M = 1 + K\,T$ effects (intercept + a spline per time point), all correlated through $\mathbf{R}$ and sampled with a non-centered Cholesky factor. The prior predictive simulation indicate a very flexible curves per time × treatment (Figure 10.4)

```
# Three timepoints, complete OGTT curves only (7 samples per participant-time)
ml_dat <- glucose_dat |>
  inner_join(AOData::idkeys) |>
  dplyr::select(participant, time, min, glucose, treat) |>
  filter(time %in% c("T1", "T2", "T4")) |>
  filter(!is.na(glucose)) |>
  mutate(.by = c(participant, time), n = n()) |>
  filter(n == 7) |>
  mutate(stglucose = (glucose - mean(glucose)) / sd(glucose),
         participant = as.factor(participant),
         tx = paste0(time, treat),
        tx = factor(tx, levels = c("T1ctrl", "T1omega", "T2ctrl", "T2omega", "T4ctrl",
"T4omega"))) |>
  mutate(P  = dense_rank(participant),  # participant index
          G  = dense_rank(tx),          # population cell (time-only; swap for time
x treat)
         Tt = dense_rank(time))         # time index for the random spline
```

```
Joining with `by = join_by(participant, treat)`
```

```
mglucose <- mean(ml_dat$glucose)
sglucose <- sd(ml_dat$glucose)


# Basis without intercept column
B2 <- bs(ml_dat$min, df = 5)

dat2 <- list(
  y   = ml_dat$stglucose,
  N   = nrow(ml_dat),
  K   = ncol(B2),
  B   = B2,
  N_G = max(ml_dat$G),
  G   = ml_dat$G,
```

```
  N_P = max(ml_dat$P),
  P   = ml_dat$P,
  N_T = max(ml_dat$Tt),
  Tt  = ml_dat$Tt,
  prior_only = 1
)


mo2 <- cmdstan_model(stan_file = "models/mo2.stan",
                     exe_file = "models/mo2.exe")



fit_wrap(mo2, fit = "fit.mo2.pp", DATA = dat2)

draws2 <- fit.mo2.pp$draws(format = "df")

min_seq2 <- seq(from = 1, to = 120, by = 1)
B2_pred_long <- predict(B2, min_seq2) |>
  data.frame() |>
  cbind(min_seq2) |>
  pivot_longer(starts_with("X"), values_to = "B")

ag_long <- draws2 |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws2 |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Map cell index g back to the time label
time_levels <- ml_dat |> distinct(time, G) |> arrange(G)

pop_by_time <- B2_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq2),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  mutate(mu = a_g + mu_spl) |>
  summarise(.by = c(g, min_seq2),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(time_levels, by = c("g" = "G")) |>
  mutate(time = if_else(g %in% c(1,2), "T1",
    if_else(g %in% c(3, 4), "T2", "T3")),
    treat = if_else(g %in% c(1, 3, 5), "ctrl", "omega"))

pop_by_time |>
```

```
  ggplot(aes(min_seq2, m,
             color = treat,

             group = paste(time, treat))) +

  geom_line(linewidth = 1) +
  alpha_theme() +
  labs(x = "Minute", y = "Standardized glucose",
       color = "Treatment") +
        facet_wrap(~ time)
```

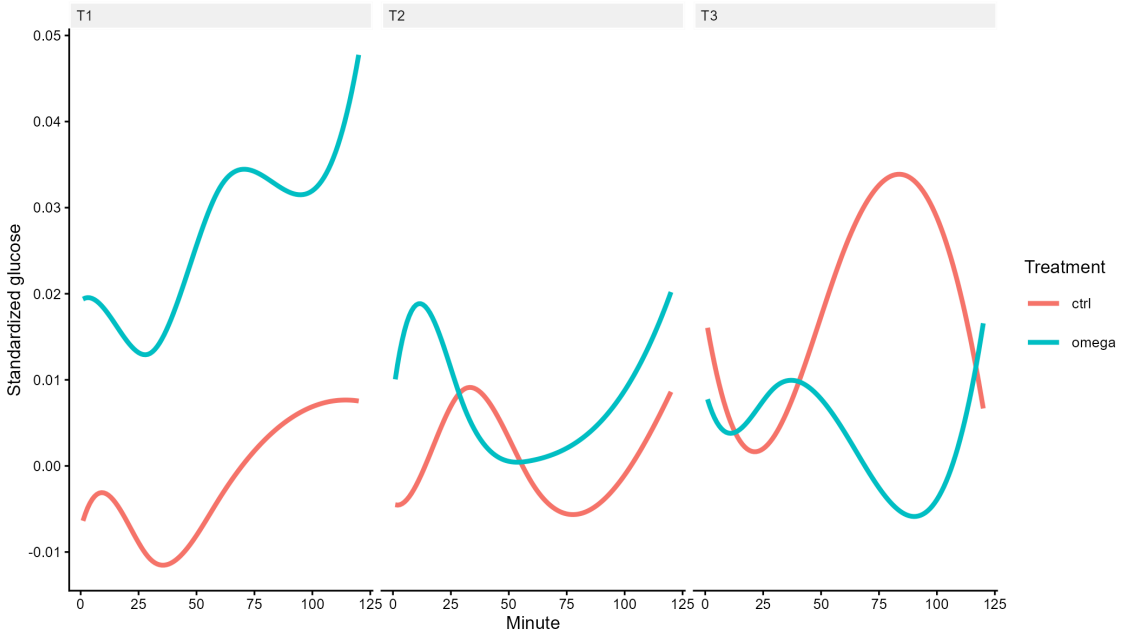

Figure 10.4: Prior predictive simulation using model `mo2` showing average group &times time curves.

```
dat2$prior_only <- 0

fit_wrap(mo2, fit = "fit.mo2", DATA = dat2)

# Population mean curve per time point (cell mean a_g + cell spline b_g)
draws2 <- fit.mo2$draws(format = "df")

min_seq2 <- seq(from = 1, to = 120, by = 1)
B2_pred_long <- predict(B2, min_seq2) |>
  data.frame() |>
  cbind(min_seq2) |>
  pivot_longer(starts_with("X"), values_to = "B")

ag_long <- draws2 |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws2 |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Map cell index g back to the time label
time_levels <- ml_dat |> distinct(time, G) |> arrange(G)

pop_by_time <- B2_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq2),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  mutate(mu = a_g + mu_spl,
         # back tranmsform the standardization
         mu = mu * sglucose + mglucose
         ) |>
  summarise(.by = c(g, min_seq2),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(time_levels, by = c("g" = "G")) |>
  mutate(time = if_else(g %in% c(1,2), "Baseline",
    if_else(g %in% c(3, 4), "Supplementation", "Combined")),
    treat = if_else(g %in% c(1, 3, 5), "Placebo", "Krill oil"))  |>
  mutate(time = factor(time, levels = c("Baseline", "Supplementation", "Combined")))


pop_by_time |>
ggplot(aes(min_seq2, m, color = treat, fill = treat, group = paste(time, treat))) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), alpha = 0.2, color = NA) +
  geom_line(linewidth = 1) +
  alpha_theme() +
  labs(x = "Minute", y = "Blood glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
        facet_wrap(~ time) +
  scale_color_manual(values = colscale[c(2, 4)]) +
  theme(axis.title.y = ggtext::element_markdown())
```

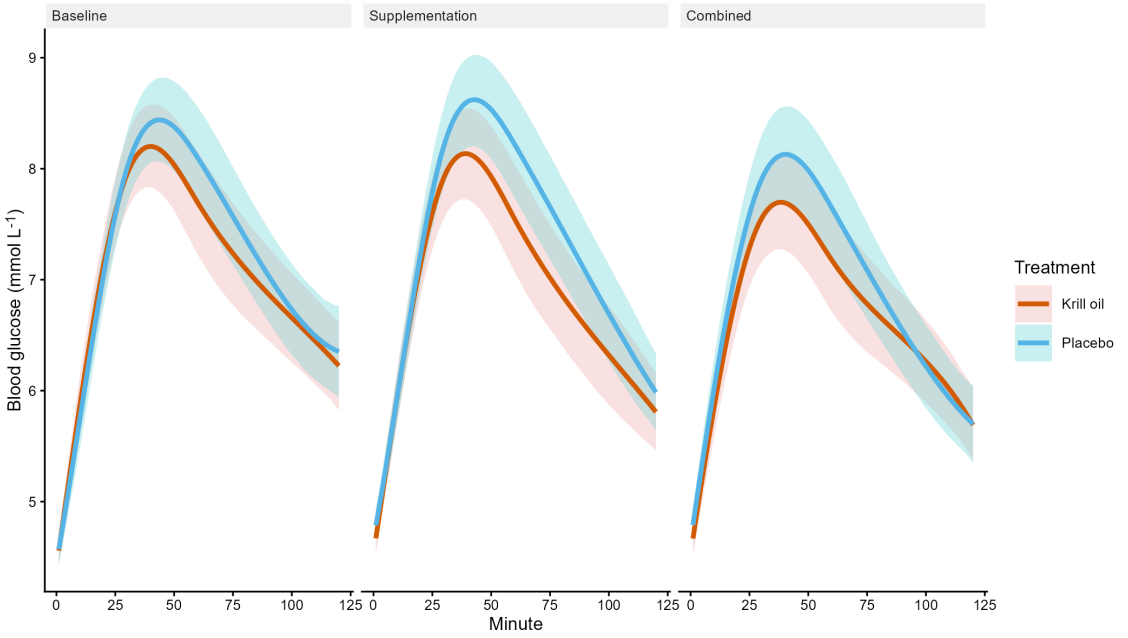

Figure 10.5: Posterior implications of model `mo2` showing average group &times time curves with 95% percentile intervals at the three sampling time points, before the intervention (Baseline), after the supplementation-only period (Supplementation) and after the combined supplementation and resistance-training period (Combined).

10.3.3 Missing data model

We further extend the multilevel-model including the time- and treatment effects to a full model including all baseline data and missing data imputation. Similarly to analyses of the metabolic syndrome criteria we want to draw conclusions from estimates that include the uncertainty associated with missing data.

Model `mo3` (`models/mo3.stan`) use the same mechanism for data imputation as used in the models of the metabolic syndrome model sets. Missing data are treated as parameters and are informed by the likelihood. The relevant additions to the model compared to Equation (10.2) therefore include:

$$ $$ {#eq-mo3}

Where the Normal $(0, 2)$ indicate a weak prior (on the standardized scale) for missing values. Again, the prior predictive check indicate that the model allows for very flexible curves per participants and groups (Figure 10.6).

```
# Extract the full data sets.
full_dat <- glucose_dat |>
  inner_join(AOData::idkeys) |>
  filter(participant %in% part_full) |>

  dplyr::select(participant, time, min, glucose, treat) |>
  # Wider to find NA -> longer -> complete data set
  pivot_wider(values_from = glucose, names_from = time) |>
  pivot_longer(values_to = "glucose",
               names_to = "time",
               cols = T1:T4) |>
  arrange(participant, time, min) |>
  # Indexing
  mutate(
    y_obs = (glucose - mean(glucose, na.rm = TRUE)) / sd(glucose, na.rm = TRUE),
    y_obs = if_else(is.na(glucose), -999, y_obs),
    y_miss = if_else(is.na(glucose), 0, 1),
    tx = paste0(time, treat),
     tx = factor(tx, levels = c("T1ctrl", "T1omega", "T2ctrl", "T2omega", "T4ctrl",
```

```
"T4omega")),

    G = dense_rank(tx),
    P = dense_rank(participant),
    Tt = dense_rank(time)
  )
```

```
Joining with `by = join_by(participant, treat)`
```

```
# For post-fitting analysis, mean and SD of glucose
mglucose <- mean(full_dat$glucose, na.rm = TRUE)
sdglucose <- sd(full_dat$glucose, na.rm = TRUE)


B <- bs(full_dat$min, df = 5)


dat <- list(
  N = nrow(full_dat),
  y_obs = full_dat$y_obs,
  y_miss = full_dat$y_miss,
  K = ncol(B),
  B = B,
  N_G = length(unique(full_dat$G)),
  G = full_dat$G,
  N_P = length(unique(full_dat$P)),
  P = full_dat$P,
  N_T = length(unique(full_dat$Tt)),
  Tt = full_dat$Tt,
  prior_only = 1
)


# Get model code/fits
mo3 <- cmdstan_model(stan_file = "models/mo3.stan",
                     exe_file = "models/mo3.exe")


fit_wrap(mo3, fit = "fit.mo3.pp", DATA = dat)


draws <- fit.mo3.pp$draws(format = "df")


# Basis evaluated on a dense grid (B was built as bs(full_dat$min, df = 5))
min_seq <- seq(from = 0, to = 120, by = 1)

B_pred_long <- predict(B, min_seq) |>
  data.frame() |>
  cbind(min_seq) |>
  pivot_longer(starts_with("X"), values_to = "B")

# Map cell index g and time index Tt back to time / treatment labels, and
# record which cell g each participant (P) belongs to at each time.
cell_key <- full_dat |>
  distinct(time, treat, G, Tt) |>
  rename(g = G)

part_key <- full_dat |>
  distinct(P, Tt, G, time, treat) |>
```

```
  rename(p = P, g = G)


# Population-average curves (full posterior -> credible intervals)

ag_long <- draws |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  # Back transform to original scale
  mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  summarise(.by = c(g, min_seq),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(cell_key, by = "g")


# Participant-level curves (posterior-mean random effects)

# Posterior-mean population coefficients per cell
a_g_mean <- ag_long |>
  summarise(.by = g, a_g = mean(a_g))

b_g_mean <- bg_long |>
  summarise(.by = c(g, name), b_g = mean(b_g))

# Posterior-mean participant random effects r_p[p, m]
rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Intercept (column 1) and time-specific spline deviations (columns 2..M)
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_p = val)
```

```r
rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(idx  = m - 1,
         Tt   = ((idx - 1) %/% 5) + 1,   # which timepoint block
         k    = ((idx - 1) %%  5) + 1,   # which spline column within the block
         name = paste0("X", k)) |>
  dplyr::select(p, Tt, name, dev = val)

# Population cell spline contribution over the grid
pop_spl <- B_pred_long |>
  inner_join(b_g_mean, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(g, min_seq), mu_pop_spl = sum(B * b_g)) |>
  inner_join(a_g_mean, by = "g") |>
  mutate(mu_pop = a_g + mu_pop_spl)

# Participant spline-deviation contribution over the grid (per p and timepoint)
dev_spl <- B_pred_long |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, Tt, min_seq), mu_dev_spl = sum(B * dev)) |>
  inner_join(rp_int, by = "p") |>
  mutate(mu_dev = a_p + mu_dev_spl)

# Combine: each participant p at time Tt sits in the cell g for their treatment
part_curves <- part_key |>
  inner_join(pop_spl, by = "g", relationship = "many-to-many") |>
  inner_join(dev_spl, by = c("p", "Tt", "min_seq")) |>
  mutate(mu = (mu_pop + mu_dev) * sdglucose + mglucose)


# Plot

ggplot(pop_curve, aes(min_seq, m, color = treat, fill = treat)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = interaction(p, treat)),
            alpha = 0.25, linewidth = 0.3) +
  geom_line(linewidth = 1) +
  facet_wrap(~ time) +
  alpha_theme() +
  scale_color_manual(values = colscale) +
  scale_fill_manual(values = colscale) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown())
```

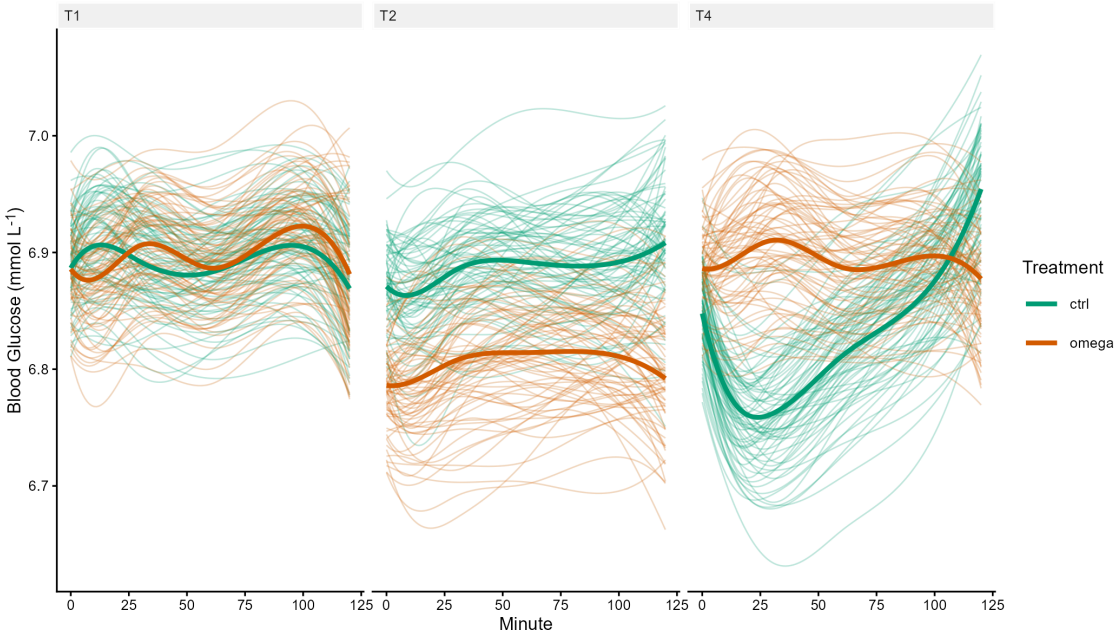

Figure 10.6: Prior predictive simulation of the `mo3` model. The plot shows individual curves and implied group means for each time point.

Next we fit the full data set to the model to get the posterior distribution. Posterior predictive checks indicate that the model capture relevant information from the data (Figure 10.7).

```
dat$prior_only <- 0

fit_wrap(mo3, fit = "fit.mo3", DATA = dat)
```

```
# Posterior draws of the parameters needed for curves
draws <- fit.mo3$draws(variables = c("a_g", "b_g", "r_p"), format = "df")

# Basis evaluated on a dense grid (B was built as bs(full_dat$min, df = 5))
min_seq <- seq(from = 0, to = 120, by = 1)

B_pred_long <- predict(B, min_seq) |>
  data.frame() |>
  cbind(min_seq) |>
  pivot_longer(starts_with("X"), values_to = "B")

cell_key <- full_dat |>
  distinct(time, treat, G, Tt) |>
  rename(g = G)

part_key <- full_dat |>
  distinct(P, Tt, G, time, treat) |>
  rename(p = P, g = G)

# Population-average curves (full posterior -> credible intervals)
ag_long <- draws |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  summarise(.by = c(g, min_seq),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(cell_key, by = "g")

# Participant-level curves (posterior-mean random effects)
a_g_mean <- ag_long |>
  summarise(.by = g, a_g = mean(a_g))

b_g_mean <- bg_long |>
  summarise(.by = c(g, name), b_g = mean(b_g))

rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_p = val)

rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(idx  = m - 1,
         Tt   = ((idx - 1) %/% dat$K) + 1,   # which timepoint block
         k    = ((idx - 1) %%  dat$K) + 1,   # which spline column within the block
         name = paste0("X", k)) |>
  dplyr::select(p, Tt, name, dev = val)

pop_spl <- B_pred_long |>
  inner_join(b_g_mean, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(g, min_seq), mu_pop_spl = sum(B * b_g)) |>
  inner_join(a_g_mean, by = "g") |>
  mutate(mu_pop = a_g + mu_pop_spl)

dev_spl <- B_pred_long |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, Tt, min_seq), mu_dev_spl = sum(B * dev)) |>
  inner_join(rp_int, by = "p") |>
```

```
  mutate(mu_dev = a_p + mu_dev_spl)

part_curves <- part_key |>
  inner_join(pop_spl, by = "g", relationship = "many-to-many") |>
  inner_join(dev_spl, by = c("p", "Tt", "min_seq")) |>
  mutate(mu = (mu_pop + mu_dev) * sdglucose + mglucose)


# Population ribbon + per-participant spaghetti, per time x treatment
p1 <- ggplot(pop_curve, aes(min_seq, m, color = treat, fill = treat)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = p),
            alpha = 0.2, linewidth = 0.3) +
  geom_ribbon(aes(ymin = lwr, ymax = upr),
              alpha = 0.2, color = NA) +
  geom_line(linewidth = 1) +
  facet_wrap(~ time) +
  alpha_theme() +
  scale_color_manual(values = colscale[c(2, 4)]) +
  scale_fill_manual(values = colscale[c(2, 4)]) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown())


# Individual curves examples (observed data as points, missing are not shown)
ex_p <- c(1, 2, 3)

p2 <- ggplot(filter(pop_curve, treat %in% unique(filter(part_key, p %in% ex_p)$treat)),
             aes(min_seq, m)) +
  geom_ribbon(aes(ymin = lwr, ymax = upr, group = treat),
              alpha = 0.2) +
  geom_line(aes(group = treat, lty = treat), linewidth = 1) +
  geom_line(data = filter(part_curves, p %in% ex_p),
            aes(min_seq, mu, group = p, color = as.factor(p)),
            alpha = 0.5, linewidth = 1) +
  geom_point(data = filter(full_dat, P %in% ex_p, !is.na(glucose)),
             aes(min, glucose, color = as.factor(P)),
             size = 2) +
  facet_wrap(~ time) +
  alpha_theme() +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Participant ID",
       lty = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown()) +
  scale_color_manual(values = colscale)


# Residuals vs. predicted (observed data points only)
obs_idx <- which(full_dat$y_miss == 1)

y_pred <- fit.mo3$draws("y_pred", format = "matrix")[, obs_idx]

resid  <- sweep(-y_pred, 2, full_dat$y_obs[obs_idx], "+")

p3 <- data.frame(m_fitted = colMeans(y_pred),
                 s_fitted = apply(y_pred, 2, sd),
                 m_resid  = colMeans(resid),
```

```
                s_resid  = apply(resid, 2, sd)) |>
  ggplot(aes(m_fitted, m_resid)) +
  geom_errorbar(aes(ymin = m_resid - s_resid,
                    ymax = m_resid + s_resid),
                alpha = 0.3) +
  geom_errorbar(aes(xmin = m_fitted - s_fitted,
                    xmax = m_fitted + s_fitted),
                orientation = "y",
                alpha = 0.3) +
  geom_point(alpha = 0.3,
             size = 3) +
  geom_smooth(color = colscale[3]) +
  geom_hline(yintercept = 0, lty = 2, alpha = 0.4) +
  alpha_theme() +
  labs(x = "Predicted values (standardized glucose)",
       y = "Observed - Predicted values")


# Simulated vs. observed distributions (observed data points only)
p4 <- fit.mo3$draws("y_rep", format = "df") |>
  filter(.draw %in% 1:40) |>
  dplyr::select(.draw, starts_with("y_rep")) |>
  pivot_longer(cols = starts_with("y_rep")) |>
  mutate(i = as.integer(gsub("y_rep\\[(\\d+)\\]", "\\1", name))) |>
  filter(i %in% obs_idx) |>
  ggplot(aes(value, group = .draw)) +
  geom_line(stat = "density", alpha = 0.3) +
  geom_line(data = filter(full_dat, y_miss == 1),
            aes(y_obs, group = NULL),
            color = colscale[3],
            linewidth = 1,
            stat = "density") +
  alpha_theme(yaxis = FALSE) +
  labs(x = "Standardized glucose (simulated and observed)",
       y = "")
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
cowplot::plot_grid(p1, p2, p3, p4,
                   ncol = 2,
                   labels = c("A", "B", "C", "D"))
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

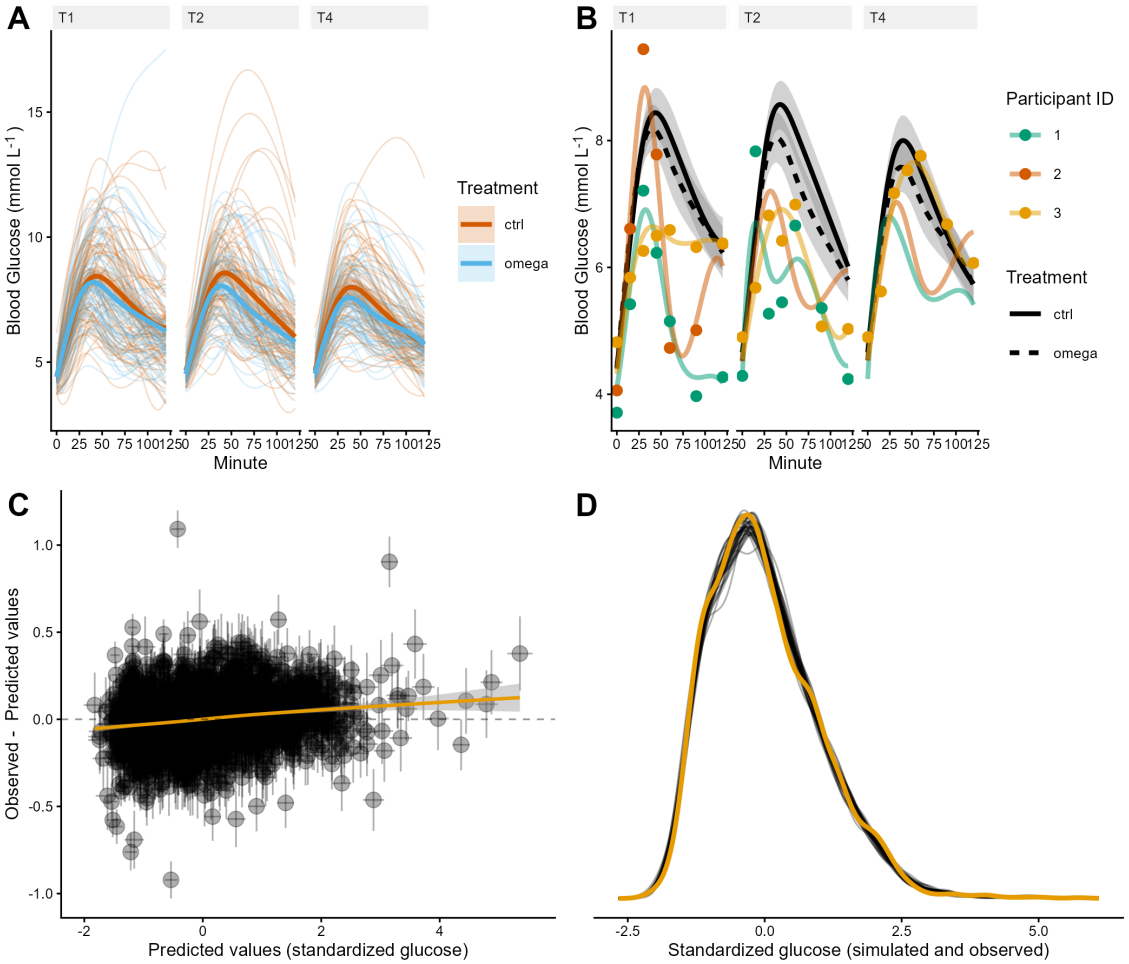

Figure 10.7: Posterior predictive checks of model `mo3`. Population average curves per time × treatment with participant-level curves (A), three example participants with observed data (B), residuals vs. predicted values for observed data points (C) and distributions of simulated (yrep) data sets and the observed data (D).

10.3.4 Inference

We extract two sets of fitted curves from model `mo3`, the population-average glucose curve for every time x treatment combination (`a_g` + cell spline `b_g`), with 95% percentile intervals from the full posterior, and the participant-level curves, which add each participant's random intercept (`r_p[p, 1]`) and time-specific random spline deviations to the population cell they belong to.

```
draws <- fit.mo3$draws(format = "df")

# Basis evaluated on a dense grid (B was built as bs(full_dat$min, df = 5))
min_seq <- seq(from = 0, to = 120, by = 1)

B_pred_long <- predict(B, min_seq) |>
  data.frame() |>
  cbind(min_seq) |>
  pivot_longer(starts_with("X"), values_to = "B")

# Map cell index g and time index Tt back to time / treatment labels, and
# record which cell g each participant (P) belongs to at each time.
cell_key <- full_dat |>
  distinct(time, treat, G, Tt) |>
```

```
  rename(g = G)

part_key <- full_dat |>
  distinct(P, Tt, G, time, treat) |>
  rename(p = P, g = G)


# Population-average curves (full posterior -> credible intervals)

ag_long <- draws |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  # This reverse the standardization (m and s are from the full data set above)
  mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  summarise(.by = c(g, min_seq),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(cell_key, by = "g")




# Group-level predictions - These are implied group level glucose curves
# at the three time points

pred <- B_pred_long |>

  inner_join(bg_long, by = "name", relationship = "many-to-many") |>

  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  # back transform the standardization
    mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  inner_join(cell_key) |>
  mutate(time_treat = paste0(time,"_",treat)) |>
  dplyr::select(.draw, min_seq, mu, time_treat)
```

```
Joining with `by = join_by(g)`
```

```
# Calculate contrasts at measured time points

within_contrasts <- pred |>
  pivot_wider(values_from = mu,
              names_from = time_treat) |>
  # Calculate contrasts
  mutate(ctrl_t2 = T2_ctrl - T1_ctrl,
         ctrl_t4 = T4_ctrl - T1_ctrl,
         omega_t2 = T2_omega - T1_omega,
         omega_t4 =  T4_omega - T1_omega,
         ) |>
  select(.draw, min_seq, ctrl_t2:omega_t4) |>

  pivot_longer(cols = ctrl_t2:omega_t4) |>

  filter(min_seq %in% c(15, 30, 45, 60, 60, 90, 120)) |>

  summarise(.by = c(name,min_seq),
            m = median(value),
            lwr50 = quantile(value, 0.25),
            upr50 = quantile(value, 0.75),
            lwr95 = quantile(value, 0.025),
            upr95 = quantile(value, 0.975)) |>

  separate(name, into = c("treat", "time"), sep = "_") |>
  mutate(time = case_when(time == "t2" ~ "Supplementation vs. Baseline",
                          time == "t4" ~ "Combined vs. Baseline"),
         time = factor(time, levels = c("Supplementation vs. Baseline",
                                        "Combined vs. Baseline")),
         treat = factor(treat, levels = c("ctrl", "omega"),
                        labels = c("Placebo change from baseline", "Krill oil change
from baseline")))


pd <- position_dodge(width = 3)

p_wcontr <- within_contrasts |>
  ggplot(aes(min_seq, m, color = time, fill = time)) +

  geom_hline(yintercept = 0, lty = 2, color = "gray") +

  geom_errorbar(aes(ymin = lwr50,
                    ymax = upr50),
                linewidth = 1.2, width = 0,
                position = pd) +
  geom_errorbar(aes(ymin = lwr95,
                    ymax = upr95),
                linewidth = 0.6, width = 0,
                position = pd) +
  geom_point(shape = 21, color = "black",
                position = pd) +

  scale_color_manual(values = colscale[c(3,6)]) +
 scale_fill_manual(values = colscale[c(3,6)]) +
    scale_x_continuous(limits = c(0, 125),
                                expand = c(0, 0)) +

  labs(x = "Time (min)",
```

```
       y = " \u0394Blood\u0020 glucose<br>(mmol L<sup>\u22121</sup>)") +


  facet_wrap(~ treat) +
  alpha_theme() +
    theme(axis.title.y = ggtext::element_markdown(),
        legend.title = element_blank(),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        axis.text.x = element_blank(),
        axis.title.x = element_blank(),
        strip.text = element_markdown(size = 7))




p_pred <- pred |>
  summarise(.by = c(time_treat, min_seq),
            m = median(mu),
            lwr50 = quantile(mu, 0.25),
            upr50 = quantile(mu, 0.75),
            lwr95 = quantile(mu, 0.025),
            upr95 = quantile(mu, 0.975)) |>

  separate(time_treat, into = c("time", "treat"), sep = "_") |>
  mutate(time = case_when(time == "T1" ~ "Baseline",
                          time == "T2" ~ "Supplementation",
                          time == "T4" ~ "Combined"),
         time = factor(time, levels = c("Baseline", "Supplementation",
                                        "Combined")),
         treat = factor(treat, levels = c("ctrl", "omega"),
                        labels = c("Placebo predictions", "Krill oil predictions")))
|>

  ggplot(aes(min_seq, m, lty = time, color = treat)) +
  geom_line() +
  facet_wrap(~ treat, ncol = 2) +
  alpha_theme() +
  labs(x = "Time (min)",
       y = "Blood glucose<br>(mmol L<sup>-1</sup>)") +

  scale_x_continuous(limits = c(0, 125),
                                expand = c(0, 0)) +

  scale_color_manual(values = colscale[c(4,2)],
                     guide = "none") +
  theme(axis.title.y = ggtext::element_markdown(),
        legend.title = element_blank(),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        axis.text.x = element_blank(),
        axis.title.x = element_blank())



# Contrasts - These are differences in delta from baseline
```

```
delta_contrasts <- pred |>
  pivot_wider(values_from = mu,
              names_from = time_treat)  |>
  # Calculate contrasts
  mutate(ctrl_t2 = T2_ctrl - T1_ctrl,
         ctrl_t4 = T4_ctrl - T1_ctrl,
         omega_t2 = T2_omega - T1_omega,
         omega_t4 =  T4_omega - T1_omega,
         omega_ctrl_t2 = omega_t2 - ctrl_t2,
         omega_ctrl_t4 = omega_t4 - ctrl_t4,
         ) |>
  pivot_longer(names_to = "contrast",
               values_to = "value",
               cols = T1_ctrl:omega_ctrl_t4) |>
  summarise(.by = c(min_seq, contrast),
            m = mean(value),
            lwr50 = quantile(value, 0.25),
            upr50 = quantile(value, 0.75),
            lwr95 = quantile(value, 0.025),
            upr95 = quantile(value, 0.975)) |>
  filter(contrast %in% c("omega_ctrl_t2", "omega_ctrl_t4")) |>
  mutate(contrast = case_when(contrast == "omega_ctrl_t2" ~ "Placebo vs. Krill oil:
Supplementation period",
                              contrast == "omega_ctrl_t4" ~ "Placebo vs. Krill oil:
Combined period"),
         contrast = factor(contrast, levels = c("Placebo vs. Krill oil: Supplementation
period",
                                                "Placebo vs. Krill oil: Combined
period"))) |>

  ggplot(aes(min_seq, m, fill = contrast)) +
    scale_x_continuous(limits = c(0, 125),
                       expand = c(0, 0),
                       breaks = c(25, 50, 75, 100)) +
  geom_ribbon(aes(ymin = lwr50, ymax = upr50),
              alpha = 0.3) +
    geom_ribbon(aes(ymin = lwr95, ymax = upr95),
              alpha = 0.3) +
  geom_line() +
  facet_wrap(~ contrast) +
  alpha_theme() +
  scale_fill_manual(values = colscale[c(3,6)]) +
  geom_hline(yintercept = 0, lty = 2, alpha = 0.5) +
  theme(legend.position = "none") +
  labs(x = "Time (min)",
       y = "&Delta;Krill oil \u2212 &Delta;Placebo<br>(mmol L<sup>-1</sup>)")




# Combine the full figure

ogtt <- cowplot::plot_grid(p_pred,
                   p_wcontr,
                   delta_contrasts,
                   ncol = 1,
                   rel_heights = c(1, 1, 1.4),
                   align = "v")
```

```
ggsave(
  filename = "figures/preliminary-figures/ogtt.pdf",
  ogtt,
  device = cairo_pdf,
  width = 18,
  height = 10,
  units = "cm"
)

ogtt
```

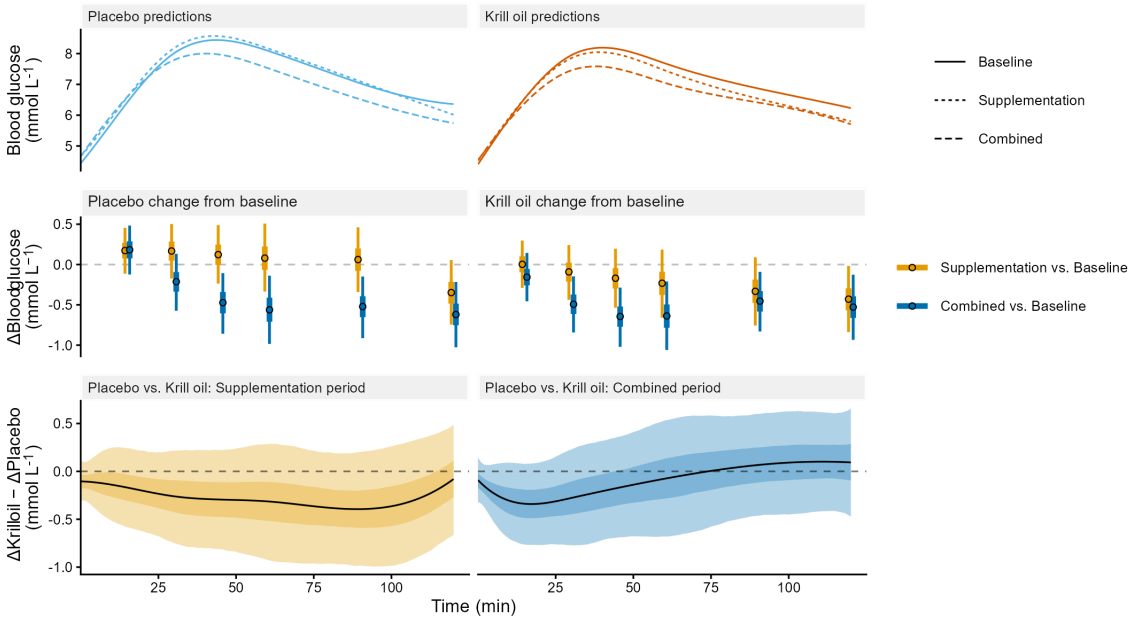

Figure 10.8: Predictions (top panels), within-group time-contrasts (middle panels) and between-group contrasts of changes (Δ) from baseline. Error bars and ribbons represent 50 and 95% intervals in the middle and lower panels.

```
# Participant-level curves (posterior-mean random effects)

# Posterior-mean population coefficients per cell
a_g_mean <- ag_long |>
  summarise(.by = g, a_g = mean(a_g))

b_g_mean <- bg_long |>
  summarise(.by = c(g, name), b_g = mean(b_g))

# Posterior-mean participant random effects r_p[p, m]
rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Intercept (column 1) and time-specific spline deviations (columns 2..M)
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_p = val)

rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(idx  = m - 1,
         Tt   = ((idx - 1) %/% 5) + 1,   # which timepoint block
         k    = ((idx - 1) %%  5) + 1,   # which spline column within the block
         name = paste0("X", k)) |>
  dplyr::select(p, Tt, name, dev = val)

# Population cell spline contribution over the grid
pop_spl <- B_pred_long |>
  inner_join(b_g_mean, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(g, min_seq), mu_pop_spl = sum(B * b_g)) |>
  inner_join(a_g_mean, by = "g") |>
  mutate(mu_pop = a_g + mu_pop_spl)

# Participant spline-deviation contribution over the grid (per p and timepoint)
dev_spl <- B_pred_long |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, Tt, min_seq), mu_dev_spl = sum(B * dev)) |>
  inner_join(rp_int, by = "p") |>
  mutate(mu_dev = a_p + mu_dev_spl)

# Combine: each participant p at time Tt sits in the cell g for their treatment
part_curves <- part_key |>
  inner_join(pop_spl, by = "g", relationship = "many-to-many") |>
  inner_join(dev_spl, by = c("p", "Tt", "min_seq")) |>
  mutate(mu = (mu_pop + mu_dev) * sdglucose + mglucose)


# Plot

ggplot(pop_curve, aes(min_seq, m, color = treat, fill = treat)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = interaction(p, treat)),
            alpha = 0.25, linewidth = 0.3) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), alpha = 0.2, color = NA) +
  geom_line(linewidth = 1) +
  facet_wrap(~ time) +
  alpha_theme() +
  scale_color_manual(values = colscale) +
  scale_fill_manual(values = colscale) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown())
```

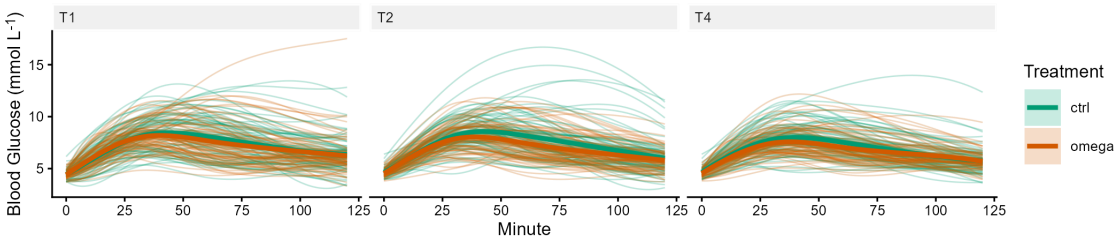

Figure 10.9: Predictions from the OGTT model `mo3` at each time point with participant-level predictions.

10.4 Model expansion: Participant-level time-specific intercepts.

In model `mo3` the participant-level intercept (a_p) is common to all three time points, while the participant-level spline deviations are time-specific. As the basis B has no intercept column, all basis functions are zero at the start of the test (min 0). The participant-level deviation from the population curve in fasting glucose is therefore constrained to be equal at all time points in `mo3`. Model `mo4` (`models/mo4.stan`) relaxes this assumption by decomposing the participant-level intercept at each time point into a shared intercept (a_p) and a time-specific deviation ($\delta_{t,p}$), otherwise the model is identical to `mo3` (including missing data imputation):

$$\begin{aligned} \mu_i &= \left(a_{g[i]} + \sum_k b_{k,g[i]}\, B_{k,i} \right) + \left(a_{p[i]} + \delta_{t[i],p[i]} + \sum_k \beta_{k,t[i],p[i]}\, B_{k,i} \right) \\ (a_p, \beta_p)^\top &\sim \text{MVNormal}\ (\mathbf{0}, \mathbf{S}\,\mathbf{R}\,\mathbf{S}) \\ \delta_{t,p} &\sim \text{Normal}\ (0, \tau_\delta) \\ \tau_\delta &\sim \text{Exponential}\ (2) \end{aligned} \tag{10.3}$$

The correlated participant block (a_p and the time-specific spline deviations) is the same as in `mo3`, and the time-specific intercept deviations are independent of this block with a common standard deviation (τ_δ). Model `mo3` is the special case $\tau_\delta = 0$, and the Exponential (2) prior regularizes τ_δ towards this simpler model. The implied correlation between a participant's intercepts at two time points is $\rho = \sigma_a^2/(\sigma_a^2 + \tau_\delta^2)$, where σ_a is the standard deviation of a_p.

We chose this decomposition over a fully correlated block with one intercept per time point, as an LKJ prior on the resulting 18×18 correlation matrix implies a marginal prior on each correlation that strongly favors values close to zero (approximately ± 0.45 95% prior interval for LKJ(1)), regardless of the choice of the LKJ shape parameter. Other priors are the same as in Equation (10.2) and **?@eq-mo3**. The prior predictive simulation is shown in Figure 10.10.

```
# Same data as for mo3 (full_dat, B, dat), prior only
dat4 <- dat
dat4$prior_only <- 1

mo4 <- cmdstan_model(stan_file = "models/mo4.stan",
                     exe_file = "models/mo4.exe")

fit_wrap(mo4, fit = "fit.mo4.pp", DATA = dat4)

draws <- fit.mo4.pp$draws(variables = c("a_g", "b_g", "r_p", "delta"), format = "df")

# Basis evaluated on a dense grid (B was built as bs(full_dat$min, df = 5))
min_seq <- seq(from = 0, to = 120, by = 1)

B_pred_long <- predict(B, min_seq) |>
  data.frame() |>
  cbind(min_seq) |>
  pivot_longer(starts_with("X"), values_to = "B")

cell_key <- full_dat |>
  distinct(time, treat, G, Tt) |>
  rename(g = G)

part_key <- full_dat |>
  distinct(P, Tt, G, time, treat) |>
  rename(p = P, g = G)

# Population-average curves (full posterior -> credible intervals)
ag_long <- draws |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
```

```
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  summarise(.by = c(g, min_seq),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(cell_key, by = "g")

# Participant-level curves (posterior-mean random effects)
a_g_mean <- ag_long |>
  summarise(.by = g, a_g = mean(a_g))

b_g_mean <- bg_long |>
  summarise(.by = c(g, name), b_g = mean(b_g))

rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Time-specific participant intercept deviations delta[t, p]
delta_mean <- draws |>
  dplyr::select(starts_with("delta[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "delta") |>
  mutate(Tt = as.integer(gsub("delta\\[(\\d+),(\\d+)\\]", "\\1", par)),
         p  = as.integer(gsub("delta\\[(\\d+),(\\d+)\\]", "\\2", par))) |>
  dplyr::select(p, Tt, delta)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Shared intercept (column 1) + time-specific deviation, and time-specific
# spline deviations (columns 2..M, blocks of K per time point as in mo3)
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_shared = val) |>
  inner_join(delta_mean, by = "p") |>
```

```
  mutate(a_p = a_shared + delta) |>
  dplyr::select(p, Tt, a_p)

rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(idx  = m - 1,
         Tt   = ((idx - 1) %/% dat$K) + 1,
         k    = ((idx - 1) %%  dat$K) + 1,
         name = paste0("X", k)) |>
  dplyr::select(p, Tt, name, dev = val)

pop_spl <- B_pred_long |>
  inner_join(b_g_mean, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(g, min_seq), mu_pop_spl = sum(B * b_g)) |>
  inner_join(a_g_mean, by = "g") |>
  mutate(mu_pop = a_g + mu_pop_spl)

dev_spl <- B_pred_long |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, Tt, min_seq), mu_dev_spl = sum(B * dev)) |>
  inner_join(rp_int, by = c("p", "Tt")) |>
  mutate(mu_dev = a_p + mu_dev_spl)

part_curves <- part_key |>
  inner_join(pop_spl, by = "g", relationship = "many-to-many") |>
  inner_join(dev_spl, by = c("p", "Tt", "min_seq")) |>
  mutate(mu = (mu_pop + mu_dev) * sdglucose + mglucose)

# Plot
ggplot(pop_curve, aes(min_seq, m, color = treat, fill = treat)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = interaction(p, treat)),
            alpha = 0.25, linewidth = 0.3) +
  geom_line(linewidth = 1) +
  facet_wrap(~ time) +
  alpha_theme() +
  scale_color_manual(values = colscale) +
  scale_fill_manual(values = colscale) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown())
```

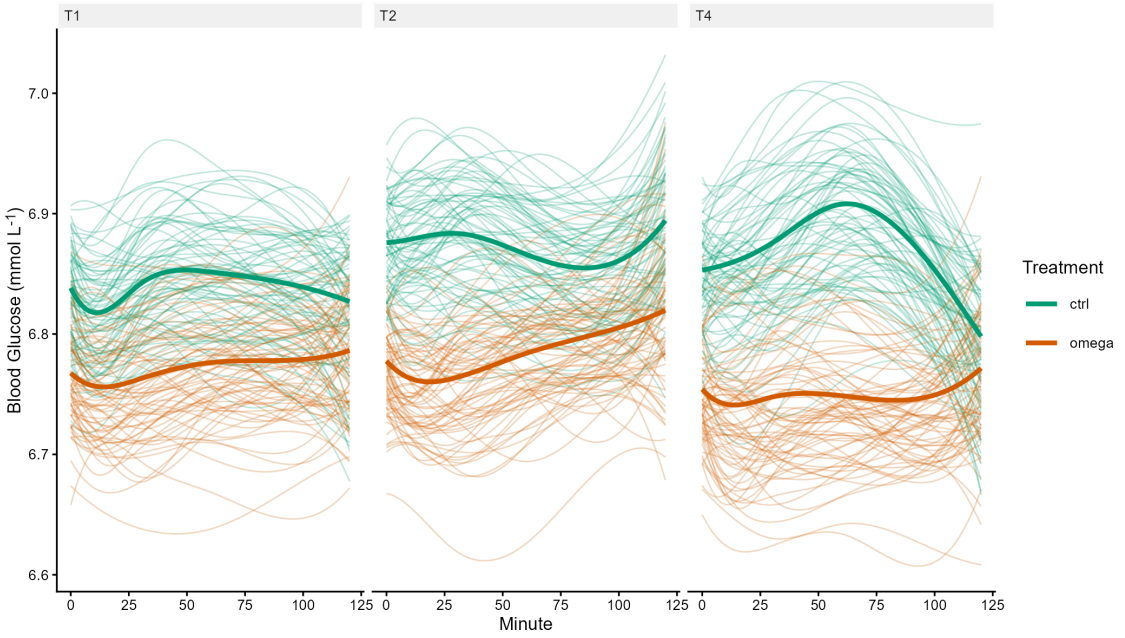

Figure 10.10: Prior predictive simulation of the `mo4` model. The plot shows individual curves and implied group means for each time point.

Next we fit the model to the full data set and perform the same posterior predictive checks as for `mo3` (Figure 10.11).

```
dat4$prior_only <- 0

fit_wrap(mo4, fit = "fit.mo4", DATA = dat4)
```

```
# Posterior draws of the parameters needed for curves
draws <- fit.mo4$draws(variables = c("a_g", "b_g", "r_p", "delta"), format = "df")

# Population-average curves (full posterior -> credible intervals)
ag_long <- draws |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
pop_curve <- B_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
```

```
  inner_join(ag_long, by = c(".draw", "g")) |>
  mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  summarise(.by = c(g, min_seq),
            m   = mean(mu),
            lwr = quantile(mu, 0.025),
            upr = quantile(mu, 0.975)) |>
  left_join(cell_key, by = "g")

# Participant-level curves (posterior-mean random effects)
a_g_mean <- ag_long |>
  summarise(.by = g, a_g = mean(a_g))

b_g_mean <- bg_long |>
  summarise(.by = c(g, name), b_g = mean(b_g))

rp_mean <- draws |>
  dplyr::select(starts_with("r_p[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "val") |>
  mutate(p = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\1", par)),
         m = as.integer(gsub("r_p\\[(\\d+),(\\d+)\\]", "\\2", par)))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
delta_mean <- draws |>
  dplyr::select(starts_with("delta[")) |>
  summarise(across(everything(), mean)) |>
  pivot_longer(everything(), names_to = "par", values_to = "delta") |>
  mutate(Tt = as.integer(gsub("delta\\[(\\d+),(\\d+)\\]", "\\1", par)),
         p  = as.integer(gsub("delta\\[(\\d+),(\\d+)\\]", "\\2", par))) |>
  dplyr::select(p, Tt, delta)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Shared intercept (column 1) + time-specific deviation, and time-specific
# spline deviations (columns 2..M, blocks of K per time point as in mo3)
rp_int <- rp_mean |> filter(m == 1) |> dplyr::select(p, a_shared = val) |>
  inner_join(delta_mean, by = "p") |>
  mutate(a_p = a_shared + delta) |>
  dplyr::select(p, Tt, a_p)

rp_spl <- rp_mean |> filter(m >= 2) |>
  mutate(idx  = m - 1,
         Tt   = ((idx - 1) %/% dat$K) + 1,
         k    = ((idx - 1) %%  dat$K) + 1,
         name = paste0("X", k)) |>
  dplyr::select(p, Tt, name, dev = val)

pop_spl <- B_pred_long |>
  inner_join(b_g_mean, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(g, min_seq), mu_pop_spl = sum(B * b_g)) |>
  inner_join(a_g_mean, by = "g") |>
  mutate(mu_pop = a_g + mu_pop_spl)

dev_spl <- B_pred_long |>
  inner_join(rp_spl, by = "name", relationship = "many-to-many") |>
  summarise(.by = c(p, Tt, min_seq), mu_dev_spl = sum(B * dev)) |>
  inner_join(rp_int, by = c("p", "Tt")) |>
  mutate(mu_dev = a_p + mu_dev_spl)
```

```
part_curves <- part_key |>
  inner_join(pop_spl, by = "g", relationship = "many-to-many") |>
  inner_join(dev_spl, by = c("p", "Tt", "min_seq")) |>
  mutate(mu = (mu_pop + mu_dev) * sdglucose + mglucose)


# Population ribbon + per-participant spaghetti, per time x treatment
p1 <- ggplot(pop_curve, aes(min_seq, m, color = treat, fill = treat)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = p),
            alpha = 0.2, linewidth = 0.3) +
  geom_ribbon(aes(ymin = lwr, ymax = upr),
              alpha = 0.2, color = NA) +
  geom_line(linewidth = 1) +
  facet_wrap(~ time) +
  alpha_theme() +
  scale_color_manual(values = colscale[c(2, 4)]) +
  scale_fill_manual(values = colscale[c(2, 4)]) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown())


# Individual curves examples (observed data as points, missing are not shown)
ex_p <- c(1, 2, 3)

p2 <- ggplot(filter(pop_curve, treat %in% unique(filter(part_key, p %in% ex_p)$treat)),
             aes(min_seq, m)) +
  geom_ribbon(aes(ymin = lwr, ymax = upr, group = treat),
              alpha = 0.2) +
  geom_line(aes(group = treat, lty = treat), linewidth = 1) +
  geom_line(data = filter(part_curves, p %in% ex_p),
            aes(min_seq, mu, group = p, color = as.factor(p)),
            alpha = 0.5, linewidth = 1) +
  geom_point(data = filter(full_dat, P %in% ex_p, !is.na(glucose)),
             aes(min, glucose, color = as.factor(P)),
             size = 2) +
  facet_wrap(~ time) +
  alpha_theme() +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Participant ID",
       lty = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown()) +
  scale_color_manual(values = colscale)


# Residuals vs. predicted (observed data points only)
obs_idx <- which(full_dat$y_miss == 1)

y_pred <- fit.mo4$draws("y_pred", format = "matrix")[, obs_idx]

resid  <- sweep(-y_pred, 2, full_dat$y_obs[obs_idx], "+")

p3 <- data.frame(m_fitted = colMeans(y_pred),
                 s_fitted = apply(y_pred, 2, sd),
                 m_resid  = colMeans(resid),
                 s_resid  = apply(resid, 2, sd)) |>
```

```
  ggplot(aes(m_fitted, m_resid)) +
  geom_errorbar(aes(ymin = m_resid - s_resid,
                    ymax = m_resid + s_resid),
                alpha = 0.3) +
  geom_errorbar(aes(xmin = m_fitted - s_fitted,
                    xmax = m_fitted + s_fitted),
                orientation = "y",
                alpha = 0.3) +
  geom_point(alpha = 0.3,
             size = 3) +
  geom_smooth(color = colscale[3]) +
  geom_hline(yintercept = 0, lty = 2, alpha = 0.4) +
  alpha_theme() +
  labs(x = "Predicted values (standardized glucose)",
       y = "Observed - Predicted values")


# Simulated vs. observed distributions (observed data points only)
p4 <- fit.mo4$draws("y_rep", format = "df") |>
  filter(.draw %in% 1:40) |>
  dplyr::select(.draw, starts_with("y_rep")) |>
  pivot_longer(cols = starts_with("y_rep")) |>
  mutate(i = as.integer(gsub("y_rep\\[(\\d+)\\]", "\\1", name))) |>
  filter(i %in% obs_idx) |>
  ggplot(aes(value, group = .draw)) +
  geom_line(stat = "density", alpha = 0.3) +
  geom_line(data = filter(full_dat, y_miss == 1),
            aes(y_obs, group = NULL),
            color = colscale[3],
            linewidth = 1,
            stat = "density") +
  alpha_theme(yaxis = FALSE) +
  labs(x = "Standardized glucose (simulated and observed)",
       y = "")
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
cowplot::plot_grid(p1, p2, p3, p4,
                   ncol = 2,
                   labels = c("A", "B", "C", "D"))
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

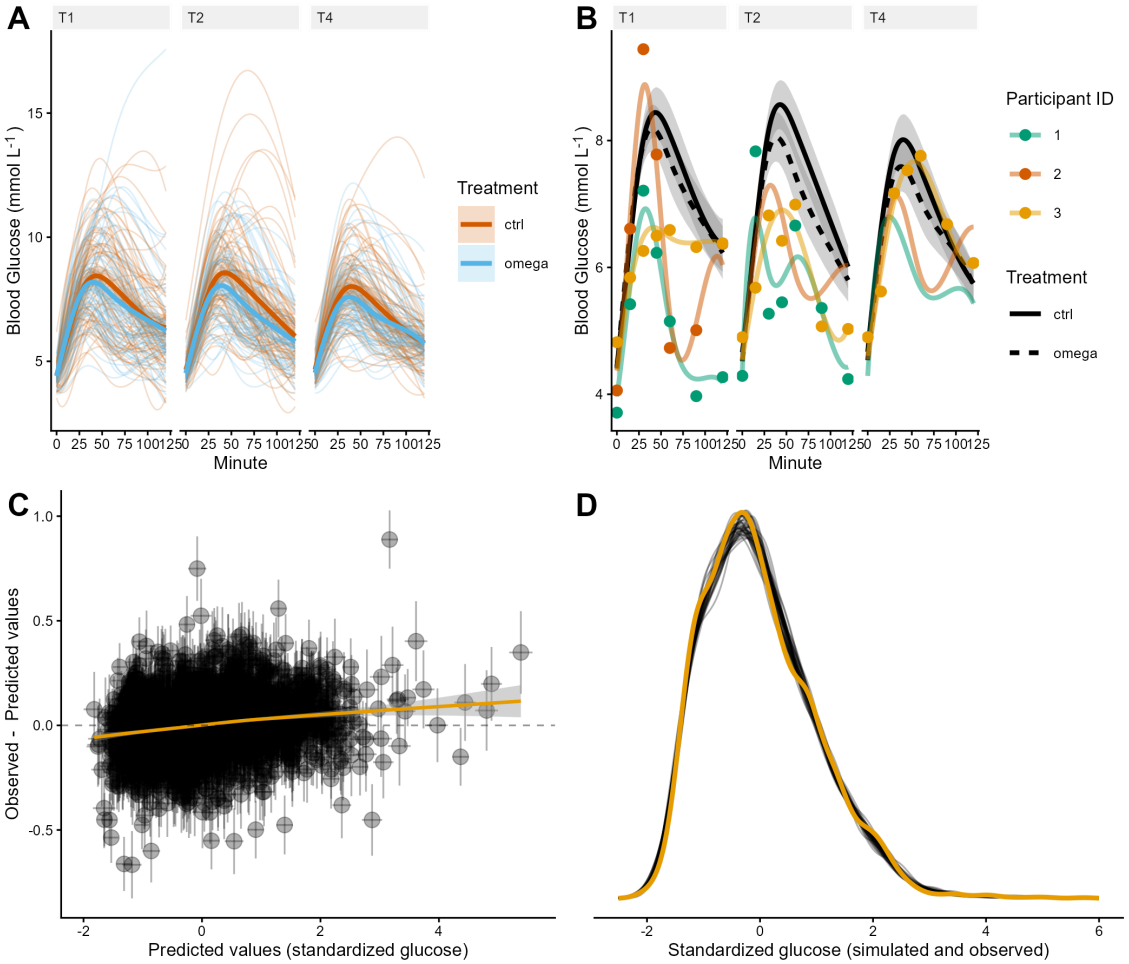

Figure 10.11: Posterior predictive checks of model `mo4`. Population average curves per time × treatment with participant-level curves (A), three example participants with observed data (B), residuals vs. predicted values for observed data points (C) and distributions of simulated (yrep) data sets and the observed data (D).

If time-specific participant intercepts are not needed, the standard deviation of the time-specific deviations (τ_δ) should be close to zero and the implied correlation between a participant's intercepts at two time points (ρ) close to one, i.e. participants keep their relative fasting level across time points. Prior and posterior summaries of these parameters are shown in Table 10.1.

```
int_pars <- c("sd_p[1]", "tau_delta", "rho_int")

bind_rows(
  fit.mo4.pp$summary(variables = int_pars,
                     mean,
                     ~ quantile(.x, probs = c(0.025, 0.975))) |>
    mutate(dist = "Prior"),
  fit.mo4$summary(variables = int_pars,
                  mean,
                  ~ quantile(.x, probs = c(0.025, 0.975))) |>
    mutate(dist = "Posterior")) |>
  mutate(variable = factor(variable,
                           levels = int_pars,
                           labels = c("SD shared intercept",
                                      "SD time-specific deviations",
                                      "Implied intercept correlation"))) |>
```

```
  pivot_wider(names_from = dist,
              values_from = c(mean, `2.5%`, `97.5%`),
              names_glue = "{dist}_{.value}") |>
  dplyr::select(variable, starts_with("Prior"), starts_with("Posterior")) |>
  gt::gt() |>
  gt::fmt_number(columns = -variable, decimals = 2) |>
  gt::tab_spanner_delim(delim = "_") |>
  gt::cols_label(variable = "")
```

Table 10.1: Prior and posterior summaries (mean and 95% percentile intervals) of the standard deviation of the shared participant-level intercept (σ_a), the standard deviation of time-specific intercept deviations (τ_δ) and the implied correlation between a participant's intercepts at two time points (ρ) in model `mo4`.

	Prior			Posterior		
	mean	2.5%	97.5%	mean	2.5%	97.5%
SD shared intercept	1.00	0.03	3.69	0.20	0.16	0.24
SD time-specific deviations	0.49	0.01	1.82	0.12	0.07	0.16
Implied intercept correlation	0.66	0.00	1.00	0.73	0.56	0.89

10.4.1 Inference (`mo4`)

We perform the same inference as for `mo3` using model `mo4`. As the population-level part of the models is identical, differences between the models' implied group-level curves and contrasts are due to how participant-level deviations are partitioned across time points.

```
draws <- fit.mo4$draws(variables = c("a_g", "b_g"), format = "df")

ag_long <- draws |>
  dplyr::select(.draw, starts_with("a_g[")) |>
  pivot_longer(starts_with("a_g["), names_to = "par", values_to = "a_g") |>
  mutate(g = as.integer(gsub("a_g\\[(\\d+)\\]", "\\1", par))) |>
  dplyr::select(.draw, g, a_g)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
bg_long <- draws |>
  dplyr::select(.draw, starts_with("b_g[")) |>
  pivot_longer(starts_with("b_g["), names_to = "par", values_to = "b_g") |>
  mutate(k = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\1", par)),
         g = as.integer(gsub("b_g\\[(\\d+),(\\d+)\\]", "\\2", par)),
         name = paste0("X", k))
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
# Group-level predictions - These are implied group level glucose curves
# at the three time points

pred <- B_pred_long |>
  inner_join(bg_long, by = "name", relationship = "many-to-many") |>
```

```
  summarise(.by = c(.draw, g, min_seq),
            mu_spl = sum(B * b_g)) |>
  inner_join(ag_long, by = c(".draw", "g")) |>
  # back transform the standardization
  mutate(mu = (a_g + mu_spl) * sdglucose + mglucose) |>
  inner_join(cell_key) |>
  mutate(time_treat = paste0(time,"_",treat)) |>
  dplyr::select(.draw, min_seq, mu, time_treat)
```

```
Joining with `by = join_by(g)`
```

```
# Calculate contrasts at measured time points

within_contrasts <- pred |>
  pivot_wider(values_from = mu,
              names_from = time_treat) |>
  mutate(ctrl_t2 = T2_ctrl - T1_ctrl,
         ctrl_t4 = T4_ctrl - T1_ctrl,
         omega_t2 = T2_omega - T1_omega,
         omega_t4 =  T4_omega - T1_omega) |>
  select(.draw, min_seq, ctrl_t2:omega_t4) |>
  pivot_longer(cols = ctrl_t2:omega_t4) |>
  filter(min_seq %in% c(15, 30, 45, 60, 90, 120)) |>
  summarise(.by = c(name,min_seq),
            m = median(value),
            lwr50 = quantile(value, 0.25),
            upr50 = quantile(value, 0.75),
            lwr95 = quantile(value, 0.025),
            upr95 = quantile(value, 0.975)) |>
  separate(name, into = c("treat", "time"), sep = "_") |>
  mutate(time = case_when(time == "t2" ~ "Supplementation vs. Baseline",
                          time == "t4" ~ "Combined vs. Baseline"),
         time = factor(time, levels = c("Supplementation vs. Baseline",
                                        "Combined vs. Baseline")),
         treat = factor(treat, levels = c("ctrl", "omega"),
                        labels = c("Placebo change from baseline", "Krill oil change
from baseline")))


pd <- position_dodge(width = 3)

p_wcontr <- within_contrasts |>
  ggplot(aes(min_seq, m, color = time, fill = time)) +
  geom_hline(yintercept = 0, lty = 2, color = "gray") +
  geom_errorbar(aes(ymin = lwr50,
                    ymax = upr50),
                linewidth = 1.2, width = 0,
                position = pd) +
  geom_errorbar(aes(ymin = lwr95,
                    ymax = upr95),
                linewidth = 0.6, width = 0,
                position = pd) +
  geom_point(shape = 21, color = "black",
             position = pd) +
  scale_color_manual(values = colscale[c(3,6)]) +
  scale_fill_manual(values = colscale[c(3,6)]) +
  scale_x_continuous(limits = c(0, 125),
                     expand = c(0, 0)) +
  labs(x = "Time (min)",
       y = " ΔBlood  glucose<br>(mmol L<sup>-1</sup>)") +
```

```
  facet_wrap(~ treat) +
  alpha_theme() +
  theme(axis.title.y = ggtext::element_markdown(),
        legend.title = element_blank(),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        axis.text.x = element_blank(),
        axis.title.x = element_blank(),
        strip.text = element_markdown(size = 7))


p_pred <- pred |>
  summarise(.by = c(time_treat, min_seq),
            m = median(mu),
            lwr50 = quantile(mu, 0.25),
            upr50 = quantile(mu, 0.75),
            lwr95 = quantile(mu, 0.025),
            upr95 = quantile(mu, 0.975)) |>
  separate(time_treat, into = c("time", "treat"), sep = "_") |>
  mutate(time = case_when(time == "T1" ~ "Baseline",
                          time == "T2" ~ "Supplementation",
                          time == "T4" ~ "Combined"),
         time = factor(time, levels = c("Baseline", "Supplementation",
                                        "Combined")),
         treat = factor(treat, levels = c("ctrl", "omega"),
                        labels = c("Placebo predictions", "Krill oil predictions")))
|>
  ggplot(aes(min_seq, m, lty = time, color = treat)) +
  geom_line() +
  facet_wrap(~ treat, ncol = 2) +
  alpha_theme() +
  labs(x = "Time (min)",
       y = "Blood glucose<br>(mmol L<sup>-1</sup>)") +
  scale_x_continuous(limits = c(0, 125),
                     expand = c(0, 0)) +
  scale_color_manual(values = colscale[c(4,2)],
                     guide = "none") +
  theme(axis.title.y = ggtext::element_markdown(),
        legend.title = element_blank(),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        axis.text.x = element_blank(),
        axis.title.x = element_blank())


# Contrasts - These are differences in delta from baseline

delta_contrasts <- pred |>
  pivot_wider(values_from = mu,
              names_from = time_treat)  |>
  mutate(ctrl_t2 = T2_ctrl - T1_ctrl,
         ctrl_t4 = T4_ctrl - T1_ctrl,
         omega_t2 = T2_omega - T1_omega,
         omega_t4 =  T4_omega - T1_omega,
         omega_ctrl_t2 = omega_t2 - ctrl_t2,
         omega_ctrl_t4 = omega_t4 - ctrl_t4) |>
  pivot_longer(names_to = "contrast",
               values_to = "value",
               cols = T1_ctrl:omega_ctrl_t4) |>
  summarise(.by = c(min_seq, contrast),
```

```
            m = mean(value),
            lwr50 = quantile(value, 0.25),
            upr50 = quantile(value, 0.75),
            lwr95 = quantile(value, 0.025),
            upr95 = quantile(value, 0.975)) |>
  filter(contrast %in% c("omega_ctrl_t2", "omega_ctrl_t4")) |>
  mutate(contrast = case_when(contrast == "omega_ctrl_t2" ~ "Placebo vs. Krill oil:
Supplementation period",
                              contrast == "omega_ctrl_t4" ~ "Placebo vs. Krill oil:
Combined period"),
         contrast = factor(contrast, levels = c("Placebo vs. Krill oil: Supplementation
period",
                                                "Placebo vs. Krill oil: Combined
period"))) |>
  ggplot(aes(min_seq, m, fill = contrast)) +
  scale_x_continuous(limits = c(0, 125),
                     expand = c(0, 0),
                     breaks = c(25, 50, 75, 100)) +
  geom_ribbon(aes(ymin = lwr50, ymax = upr50),
              alpha = 0.3) +
  geom_ribbon(aes(ymin = lwr95, ymax = upr95),
              alpha = 0.3) +
  geom_line() +
  facet_wrap(~ contrast) +
  alpha_theme() +
  scale_fill_manual(values = colscale[c(3,6)]) +
  geom_hline(yintercept = 0, lty = 2, alpha = 0.5) +
  theme(legend.position = "none") +
  labs(x = "Time (min)",
       y = "&Delta;Krill oil − &Delta;Placebo<br>(mmol L<sup>-1</sup>)")


# Combine the full figure

cowplot::plot_grid(p_pred,
                   p_wcontr,
                   delta_contrasts,
                   ncol = 1,
                   rel_heights = c(1, 1, 1.4),
                   align = "v")
```

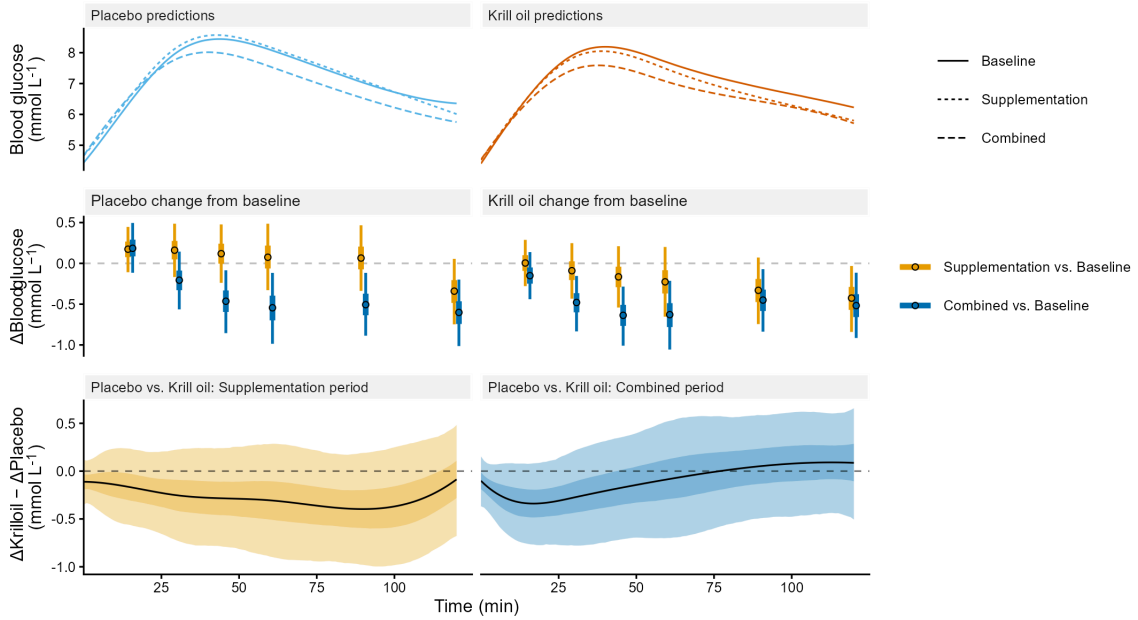

Figure 10.12: Model `mo4` predictions (top panels), within-group time-contrasts (middle panels) and between-group contrasts of changes (Δ) from baseline. Error bars and ribbons represent 50 and 95% intervals in the middle and lower panels.

```
# Population curves (pop_curve) and participant-level curves (part_curves)
# from the mo4 posterior were calculated for the posterior predictive checks above.

ggplot(pop_curve, aes(min_seq, m, color = treat, fill = treat)) +
  geom_line(data = part_curves,
            aes(min_seq, mu, group = interaction(p, treat)),
            alpha = 0.25, linewidth = 0.3) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), alpha = 0.2, color = NA) +
  geom_line(linewidth = 1) +
  facet_wrap(~ time) +
  alpha_theme() +
  scale_color_manual(values = colscale) +
  scale_fill_manual(values = colscale) +
  labs(x = "Minute",
       y = "Blood Glucose (mmol L<sup>-1</sup>)",
       color = "Treatment", fill = "Treatment") +
  theme(axis.title.y = ggtext::element_markdown())
```

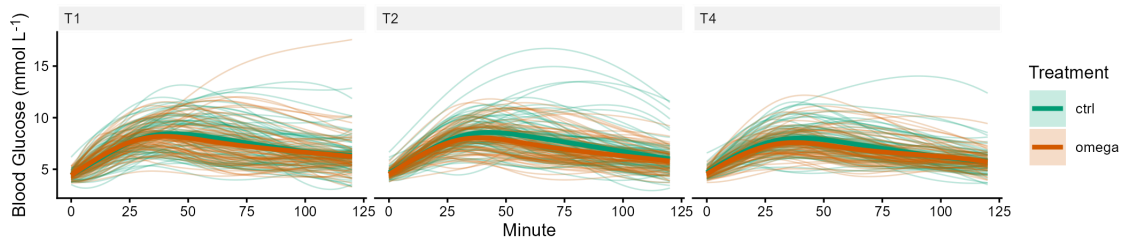

Figure 10.13: Predictions from the OGTT model `mo4` at each time point with participant-level predictions.

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